Applications of the Quasi Solution Method to Interfacial and Other Nonlinear Problems

DISSERTATION

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By

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Abstract

This thesis concerns analysis of two nonlinear problems: One concerns construction of accurate approximate Tritronquée solution to the Peinlevé-I (P1) equation with rigorous error bound in some domain that allows accurate confirmation of the pole location closest to the origin. The second part concerns determination of both the shape and velocity of steadily moving bubbles in a Hele-Shaw cell. We rigorously prove the existence of bubble solutions for a range of bubble size when neither surface tension nor bubble size are small.

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Fields of Study

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Chapter 1
Introduction

Nonlinear problems arise very frequently in nature. However, the methods available for these problems are rather limited. Here we present the application of a relatively new method (Quasi-Solution method [8]-[24]) to two different problems. One involves rigorous results for the classical Painlevè-1(P-1) equation involving a nonlinear second order ordinary differential equation. The second one, which is the main part of the thesis, is also a classical problem referred to as the Hele-Shaw flow originally studied more than fifty years back by Saffman and Taylor[31] and involves a nonlinear free boundary problem in fluid dynamics. Nonetheless, the quasi-solution method that has been employed gave us rigorous results in the regime that have not been studied.

The thesis is presented in the following manner. In the following chapter, we describe in detail the free boundary problem that involves a steadily moving bubbles in a Hele-Shaw cell. We present theorems that give constructive approximations to both the the shape of the bubble and its speed as a function of the bubble size parameter over a certain range. In the final chapter we present analysis of work that has already appeared in the *Journal of Differential Equations*. The main result is a constructive rigorous approximation of the Tritonqueé solution of P-1 in a certain domain and accurate prediction of the singularity closest to the origin.

We conclude this chapter by presenting the Quasi-Solution method in an abstract way: Suppose we have a nonlinear problem, written abstractly

$$\mathcal{N}[f] = 0,$$

for some nonlinear operator $\mathcal{N}$ in some suitable Banach space. Suppose we find f_0 with the property that

$$\mathcal{N}[f_0] = \mathcal{N}^0,$$

and $\mathcal{N}^0$ is small in some appropriate norm. If we decompose $f = f_0 + E$, then E satisfies

$$\mathcal{L}[E] = -\mathcal{N}^0 - r[E],$$

where $\mathcal{L}$ is the Frechèt derivative of $\mathcal{N}$ at $f = f_0$, which is assumed to exist, and where

$$r[E] = \mathcal{N}[f_0 + E] - \mathcal{N}[f_0] - \mathcal{L}[E]$$

For a regular enough nonlinearity $\mathcal{N}$, r will scale quadratically with E for small E. Now, assuming we have an explicit operator $\mathcal{L}^I$[(1)] on the appropriate space such that

$$\mathcal{E}_{\mathcal{L}} = \mathcal{L}^I\mathcal{L} - \mathcal{I}$$

is small in the corresponding operator norm where $\mathcal{I}$ is the identity operator, the question of solution to $\mathcal{N}[f] = 0$ in a neighborhood of $f = f_0$ is equivalent to determining solution E in some small ball, that satisfies

$$E = -\mathcal{E}_{\mathcal{L}}E - \mathcal{L}^I\mathcal{N}^0 - \mathcal{L}^I r[E], \tag{1.0.1}$$

If $\mathcal{L}^I$ and f_0 are appropriately chosen to ensure that $\mathcal{E}_{\mathcal{L}}$ and $\mathcal{L}^I\mathcal{N}^0$ are suitably small and $\mathcal{L}^I$ is conveniently bounded in some appropriate space, then a contraction mapping argument establishes existence of solution E in some ball.

This kind of analysis arises routinely in justifying asymptotic solutions to differential systems. For instance if we have a problem $\mathcal{N}[f, \epsilon] = 0$ and know the solution f_0 that satisfies $\mathcal{N}[f_0, 0] = 0$, then to prove that there is a solution f in the vicinity of f_0 for sufficiently small ϵ, one decomposes $f = f_0 + E$ and the equation for E is similar to (1.0.1) where remainder $R = \mathcal{N}[f_0, \epsilon]$.

Indeed, in a posteriori error analysis of finite element calculations, similar ideas have

[(1)] One may choose $\mathcal{L}^I$ as $\mathcal{L}^{-1}$ when $\mathcal{L}^{-1}$ is explicitly known.

been used[2]. This idea also forms the core of computer assisted proofs that arise in the context of ODE's or PDE's. For integro-differential equations in the context of steady periodic 2-D water waves, Kobayashi has employed this idea. However, in the analysis of Frenkel of 2-D water waves based on Nekrasov's integral formulation, f_0 contained just one term, and therefore the analysis did not require any computer assist.

What does not appear to have been realized until recently [8]-[24] is that the choice of quasi-solution f_0 in many problems arising in many diverse contexts can be made relatively simple. With right kind of inversion strategies and use of asymptotic information, the analysis of error E is simple enough to make the proofs quite transparent with minimal computer assist. In the next chapter, we employ a similar strategy for integro-differential equations that arise in steadily translating Hele-Shaw bubble over a range of bubble size parameter. A particular challenge here is to determine a formulation and a space of function that allows suitable inversion of the linearized second order integro-differential equations over a range of parameter values. We prove the existence of steadily translating bubble solution over a range of bubble size when the surface tension is not small. Computer assist is restricted to checking the values of certain explicit functions, and we have avoided numerical inversions of large matrices that is typical of many computer assisted proofs.

It is worth noting that we are unaware of any other rigorous mathematical method suitable for analysis when neither surface tension nor bubble size is small. Quasi-Solution method is observed to be a general strategy that works for any sized bubble. Unlike exponential asymptotic analysis, where control of the terms facilitated by analytic continuation, this constructive method does not require analysis in the unphysical complex-plane. Furthermore, it has the advantage of being adaptable to regularization other than surface tension.

Chapter 2
Hele-Shaw Analysis

2.1 Introduction

The problem of displacement of a more viscous fluid by a less viscous fluid in a Hele-Shaw cell is a classical problem that has garnered much attention in literature (for an extensive survey, see for instance, [17]). It is relevant to the general area of pattern formation[27] and is recognized as a canonical problem related to others that arises in porous media flow[11], dendritic crystal growth[20][27] and directional solidification[40]. More recently, it is shown that a free boundary model for streamers arising in dialectic breakdown satisfy similar equations[13]. There is also considerable recent interest in the general area of Laplacian growth, as some models are known to be connected to integrable systems[23]. Also, flow in narrow gaps, typified by Hele-Shaw flow, is technologically quite relevant in micro fluidic devices.

2.1.1 Two Phase Hele-Shaw Flow in Channel Geometry

A Hele-Shaw cell is the narrow gap between two closely placed parallel plates. The motion of two fluids with different viscosities in a Hele-Shaw cell, also referred as two phase Hele-Shaw flow, has been a subject for investigation for cells of various geometries. This research particularly focuses on two phase Hele-Shaw flows whose geometry, also referred as rectangular channel geometry, is described as follows: The plates of the cell are identical rectangles with width $2a$ and length L and are evenly spaced by a distance b. It is assumed that the geometry is suitable for Hele-Shaw approximation, that is, the gap between the

plates is very small compared to the width of the rectangles ($b/a << 1$), and the length of the rectangles is very large compared to the width ($L/a >> 1$, see Figure 2.1). It is also assumed that the cell is horizontal and pressure gradients due to gravity may be ignored.

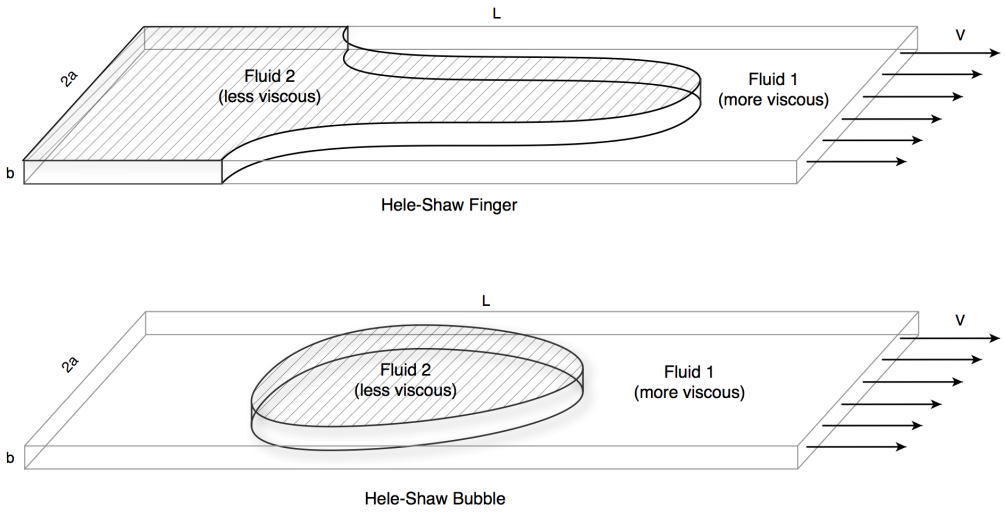


Figure 2.1: Examples of a Hele-Shaw Finger and a Hele-Shaw Bubble

2.2 Mathematical Formulation

When the flow is observed from the top (see Figure 2.2), or equivalently, projected onto an $\bar{x}\bar{y}$[(2)] plane parallel to the plates where the $\bar{x}$ coordinate is parallel to the long sides and passing through the centerline of the cell; the following mathematical equations are obtained: By averaging the Stokes equation (see, for example [21], section 330) the mean velocities of the fluids over the narrow gap are

$$\vec{\mathbf{u}}_i = -\left(\frac{b^2}{12\mu_i}\right)\nabla\bar{p}_i \text{ in } \bar{\Omega}_i(t), \tag{2.2.1}$$

where $\vec{\mathbf{u}}_i = (\bar{u}_i(\bar{x},\bar{y}), \bar{v}_i(\bar{x},\bar{y}))$ is the mean velocity across the gap, $\bar{p}_i(\bar{x},\bar{y})$ is the pressure, μ_i is the viscosity and $\bar{\Omega}_i(t)$ is the area of the fluid i for $i = 1, 2$. On introducing the velocity

[(2)]The bar dependences in the parameters will be removed after a new transformation is introduced.

potentials

$$\bar{\phi}_i = -\frac{b^2}{12\mu_i}\bar{p}_i \ , \tag{2.2.2}$$

the incompressibility condition $\nabla.\vec{\mathbf{u}}_i = 0$ implies

$$\Delta\bar{\phi}_i = 0 \text{ in } \bar{\Omega}_i(t) \ , \tag{2.2.3}$$

for $i = 1, 2$, which enables us to to introduce harmonic conjugates $\bar{\psi}_i$ of $\bar{\phi}_i$ for $i = 1, 2$. The natural conditions that the flow is continuous up to cell walls and that there is no slipping at the walls imply

$$\vec{\mathbf{u}}_1.\vec{n}_1 = 0 \text{ on } \bar{y} = \pm a, \tag{2.2.4}$$

where $\vec{n}_1$ is the outward normal vector. This readily translates into $\frac{\partial\bar{\phi}_1}{\partial n} = \frac{\partial\bar{\phi}_1}{\partial\bar{x}} = 0$ on $\bar{y} = \pm a$, or equivalently $\frac{\partial\bar{\psi}_1}{\partial s} = \frac{\partial\bar{\psi}_1}{\partial\bar{y}} = 0$ on $\bar{y} = \pm a$, and hence

$$\bar{\psi}_1 \equiv \text{ constant on } \bar{y} = \pm a, \tag{2.2.5}$$

Additionally, we assume that the viscous fluid is moving uniformly at a constant velocity V at the very far end of the channel due to a pressure gradient, or

$$\bar{\phi}_1 \sim V\bar{x} + O(1) \text{ as } \bar{x} \to \infty, \tag{2.2.6}$$

This condition implies $\bar{\psi}_1 \sim V\bar{y} + O(1)$ as $\bar{x} \to \infty$, and (2.2.5) can be calculated as

$$\bar{\psi}_1 \equiv \pm Va \text{ on } \bar{y} = \pm a, \tag{2.2.7}$$

We further assume that the interface of the two fluids $\bar{\Gamma}(t) = \bar{\Omega}_1(t) \cap \bar{\Omega}_2(t)$ is a simply connected curve and we impose two physically reasonable assumptions on $\bar{\Gamma}(t)$: **i)** the thin film effects (the phenomenon of intermingling of the two fluids at a thin layer around the interface due to the fluid motion) has no effect on the dynamics of the flow and that the fluid velocities are continuous across the interface, **ii)** the pressure drop across the two fluids equals to $\kappa \times T$, where κ is the bubble curvature, computed in the sense that it is positive when the bubble is circular, and T is the surface tension (which is also referred as McLean & Saffman boundary condition[22]). At this point it is worth highlighting

that the thin film effects can have singular impacts on the flow under some conditions(see, for example, [28][26][30] for further analysis). Let $\bar{\Gamma}(t) : \bar{F}(\bar{x}, \bar{y}, t) = 0$ describe the free boundary of the interface. The time derivative of $\bar{F}$ must vanish on the moving frame, i. e. $\bar{F}_t + \bar{F}_{\bar{x}}\bar{x}_t + \bar{F}_{\bar{y}}\bar{y}_t = 0$, implying $-\frac{\bar{F}_t}{|\nabla \bar{F}|} = \frac{\bar{F}_x x_t + \bar{F}_y y_t}{|\nabla \bar{F}|} = v_n$. Then the condition i) can be interpreted as

$$\frac{\partial \bar{\phi}_1}{\partial n_1} = \frac{\partial \bar{\phi}_2}{\partial n_2} = v_n = -\frac{\bar{F}_t}{|\nabla \bar{F}|} \text{ on } \bar{\Gamma}(t), \tag{2.2.8}$$

where $\nabla \bar{F} = (\bar{F}_{\bar{x}}, \bar{F}_{\bar{y}})$, n_i are unit normal vectors, and v_n is the velocity in the normal direction. The condition ii), thanks to (2.2.2), translates into

$$\bar{\phi}_1 - \left(\frac{\mu_2}{\mu_1}\right)\bar{\phi}_2 = \left(\frac{b^2 T}{12\mu_1 V a^2}\right)\kappa \text{ on } \bar{\Gamma}(t), \tag{2.2.9}$$

At this point we simplify the equations of the motion by choosing $a = 1$ and $V = 1$, which is equivalent to nondimensionalizing all length scales by a and the time scale by $\frac{a}{V}$, and combine all equations which determine the evolution of the flow as

$$\begin{aligned}
\Delta \bar{\phi}_i &= 0 \text{ in } \bar{\Omega}_i(t), && (2.2.10)\\
\bar{\psi}_1 &= \pm 1 \text{ on } \bar{y} = \pm 1, && (2.2.11)\\
\bar{\phi}_1 &\sim \bar{x} + O(1) \text{ as } \bar{x} \to \infty, && (2.2.12)\\
\frac{\partial \bar{\phi}_i}{\partial n_i} &= -\frac{\bar{F}_t}{|\nabla \bar{F}|} \text{ on } \bar{\Gamma}(t), && (2.2.13)\\
\bar{\phi}_1 - \left(\frac{\mu_2}{\mu_1}\right)\bar{\phi}_2 &= \bar{B}\kappa \text{ on } \bar{\Gamma}(t), && (2.2.14)
\end{aligned}$$

for $i = 1, 2$, where $\bar{B} = \frac{b^2 T}{12\mu_1 a^2}$.

2.2.1 History and Mathematical Issues

One problem of particular historical interest is the selection of a steadily translating finger or bubble [39]. In a model where surface tension or any other regularization is ignored, it is known since Saffman and Taylor that there is a continuum of solutions characterized by speed of finger[(3)][29] or bubble[29]. The selection problem remained remained open for a long time, until numerical calculations[22][41][33] in the eighties supported by formal

[(3)]In the case of a finger, this relative speed is reciprocal of the finger width relative to the channel width.

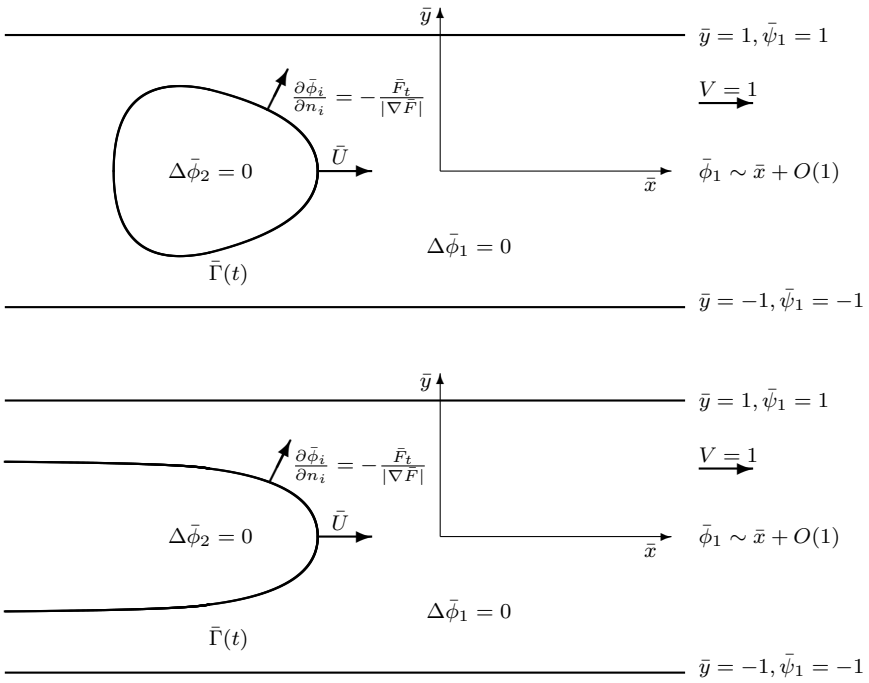


Figure 2.2: Hele-Shaw cell geometry for moving bubble and finger in the physical $\bar{x}\bar{y}$-plane

asymptotic calculations [16][32][33][34] suggested that surface tension selected a discrete set of solutions and that exponentially small terms play a vital role in this selection, though there were a few counter-claims [42][12]. Numerical and formal asymptotic results also confirm that only one of these discrete solutions is linearly stable, while others are unstable. The steady state results have since been using rigorous exponential asymptotic. There is another limit where the mathematical problem is amenable to rigorous analysis- the limit when bubbles are small relative to cell-width.

Mathematical issues arise in **(1)** proving the existence of solution to the initial value problem, **(2)** proving the existence of steadily translating flows, and **(3)** analyzing the stability of the steadily translating solutions. Existence issue was recognized to be a difficult problem when $\bar{B} \neq 0$. $\bar{B} = 0$ is completely analyzed and observed to allow many exact solutions (see, for example, [29]). However, it is also known that $\bar{B} = 0$ case is ill posed in the sense that two initial shapes that are arbitrarily close to each other end up giving

completely different shapes in finite time[18]. It is also structurally unstable since $\bar{B} = 0$ solutions need not be $\lim_{\bar{B}\to 0}$ solutions.

2.3 The Problem of Steadily Translating Symmetric Bubble with Surface Tension

It is noted that the main focus of the current work is on **(2)** in the case where $\bar{B}$ is significantly large. It is now assumed that the bubble (or finger) is translating at a constant velocity $\bar{U}$. Without loss of generality one may choose $\bar{\phi}_2 = \bar{U}\bar{x}$ and $\bar{\psi} = \bar{U}\bar{y}$. On applying the Saffman-Taylor transformation [39], [37] to the evolution equations (2.2.10)-(2.2.14)

$$\begin{gathered} x = \bar{x} - \bar{U}t \ , \ y = \bar{y} \ , \ U = \frac{\bar{U} - \frac{\mu_2}{\mu_1}\bar{U}}{1 - \frac{\mu_2}{\mu_1}\bar{U}}, \\ \psi(x,y) = \frac{\bar{\psi}_1(\bar{x},\bar{y}) - \bar{U}\bar{y}}{1 - \frac{\mu_2}{\mu_1}\bar{U}} \ , \ \phi(x,y) = \frac{\bar{\phi}_1(\bar{x},\bar{y}) - \bar{U}\bar{x}}{1 - \frac{\mu_2}{\mu_1}\bar{U}} \ , \ F(x,y,t) = \bar{F}(\bar{x},\bar{y},t) \end{gathered} \tag{2.3.1}$$

which amounts to switching from translating bubble/finger frame to steady bubble/finger frame and appropriately taking away the effects of $\bar{\phi}_2$ and $\bar{\psi}_2$ from both velocity potentials, the equations are simplified to

$$\Delta\phi = 0 \text{ in } \Omega, \tag{2.3.2}$$

$$\psi = \mp(U-1) \text{ on } y = \pm 1, \tag{2.3.3}$$

$$\phi \sim -(U-1)x + O(1) \text{ as } x \to \infty, \tag{2.3.4}$$

$$\psi = 0 \text{ on } \Gamma, \tag{2.3.5}$$

$$\phi + Ux = B\kappa \text{ on } \Gamma, \tag{2.3.6}$$

for $i = 1, 2$, where

$$\Omega = \Omega_1(t) - \bar{U}t \ , \ \Gamma = \bar{\Gamma} - \bar{U}t \ , \ B = \frac{\bar{B}}{1 - \frac{\mu_2}{\mu_1}}, \tag{2.3.7}$$

It is noted that the equations except (2.3.5) are immediate when the transformation is applied to the equations (2.2.10)-(2.2.14) and (2.3.5) is obtained in the following way: the transformation converts (2.2.13) into $\frac{\partial\phi}{\partial n} = 0 = \frac{F_t}{|\nabla F|}$, equivalently $\frac{\partial\psi}{\partial s} = 0$, i. e. ψ is constant

on Γ, and is set to be 0 without loss of generality.

We restrict ourself to the case of the *bubble*. We also limit ourselves to the problem of determining a *symmetric bubble*, where Γ remains invariant under $(x, y) \to (x, -y)$ transformation. The more general problem where symmetry is not assumed *a priori* but a consequence of surface tension effects is yet to be rigorously analyzed, though both numerical[34] and formal asymptotic computations[6] suggest this to be the case. A continuum of non-symmetric bubble solutions exist when surface tension is completely ignored[34].

2.3.1 Complex Analytic Formulation of The Problem

In this section we present how the Hele-Shaw problem described above can be interpreted as an equivalent complex analysis problem. We adopt the formulation in Tanveer[34]. we set $z = x + iy$ and we introduce the complex velocity potential $W = \phi + i\psi$ which is an analytic function of z on Ω due to (2.3.2).

Definition 2.3.1. For $0 < \rho < 1$, define the annular and the upper-half annular domains

$$\mathcal{A}_\rho = \{\zeta \in \mathbb{C} : \rho < |\zeta| < 1\} \quad , \quad D_\rho = \{\zeta \in \mathbb{C} : \rho < |\zeta| < 1 \ \ , \Im\, \zeta > 0\} \tag{2.3.8}$$

Introduce the conformal map $z = Z(\zeta)$ where $Z : D_\rho \to \Omega$ such that $Z(\mp 1)$ is mapped to the front and rear points of the bubble, $\zeta = \mp\rho$ corresponds to $\Re z = \pm\infty$. If bubble shape were known, the parameter $0 < \rho < 1$ would have to be determined as part of the problem. However, since we are considering a free boundary problem, it is convenient to choose ρ as a given 'size' parameter, since $\rho \to 0^+, 1^-$ corresponds to infinitesmially small and infinitely large sized bubbles respectively.

Then the equivalent problem is finding z, W and U where W satisfies

$$Im(W) = 0 \text{ on } -1 < \zeta < -\rho \text{ and } \rho < \zeta < 1, \tag{2.3.9}$$

$$Im(W) = 0 \text{ on } |\zeta| = 1, \tag{2.3.10}$$

$$Im(W) = -(U - 1) \text{ on } |\zeta| = \rho, \tag{2.3.11}$$

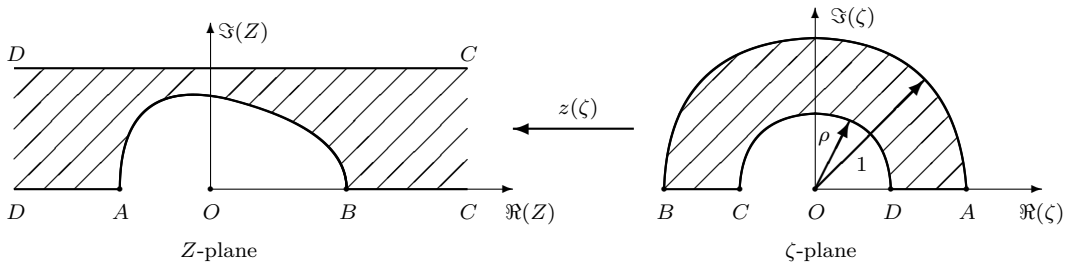


Figure 2.3: Conformal map between half of the bubble in z-plane and half annular region in the ζ-plane

and where z, W and U satisfy

$$Im(z) = 0 \text{ on } -1 < \zeta < -\rho \text{ and } \rho < \zeta < 1, \tag{2.3.12}$$

$$Im(z) = 1 \text{ on } |\zeta| = \rho, \tag{2.3.13}$$

$$z \to \mp\infty \text{ as } \zeta \to \pm\rho, \tag{2.3.14}$$

$$Re(W) + URe(z) = -\frac{B}{|z_\zeta|} Re\Big[\zeta \frac{d}{d\zeta} \ln(\zeta z_\zeta)\Big] \text{ on } |\zeta| = 1, \tag{2.3.15}$$

We would like to note that although the complex analytic version of the problem opens up possibilities for deeper complex analytic investigation (see for example [36][37]) this research requires such analysis at a minimum level. We only use the fact that z must have a specific series representation. This representation enabled computer programs to converge to a solution in a very robust way where the others observed to be incapable of, and it provided a compact representation for a quasi solution f_0.

2.4 Alternative Formulations

We seek to determine a steadily translating bubble in a Hele-Shaw cell. In the case of a bubble that is symmetric about the channel centerline, in the frame of the moving bubble, this corresponds to determining a complex analytic W function of $z = x + iy$ in the domain Ω shown in Figure 2.3 such that

$$\Im\, W = -(U-1) \text{ on } \Im z = 1 \ , \ \Im\, W = 0 \ \text{ when } z \in \{\Im\, z = 0\} \cap \partial\Omega \ , \tag{2.4.1}$$

with far-field condition

$$W = -(U-1)z + O(1) \ \text{ as } z \to \infty \ , \tag{2.4.2}$$

and following two boundary conditions on the free boundary segment Γ in the upper-half z-plane:

$$\Im\, W = 0 \ \ , \ \Re\,(W + Uz) = \mathcal{B}\kappa \ , \tag{2.4.3}$$

where κ is the interfacial curvature.

It is also clear that, modulo an constant which does not affect the shape, Z has the following series representation

$$Z(\zeta) = \frac{4}{\pi}\left[\frac{1}{2}\log\left(\frac{\zeta-\rho}{\zeta+\rho}\right) + \sum_{n=1}^{\infty} z_n\left(\zeta^n + \rho^{2n}\zeta^{-n}\right)\right] \ , \quad \text{for } z_n \in \mathbb{R} \tag{2.4.4}$$

where regularity of the interface is reflected through the behavior of z_n for large n. For instance, $n^2 z_n \in l^1(\mathbb{Z})$ ensures $Z_{\zeta\zeta}$ is continuous on $\rho \le |\zeta| \le 1$. Furthermore, from Schwartz reflection principle, it follows Z maps the annular domain $\mathcal{A}_\rho$ into the exterior of the bubble between two Hele-Shaw walls $-1 < y < 1$. Furthermore, for the mapping Z to be meaningful physically, Z in the form (2.4.4) must be univalent in $\mathcal{A}_\rho$. The complex velocity potential W can be determined exactly by applying method of images, and has the representation

$$W(\zeta) = W_0 + \frac{2}{\pi}(U-1)\left[-\log\left(\frac{\zeta-\rho}{\zeta+\rho}\right) + \sum_{n=0}^{\infty}\frac{2\rho^{2n+1}}{(2n+1)(1-\rho^{4n+2})}\left(\zeta^{2n+1} + \rho^{4n+2}\zeta^{-(2n+1)}\right)\right] \tag{2.4.5}$$

where W_0 is some real constant that will be chosen suitably. Note that W may also be expressed in terms of an elliptic function[34]:

$$W(\zeta) = W_0 - \frac{2}{\pi}(U-1)\log\left\{\frac{\theta_4\left(-\frac{i}{2}\log\zeta, \rho\right)}{\theta_4\left(-\frac{i}{2}\log\zeta - \frac{\pi}{2}, \rho\right)}\right\} \ , \tag{2.4.6}$$

where $\theta_4(t,\rho)$ is a well-known elliptic function with the following rapidly convergent representation

$$\theta_4(t,\rho) = 1 + 2\sum_{n=1}^{\infty}(-1)^n \rho^{n^2}\cos(2nt) \tag{2.4.7}$$

It is to be noted that if we introduce symbol

$$\gamma = \exp\left[-\frac{\pi}{(U-1)}(W(\zeta)-W_0)\right] = \frac{\theta_4^2\left(-\frac{i}{2}\log\zeta,\rho\right)}{\theta_4^2\left(-\frac{i}{2}\log\zeta-\frac{\pi}{2},\rho\right)} \tag{2.4.8}$$

then relating $\alpha \in (0,1)$ to $\rho \in (0,1)$ through the relation

$$\alpha = \frac{\theta_4\left(-\frac{\pi}{2},\rho\right)-\theta_4(0,\rho)}{\theta_4\left(-\frac{\pi}{2},\rho\right)+\theta_4(0,\rho)} = \frac{2\sum_{k=1}^{\infty}\rho^{(2k-1)^2}}{1+2\sum_{k=1}^{\infty}\rho^{4k^2}}\ , \tag{2.4.9}$$

the mapping

$$\chi(\zeta) = \frac{(1-\gamma)(1+\alpha^2)}{2\alpha(1+\gamma)} - \frac{1}{2\alpha(\gamma+1)}\sqrt{(\gamma-1)^2(1+\alpha^2)^2-4\alpha^2(1+\gamma)^2} \tag{2.4.10}$$

maps the annular region outside between the two circles to the region exterior of the cut between $[-\alpha,\alpha]$ but interior of the unit circle. We notice that in accordance to (2.4.5) and (2.4.10), $\zeta=\rho$ corresponds to $\Re W=+\infty$, which corresponds to $\gamma=0$, which is mapped to $\chi=(1+\alpha^2)/(2\alpha)-1/(2\alpha)(1-\alpha^2)=\alpha$. and similarly $\zeta=-\rho$ corresponds to $\chi=-\alpha$. In the unit χ plane, though we have a somewhat simpler representation of the free boundary problem, nonetheless, this is not very suitable for large bubble size.

In the domain D_ρ, we can re-state the mathematical problem in the following manner:

Problem 1: Given $0<\rho<1$, determine bubble velocity $U\in\mathbb{R}^+$ and mapping Z in the form (2.4.4) that is univalent and C^2 in $\overline{\mathcal{A}_\rho}$, such that

$$\Re\ (W+UZ) = \mathcal{B}\kappa,\ \text{on}\ |\zeta|=1, \tag{2.4.11}$$

where curvature

$$\kappa = -\frac{1}{|Z_\zeta|}Re\left[\zeta\frac{d}{d\zeta}\log\left(\zeta Z_\zeta\right)\right]. \tag{2.4.12}$$

The formulation in Problem 1 is not as convenient for applying the quasi-solution method. For one thing, it is *a priori* unclear how the restriction on U arises. Indeed, when $\mathcal{B}=0$, there is no restriction on $U\in(1,\infty)$ as mentioned earlier in the introduction. Therefore, we introduce change of variable

$$\omega(\zeta) = \log\left(\frac{\pi}{4\rho}(\zeta^2-\rho^2)Z_\zeta\right)\quad,\ Z_\zeta = \frac{4\rho}{\pi(\zeta^2-\rho^2)}e^{\omega} \tag{2.4.13}$$

In order that Z_ζ has residues at the poles $\zeta = \pm\rho$ that matches (2.4.4), it is imperative that

$$\omega(\rho) = 0 = \omega(-\rho) \tag{2.4.14}$$

We also note that the curvature κ in (2.4.12) defined on $|\zeta| = 1$ becomes

$$\kappa = -\frac{1}{|Z_\zeta|}\ \Re\left[\zeta\frac{d\omega}{d\zeta} - \frac{\zeta^2+\rho^2}{\zeta^2-\rho^2}\right] \tag{2.4.15}$$

From representation (2.4.4) and univalence of Z in the closure of $\mathcal{A}_\rho$ (which implies $Z_\zeta \neq 0$), it follows at once that ω is analytic in $\mathcal{A}_\rho$ and therefore in D_ρ with series representation

$$\omega(\zeta) = \omega_0 + \sum_{n=1}^{\infty}\omega_n\left(\zeta^n + \frac{\rho^{2n}}{\zeta^n}\right) \quad , \text{ where } \omega_n \in \mathbb{R}\ . \tag{2.4.16}$$

We will look for a solution to Problem 1 in the space of functions where ω in (2.4.16) satisfies

$$n\omega_n \in l^1(\mathbb{Z}^+) \tag{2.4.17}$$

This would ensure that $Z_{\zeta\zeta}$ is continuous in $\overline{D_\rho}$ and therefore in particular the interface curvature κ is continuous. On $|\zeta| = 1$, with $\zeta = e^{i\nu}$, if we define

$$\omega(e^{i\nu}) = \omega_0 + \phi(\nu) + i\psi(\nu) \tag{2.4.18}$$

then it follows from (2.4.16) that ϕ and ψ are even and odd 2π-periodic functions of ν with representation

$$\phi(\nu) = \sum_{n=1}^{\infty}\omega_n(1+\rho^{2n})\cos(n\nu) \quad , \ \psi(\nu) = \sum_{n=1}^{\infty}\omega_n(1-\rho^{2n})\sin(n\nu) \tag{2.4.19}$$

Using known result

$$\frac{1}{2\pi}PI\int_{-\pi}^{\pi}\cot\left(\frac{\nu'-\nu}{2}\right)\sin(n\nu')d\nu' = \cos(n\nu)\ ,$$

and relating cosine and sine series coefficient with integrals of function, we arrive at

$$\phi(\nu) = \frac{1}{2\pi}PI\int_{-\pi}^{\pi}K(\nu'-\nu)\psi(\nu')d\nu' =: \mathcal{H}[\psi](\nu), \tag{2.4.20}$$

where

$$K(\theta) = \cot\frac{\theta}{2} + \sum_{n=1}^{\infty} \frac{2\rho^n}{1-\rho^{2n}} \sin(n\theta) \ . \tag{2.4.21}$$

We now formulate a nonlinear integro-differential equation for $\psi_\nu = \Im\left(i\zeta\omega_\zeta\right)$ on $\zeta = e^{i\nu}$, which from (2.4.16) has the Fourier representation

$$\psi_\nu = \sum_{n=1}^{\infty} n\omega_n \left(1-\rho^{2n}\right)\cos(n\nu) \tag{2.4.22}$$

It is clear that that ψ_ν determines ϕ and ψ, either through series coefficients, or through integration in ν and application of annular Hilbert-Transform operator $\mathcal{H}$ defined in (2.4.21).

Definition 2.4.1. We define $\mathcal{S}_k, \mathcal{C}_k$ to be class of 2π real periodic odd and even functions respectively with representations

$$\psi(\nu) = \sum_{n=1}^{\infty} \psi_n \sin(n\nu) \quad , \quad \text{with } \|\psi\|_k = \sum_{n=1}^{\infty} n^k |\psi_n| < \infty \tag{2.4.23}$$

$$\phi(\nu) = \phi_0 + \sum_{n=1}^{\infty} \phi_n \cos(n\nu) \quad , \quad \text{with } \|\phi\|_k = |\phi_0| + \sum_{n=1}^{\infty} n^k |\phi_n| < \infty \tag{2.4.24}$$

The subspace of $\mathcal{C}_k$ with zero average, *i.e.* $\phi_0 = 0$, will be denoted by $\mathcal{C}_k^0$.

Lemma 2.4.2. If there exists solution to Problem 1 for $n\omega_n \in l^1(\mathbb{Z}^+)$, then $n^2\omega_n \in l^1(\mathbb{Z}^+)$ as well, implying existence of $\omega_{\zeta\zeta}$ on $|\zeta| = 1$.

Proof. We note that using (2.4.15) in (2.4.11) leads to following relation on the unit circle $\zeta = e^{i\nu}$:

$$\psi_\nu = \Re\left[\zeta\omega_\zeta\right] = \Re\left[\frac{1+\rho^2\zeta^2}{1-\rho^2\zeta^2}\right] - \frac{|Z_\zeta|}{\mathcal{B}}\Re\left(W + UZ\right) \tag{2.4.25}$$

We note that if $n\omega_n \in l^1(\mathbb{Z}^+)$, then using (2.4.13) and Banach algebra of product of Fourier cosine series , it follows that the right hand side of (2.4.25) on $\zeta = e^{i\nu}$ has a Fourier cosine series representation

$$r(\nu) = \sum_{n=0}^{\infty} r_n \cos(n\nu) \ \text{ with } r_n \in \mathbb{R} \ \text{ with } nr_n \in l^1(\mathbb{Z}^+) \tag{2.4.26}$$

Therefore, derivative of the right hand side of (2.4.25) exists with respect to ν on $\zeta = e^{i\nu}$, implying from (2.4.16) that $n^2\omega_n \in l^1(\mathbb{Z}^+)$, which in particular implies $\omega \in C^2(\overline{D_\rho})$. □

Remark 2.4.3. The bootstrapping argument above can be applied indefinitely to conclude $n^k\omega_n \in l^1(\mathbb{Z}^+)$ for any $k > 0$ implying that the interface is C^∞; indeed, for small enough B, it is known that the interface is analytic[38]. Here, we will only need $\omega \in C^2(\overline{D_\rho})$ in the last lemma, which implies in particular that $\omega_{\zeta\zeta}$ is continuous in the closure of D_ρ.

Definition 2.4.4. Define nonlinear operator $\mathcal{N}$ acting on functions[4] in $\mathcal{C}_1^0$ depending on scalars ω_0, U, such that

$$\begin{aligned}\mathcal{N}\left[\psi_\nu;\omega_0,U\right] = \psi_{\nu\nu} + \frac{(1-\rho^4)q_\nu}{q^2} + \left(\psi_\nu - \frac{1-\rho^4}{q}\right)\left(-\phi_\nu + \frac{q_\nu}{2q}\right) \\ - \frac{1}{B}|Z_\zeta|\Big\{U\Im\left[\zeta Z_\zeta\right] + (U-1)Q\Big\}\ ,\end{aligned} \tag{2.4.27}$$

where

$$Z_\zeta = \frac{4\rho e^{\omega_0+\phi(\nu)+i\psi(\nu)}}{\pi(e^{2i\nu}-\rho^2)}\ ,\quad \left|Z_\zeta\right| = \frac{4\rho e^{\omega_0+\phi(\nu)}}{\pi q^{1/2}}\ , \tag{2.4.28}$$

where

$$\begin{aligned} Q &= \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{2\rho^{2n+1}}{1-\rho^{4n+2}}\sin\left((2n+1)\nu\right), \\ q &= \left|\zeta^2-\rho^2\right|^2 = 1+\rho^4-2\rho^2\cos(2\nu), \end{aligned}$$

and where ψ, ϕ appearing in (2.4.27),(2.4.28) are determined from ψ_ν through the relations

$$\begin{aligned}\psi(\nu) &= \mathcal{I}\left[\psi_\nu\right](\nu) := \int_0^\nu \psi_\nu(\nu')d\nu', \\ \phi(\nu) &= \mathcal{H}\left[\psi\right](\nu)\end{aligned}$$

We also define linear functionals[5] M_1 and M_2 of of ψ_ν with representation (2.4.22) such that

$$M_1\left[\psi_\nu\right] = 2\sum_{j=1}^{\infty}\omega_{2j}\rho^{2j}\ , \tag{2.4.29}$$

$$M_2\left[\psi_\nu\right] = 2\sum_{j=1}^{\infty}\omega_{2j-1}\rho^{2j-1}\ , \tag{2.4.30}$$

Proposition 2.4.5. If Z restricted to the class of functions where corresponding ω in

[4] Note such functions are necessarily the derivative of some $\psi \in \mathcal{S}_2$

[5] These have alternate expressions in terms of integrals involving ψ_ν

(2.4.13) satisfies restriction (2.4.17), then Problem 1 is equivalent to **Problem 2**: Determine (ψ_ν, ω_0, U), where $\psi_\nu \in \mathcal{C}_1^0$, $\omega_0 \in \mathbb{R}$ and $U \in \mathbb{R}^+$ such that

$$\mathcal{N}[\psi_\nu; \omega_0, U] = 0 \tag{2.4.31}$$

$$\omega_0 + M_1[\psi_\nu] = 0 \tag{2.4.32}$$

$$M_2[\psi_\nu] = 0\ , \tag{2.4.33}$$

when $Z(\zeta)$ determined from ω through (2.4.13) is univalent, where ω is determined from ψ_ν through (2.4.16) and (2.4.22).

Proof. Taking derivative of (2.4.11) with respect to ν, multiplying through by $\frac{|Z_\zeta|}{B}$, and realizing $\frac{d}{d\nu} = i\zeta\frac{d}{d\zeta}$ for $\zeta = e^{i\nu}$, using (2.4.13) and (2.4.18), and noting that for $\zeta = e^{i\nu}$, $\frac{\zeta^2+\rho^2}{\zeta^2-\rho^2} = \frac{1-\rho^4}{q} + i\frac{q_\nu}{2q}$, we obtain (2.4.31). Clearly the steps above are reversible, and if (2.4.31) is satisfied, then (2.4.11) follows, modulo an additive constant that can be chosen to be zero without any loss of generality by choosing real constant W_0 in (2.4.5) appropriately. Now, the form of Z in (2.4.4) immediately implies the constraints (2.4.14), which is equivalent to $\frac{1}{2}\omega(\rho) + \frac{1}{2}\omega(-\rho) = 0$ and $\frac{1}{2}\omega(\rho) - \frac{1}{2}\omega(-\rho) = 0$. This is equivalent to (2.4.32)-(2.4.33) on using series representation (2.4.16). □

Remark 2.4.6. Our strategy will be to first show that (2.4.31)-(2.4.33) has a unique solution in some small neighborhood of a quasi-solution $(\psi_\nu^0, U^0, \omega_0^0)$. We will use properties of corresponding Z^0 and smallness in the estimate of $\psi_\nu - \psi_\nu^0$ to prove that the corresponding Z must be univalent, completing the proof of solution of Problem 2.

Definition 2.4.7. For given quasi-solution $\psi_\nu^0 \in \mathcal{C}_1^0$, and constants ω_0^0, U^0 with representation

$$\psi_\nu^0 = \sum_{n=1}^{\infty} n\omega_n^0 \left(1 - \rho^{2n}\right)\cos(n\nu)\ , \tag{2.4.34}$$

we define corresponding analytic function

$$\omega^0(\zeta) = \omega_0^0 + \sum_{n=1}^{\infty} \omega_n^0 \left(\zeta^n + \rho^{2n}\zeta^{-n}\right)\ . \tag{2.4.35}$$

We define corresponding Z^0 through integration of the relation

$$Z^0_\zeta(\zeta) = \frac{4\rho}{\pi(\zeta^2-\rho^2)} e^{\omega_0(\zeta)} \ . \tag{2.4.36}$$

We also define

$$u = U - U^0 \tag{2.4.37}$$

$$\Omega(\zeta) = \omega(\zeta) - \omega^0(\zeta) = w + \sum_{n=1}^{\infty} \Omega_n \left(\zeta^n + \rho^{2n}\zeta^{-n}\right) \tag{2.4.38}$$

Also, we define E, F in $\mathcal{S}_1$ and $\mathcal{C}^0_1$ respectively so that on $\zeta = e^{i\nu}$,

$$\Omega(\zeta) = w + F(\nu) + iE(\nu) \tag{2.4.39}$$

Clearly, F and E have fourier cosine and sine series representation, involving Ω_n.

Lemma 2.4.8. (ψ_ν, U, ω_0) satisfies (2.4.31)-(2.4.33) if and only if

$$(E_\nu, w, u) = \left(\psi_\nu - \psi^0_\nu, \omega_0 - \omega^0_0, U - U^0\right)$$

satisfies

$$\mathcal{L}E_\nu = -\mathcal{N}^0 - uA_1 - wA_2 - R\left[E_\nu; w, u\right] \ , \tag{2.4.40}$$

$$w + M_1\left[E_\nu\right] = -\omega^0_0 - M_1\left[\psi^0_\nu\right] \ , \tag{2.4.41}$$

$$M_2\left[E_\nu\right] = -M_2\left[\psi^0_\nu\right] \ , \tag{2.4.42}$$

where

$$\mathcal{N}^0 := \mathcal{N}\left[\psi^0_\nu, \omega^0_0, U^0\right] \ , \tag{2.4.43}$$

$$\mathcal{L}E_\nu = E_{\nu\nu} + E_\nu\left(-\phi^0_\nu + \frac{q_\nu}{2q}\right) - F_\nu\left(\psi^0_\nu - \frac{1-\rho^4}{q}\right) + A_2F + A_3E \ , \tag{2.4.44}$$

where

$$A_1 = a + c \ , \ A_2 = 2U^0a + (U^0-1)c \ , \ A_3 = U^0b \ , \tag{2.4.45}$$

with

$$a = -\frac{|Z^0_\zeta|}{B}\Im\left[\zeta Z^0_\zeta\right] \ , \ b = -\frac{|Z^0_\zeta|}{B}\Re\left[\zeta Z^0_\zeta\right] \ , \ c = -\frac{|Z^0_\zeta|}{B}Q \ , \tag{2.4.46}$$

and nonlinear terms

$$R\left[E_\nu, w, u\right] = \mathcal{N}\left[\psi_\nu^0 + E_\nu, \omega_0^0 + w, U^0 + u\right] - \mathcal{N}\left[\psi_\nu^0, \omega_0^0, U^0 + u\right] - \mathcal{L}E_\nu - uA_1 - wA_2 \tag{2.4.47}$$

Proof. The proof is obvious from inspection, after we note the cancellation of terms $\mathcal{L}E_\nu$ and $\mathcal{N}^0$ with same terms appearing in R. □

Lemma 2.4.9. The nonlinear term R defined in (2.4.47) may be expressed as

$$R = R_0 + R_1 \ , \tag{2.4.48}$$

where

$$\begin{aligned} R_0 = &-E_\nu F_\nu + \Im\left\{(b+ia)\left[\frac{U_0}{2}\left(w+F+iE\right)^2 + \left(U_0(w+F)+u\right)\left(w+F+iE\right)\right]\right\} \\ &+ \frac{1}{2}\left(aU_0 + c(U_0-1)\right)(w+F)^2 + u(a+c)(w+F) \end{aligned} \tag{2.4.49}$$

$$\begin{aligned} R_1 = &\, e_2[w+F]\left\{U\Im\left[(b+ia)e^{w+F+iE}\right] - U_0 a\right\} + U_0 a e_3[w+F] \\ &+ U_0\Im\left[(b+ia)e_3[w+F+iE]\right] + \left((1+w+F)U - U_0\right)\Im\left((b+ia)e_2[w+F+iE]\right) \\ &+ (w+F)u\Im\left[(b+ia)(w+F+iE)\right] + uce_2[w+F] + (U_0-1)ce_3[w+F] \ , \end{aligned} \tag{2.4.50}$$

where $U = U_0 + u$ and exponential tail function

$$e_j[x] := \sum_{m=j}^{\infty} \frac{x^m}{m!} \ , \tag{2.4.51}$$

Proof. Using definition of a, b and c, we notice that

$$\begin{aligned} R = &\, \mathcal{N}\left[\psi_\nu^0 + E_\nu, \omega_0^0 + w, U_0 + u\right] - \mathcal{N}\left[\psi_\nu^0, \omega_0^0, U_0\right] - \mathcal{L}E_\nu = -E_\nu F_\nu - \frac{|Z_\zeta|}{B} U\Im[\zeta Z_\zeta] \\ &- \frac{(U-1)Q|Z_\zeta|}{B} + \frac{|Z_\zeta^0|}{B} U_0\Im[\zeta Z_\zeta^0] - \frac{(U-1)Q|Z_\zeta^0|}{B} - A_2(F+w) - A_3E - uA_1 \ . \end{aligned} \tag{2.4.52}$$

We notice from definition of a, b, c in (2.4.46) and the relation (2.4.28), and recalling $\omega_0 =$

$\omega_0^0 + w$, that

$$-\frac{(U-1)|Z_\zeta|Q}{B} = c(U_0 - 1 + u)e^{w+F} \ , \ \ -\frac{U|Z_\zeta|}{B}\Im\left[\zeta Z_\zeta\right] = (U_0+u)e^{w+F}\Im\left[(b+ia)e^{w+F+iE}\right] \tag{2.4.53}$$

We then use

$$\begin{aligned} &Ue^{w+F}\Im\left[(b+ia)e^{w+F+iE}\right] = e_2[w+F]\Im\left[U(b+ia)e^{w+F+iE} - U_0 a\right] \\ &+ U_0 a e_2[w+F] + U(1+w+F)\Im\left((b+ia)e_2[w+F+iE]\right) \\ &+ (U_0+u)(1+w+F)\Im\left[(b+ia)(1+w+F+iE)\right] \ , \end{aligned}$$

$$c(U_0 - 1 + u)e^{w+F} = c(U_0 - 1 + u)e_2[w+F] + c(U_0 - 1 + u)(1+w+F) \ ,$$

and separate out the quadratic terms and other nonlinear terms to identify R_0 and R_1. $\square$

Remark 2.4.10. R_0 defined above isolates quadratic terms in w, F, E, while R_1 includes cubic and higher order nonlinearities.

Definition 2.4.11. For $f \in \mathcal{C}$ of $f \in \mathcal{S}$, we denote N-th Fourier cosine/sine mode as $F_N[f]$. For $f \in \mathcal{C}_0$, we define functionals $M_\pm : \mathcal{C}_0 \to \mathbb{R}$ as

$$M_\pm[f] = \sum_{N=1}^{\infty} \frac{\rho^N}{N}\left(1 \pm (-1)^N\right) F_N[f] \ , \tag{2.4.54}$$

and vector functional $\mathbf{m} : \mathcal{C}_0 \to \mathbb{R}^2$ through

$$\mathbf{m}[f] := [M_+[f], M_-[f]]^T \tag{2.4.55}$$

For $f, g \in \mathcal{C}_0$, we define 2×2 matrix functional

$$\mathbf{m}[f,g] := (\mathbf{m}[f], \mathbf{m}[g]) = \begin{pmatrix} M_+[f] & M_+[g] \\ M_-[f] & M_-[g] \end{pmatrix} \tag{2.4.56}$$

Proposition 2.4.12. Let $\mathcal{L}^I : \mathcal{S} \to \mathcal{C}_0$ be a linear operator, and let

$$\mathcal{E}_\mathcal{L} := \mathcal{L}^I \mathcal{L} - \mathcal{I}, \tag{2.4.57}$$

where $\mathcal{I}$ is the identity operator on $\mathcal{C}$. Let

$$\mathbf{a} = [a_2, a_1]^T := \left[\mathcal{L}^I A_2, \mathcal{L}^I A_1\right]^T \tag{2.4.58}$$

Let $M^I = \begin{pmatrix} m_{1,1} & m_{1,2} \\ m_{2,1} & m_{2,2} \end{pmatrix}$ be a 2 by 2 matrix whose entries are polynomials in ρ and let

$$\mathcal{E}_M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - M^I \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \mathbf{m}\,[a_2, a_1] \right), \tag{2.4.59}$$

Then determining solution $(\psi_\nu, \omega_0, U) \in \mathcal{C}_0 \times \mathbb{R}^2$ to (2.4.27)-(2.4.42) is equivalent to determining solution $\chi \in \mathcal{C}_0 \times \mathbb{R}^2$ to the nonlinear equation

$$\chi = \chi_0 + \epsilon\chi + \mathcal{R}\chi \tag{2.4.60}$$

where

$$\begin{aligned} \chi &= \langle E_\nu, \vec{v} \rangle,\ E_\nu \in \mathcal{C}_0 \text{ and } \vec{v} = \langle w, u \rangle^\intercal \in \mathbb{R}^2 \\ \chi_0 &= -K\mathcal{L}^I \mathcal{N}_0, \\ \epsilon &: \chi \to -K\mathcal{E}_{\mathcal{L}} E_\nu - J\mathcal{E}_M \vec{v}, \\ \mathcal{R} &: \chi \to -K\mathcal{L}^I R \end{aligned} \tag{2.4.61}$$

with

$$Kf = \langle K_1 f, K_2 f \rangle := \langle f + \mathbf{a}^T M^I \mathbf{m}[f]\ ,\ M^I \mathbf{m}[f] \rangle \tag{2.4.62}$$

and

$$J\vec{v} = \langle J_1 \vec{v}, J_2 \vec{v} \rangle := \langle \mathbf{a}^T \vec{v}\ ,\ \vec{v} \rangle \tag{2.4.63}$$

Proof. It is observed that $\mathcal{L}E_\nu = -\mathcal{N}_0 - [A_2, A_1] \cdot \vec{v} - R$ and thus

$$E_\nu = -\mathcal{E}_{\mathcal{L}} E_\nu - \mathcal{L}^I \mathcal{N}_0 - \mathbf{a}^T \vec{v} - \mathcal{L}^I R, \tag{2.4.64}$$

(2.4.41) and (2.4.42) are equivalent to $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \vec{v} + \vec{m} \cdot E_\nu = -\langle \omega_0^0, 0 \rangle^\intercal - \vec{m} \cdot \psi^0$, and the form of ω^0 in (2.4.13) with form of Z^0 in (2.4.4) ensures that $-\langle \omega_0^0, 0 \rangle^\intercal - \vec{m} \cdot \psi^0 = 0$. Substituting

this and (2.4.64) into the last equation yields

$$\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \vec{m} \cdot \vec{l}_A \right) \vec{v} = \vec{m} \cdot \mathcal{E}_{\mathcal{L}} E_\nu + \vec{m} \cdot \mathcal{L}^I \mathcal{N}_0 + \vec{m} \cdot \mathcal{L}^I R,$$

and hence,

$$\vec{v} = \mathcal{E}_M \vec{v} + M^I \cdot \vec{m} \cdot \mathcal{E}_{\mathcal{L}} E_\nu + M^I \cdot \vec{m} \cdot \mathcal{L}^I \mathcal{N}_0 + M^I \cdot \vec{m} \cdot \mathcal{L}^I R, \tag{2.4.65}$$

The result is obtained by substituting (2.4.65) into (2.4.64) and regrouping the terms. □

Definition 2.4.13. For $\chi = \langle f, \vec{v} \rangle \in \mathcal{C}_0 \times \mathbb{R}^2$ define norm

$$\|\chi\|_1 = \|f\|_1 + \|\vec{v}\|_1 \tag{2.4.66}$$

We denote the corresponding Banach space $\mathbb{S}$

Theorem 2.4.14. Assume for $0 < \delta \leq \hat{\delta} \leq \frac{1}{100}$, when $\|\chi\|_1, \|\chi_1\|_1, \|\chi_2\|_1 \leq \delta$,

$$\|\epsilon\chi\|_1 \leq k_1 \|\chi\|_1 \ , \text{ for } k_1 < 1 \tag{2.4.67}$$

$$\|\mathcal{R}[\chi]\|_1 \leq k_2 \|\chi\|_1^2 \ \ , \ \|\mathcal{R}[\chi_1] - \mathcal{R}[\chi_2]\|_1 \leq 2k_2\delta \|\chi_1 - \chi_2\|_1 \ . \tag{2.4.68}$$

If

$$\|\chi_0\|_1 \leq \delta \left(1 - k_1 - k_2\delta\right) \tag{2.4.69}$$

for some small $\delta \leq \hat{\delta}$ that ensures

$$k_1 + 2\delta k_2 < 1 \tag{2.4.70}$$

then, there exists a unique unique solution χ satisfying (2.4.60) in a ball $\mathcal{B}_{\delta_0}$ centered at χ_0 where

$$\delta_0 = k_1\delta + k_2\delta^2 \tag{2.4.71}$$

Proof. First, we notice that if $\chi \in \mathcal{B}_{\delta_0}$ implies

$$\|\chi\|_1 \leq \|\chi_0\|_1 + \|\chi - \chi_0\|_1 \leq \delta(1 - k_1 - k_2\delta) + k_1\delta + k_2\delta^2 \leq \delta$$

Therefore, when $\chi \in \mathcal{B}_{\delta_0}(\chi_0)$,

$$\|\mathcal{M}[\chi] - \chi_0\|_1 \leq k_1\delta + k\delta^2 \leq \delta_0 \tag{2.4.72}$$

implying $\mathcal{M} : \mathcal{B}_{\delta_0} \to \mathcal{B}_{\delta_0}$. Furthermore, for χ_1, χ_2 in this ball,

$$\|\mathcal{M}[\chi_1] - \mathcal{M}[\chi_2]\|_1 \leq (k_1 + 2k\delta)\,\|\chi_1 - \chi_2\|_1 \tag{2.4.73}$$

implying from (2.4.70) that $\mathcal{M}$ is contractive. Therefore from Banach contraction theorem, a unique solution exists to $\chi = \mathcal{M}[\chi]$ in $\mathcal{B}_{\delta_0}(\chi_0) \subset \mathbb{S}$, and the theorem follows. □

Remark 2.4.15. The Theorem above relies on smallness of χ_0 and the property that $k_1 < 1$. The first only depends on a good guess for a quasi-solution–something that in principle can always be arranged by using a good enough basis representation. The property $k_1 < 1$ depends on how close $\mathcal{L}^I$ and M^I are to the actual inverses of $\mathcal{L}$ and M, if indeed each exist, since this determines contractivity properties of ϵ are. For instance if $\mathcal{L}^I$ and M^I are exactly $\mathcal{L}^{-1}$ and M^{-1}, then $\mathcal{E}_L$ and $\mathcal{E}_M$ are each zero, implying $\epsilon = 0$. The choice of $\mathcal{L}^I$ depends on the range of ρ and B chosen. If numerical evidence shows M is invertible, we can construct M^I by using the empirical inverse of M. When a range of ρ values are involved, then a low order polynomial fit in ρ is used to obtain a formula for M^I empirically. The process of obtaining empirical M^I and $\mathcal{L}^I$ need not be rigorous, since the smallness estimate on operator ϵ will suffice.

Corollary 2.4.16. If $k_1 < 1$ and $0 < \|\chi_0\| < \frac{(1-k_1)^2}{4k_2}$, then Theorem 2.4.14 applies for any $\delta \in \left(\delta_l, \frac{1-k_1}{2k_2}\right)$ where

$$\delta_l = \frac{(1-k_1) - \sqrt{(1-k_1)^2 - 4k_2\|\chi_0\|}}{2k_2} \tag{2.4.74}$$

in which case,

$$\delta_0 = k_1\delta + k_2\delta^2 \tag{2.4.75}$$

Proof. Clearly, from expression above since $\delta < \frac{1-k_1}{2k_2}$, the contractivity condition $k_1 + 2\delta k_2 < 1$ is satisfied. Furthermore, the condition $\|\chi_0\| \leq \delta(1 - k_1 - k_2\delta)$ on quadratic factorization translates to

$$k_2(\delta_r - \delta)(\delta - \delta_l) \geq 0 \tag{2.4.76}$$

where δ_l is as given above and

$$\delta_r = \frac{(1-k_1) + \sqrt{(1-k_1)^2 - 4k_2\|\chi_0\|}}{2k_2} \tag{2.4.77}$$

Since $\delta_l \leq \delta < \frac{k_1}{2k_2} \leq \delta_r$ it follows that (2.4.76) is satisfied and all conditions for existence of solution in Theorem 2.4.14 is satisfied. □

Remark 2.4.17 (Main Result)**.** We note in the case when $B = 0.3$ and the quasi-solution as introduced in 2.8.1 we have $k_1 = \frac{2}{5}$, and $\|\chi_0\| < 5 \times 10^{-4}$, which gives $0 \leq \delta_l < \frac{3}{10k_2} - \frac{\sqrt{\frac{9}{25} - 2\times 10^{-3}k_2}}{2k_2}$. A local solution is guaranteed if $k_2 < 180$. Alternatively, for any $k_2 > 0$ were specified, if $k_1 < 0$, we can always find a solution if $\|\chi_0\|$ is sufficiently small.

The proof follows by checking $k_1 = \frac{2}{5}$, $\|\chi_0\| < 5 \times 10^{-4}$ and $k_2 < 180$, which ensure conditions of general theorem 2.4.14, and this is what is accomplished in the remaining part of this chapter.

2.5 Preliminaries

In this section, we present a general methodology how to manipulate a quasi solution to obtain the bound conditions in Theorem 2.4.14 **uniformly** in ρ for all $\rho \in [\rho_0, \rho_1]$. In the next section, we present an explicit quasi solution for the case when the surface tension parameter $B = 0.3$, and we obtain explicit results for the bubble size parameter ρ over the range $\rho \in [0.5, 0.7]$.

We work with objects of the form $f = \sum_{n=0}^{\infty} f_n \cos(n\nu)$ or $f = \sum_{n=0}^{\infty} f_n \sin(n\nu)$ where f_n's are functions in ρ for $\rho \in [\rho_0, \rho_1]$ and we are mainly required to find bounds on $\| \|_1$ norm of such objects. In order to provide a uniform treatment for $\rho \in [\rho_0, \rho_1]$, we introduce a new norm, which we will refer as Chebyshev norm, which is larger than $\| \|_1$ norm but conveniently close and is explicitly calculable when the function is known.

Remark 2.5.1. Let $0 < \rho_0 < \rho_1 < 1$ be real numbers. It is well known from Theory of Chebyshev approximation that when $f(\rho) \in C[\rho_0, \rho_1]$, then

$$f \sim \sum_{n=0}^{\infty} f_n T_n(\tau) \ , \ \tau = \frac{\rho - \left(\frac{\rho_1+\rho_0}{2}\right)}{\left(\frac{\rho_1-\rho_0}{2}\right)} \tag{2.5.1}$$

where T_n is the n-th Chebyshev-T polynomial and where

$$f_n = \frac{(2 - 1_{n=0})}{\pi} \int_{-1}^{1} f\left(\left(\frac{\rho_1 - \rho_0}{2}\right)\tau + \left(\frac{\rho_1 + \rho_0}{2}\right)\right) T_n(\tau) \frac{d\tau}{\sqrt{1-\tau^2}} \tag{2.5.2}$$

Since $|T_n(\tau)| \leq 1$ for all $n \geq 0$ and $\tau \in [-1, 1]$, it is straightforward to observe

$$|f(\rho)| = \left|\sum_{n=0}^{\infty} f_n T_n(\tau)\right| \leq \sum_{n=0}^{\infty} |f_n|$$

Hence the quantity $\sum_{n=0}^{\infty} |f_n|$, which may be referred as Chebyshev norm of f, provides an upper bound on $|f|$. It is observed that for a large class of $C[\rho_0, \rho_1]$ functions, which includes the ones that are used to construct quasi solutions for our problems, the Chebyshev norm is reasonably close to the actual maximum and easy to obtain, and hence we use the Chebyshev norm to obtain uniform upper bounds. We also note that Chebyshev approximation is equivalent to the following Fourier approximation with the appropriate change of variables

$$f(\rho) \sim \sum_{n=0}^{\infty} f_n \cos(n\theta) \ , \ \cos(\theta) = \tau = \frac{\rho - \left(\frac{\rho_1+\rho_0}{2}\right)}{\left(\frac{\rho_1-\rho_0}{2}\right)} \ , \ \theta \in [\pi, 2\pi]$$

Now we introduce the Chebyshev norm for the spaces of our interest.

Definition 2.5.2. Let $0 < \rho_0 < \rho_1 < 1$ be fixed real numbers. We define the the Chebyshev norm on $C[\rho_0, \rho_2]$ as

$$\|f\|_T := \sum_{n=0}^{\infty} |\mathcal{T}_n(f)| \text{ for } f \in C[\rho_0, \rho_1], \tag{2.5.3}$$

where $\mathcal{T}_n(f)$, which is also referred as the n-th Chebyshev mode of f, is defined as

$$\begin{aligned} \mathcal{T}_n(f) &:= \frac{(2 - 1_{n=0})}{\pi} \int_{-1}^{1} f\left(\left(\frac{\rho_1 - \rho_0}{2}\right)\tau + \left(\frac{\rho_1 + \rho_0}{2}\right)\right) T_n(\tau) \frac{d\tau}{\sqrt{1-\tau^2}} \\ &= \frac{(2 - 1_{n=0})}{\pi} \int_{\pi}^{2\pi} f\left(\left(\frac{\rho_1 - \rho_0}{2}\right)\cos(\theta) + \left(\frac{\rho_1 + \rho_0}{2}\right)\right) \cos(n\theta) d\theta \end{aligned} \tag{2.5.4}$$

Let $f \in \mathcal{C}$ (or $f \in \mathcal{S}$) where $f = \sum_{n=0}^{\infty} f_n \cos(n\nu)$ (or $f = \sum_{n=1}^{\infty} f_n \sin(n\nu)$), we define its Chebyshev norm as

$$\|f\|_T = \sum_{n=0}^{\infty} \|f_n\|_T, \tag{2.5.5}$$

Corollary 2.5.3. Let P be a real polynomial in ρ, then its Chebyshev norm can be calcu-

lated as

$$\|P\|_T = \sum_{n=0}^{\deg(P)} \left| \mathcal{T}_n \left(P \left(\frac{(\rho_1 - \rho_0)}{2} \cos(\theta) + \frac{(\rho_1 + \rho_0)}{2} \right) \right) \right|, \tag{2.5.6}$$

Let $f^d \in \mathcal{S}$ (or $\mathcal{C}_0$) such that $f^d = \sum_{n=0}^{N'} f_n^d \cos(n\nu)$ (or $f^d = \sum_{n=1}^{N'} f_n^d \sin(n\nu)$) where f_n^d is a polynomial in ρ for all $n = 0, \ldots, N'$, then the Chebyshev norm of f^d is explicitly calculated as

$$\left\| f^d \right\|_T = \sum_{n=0}^{N'} \left\| f_n^d \right\|_T \tag{2.5.7}$$

We approximate the continuous functions on $[\rho_0, \rho_1]$ with Chebyshev polynomials, likewise we approximate to the elements in $\mathcal{C}$ or $\mathcal{S}$ $f = \sum_{n=0}^{\infty} f_n \cos(n\nu)$ (or $f^d = \sum_{n=1}^{N'} f_n^d \sin(n\nu)$) by $f = \sum_{n=0}^{N'} f_n \cos(n\nu)$ (or $f^d = \sum_{n=1}^{N'} f_n^d \sin(n\nu)$) where f_n^d are polynomials. In this way we immediately obtain

$$\|f\|_1 \leq \left\| f^d \right\|_1 + \left\| f - f^d \right\|_1 \leq \left\| f^d \right\|_T + \left\| f - f^d \right\|_1 \tag{2.5.8}$$

uniformly in $\rho \in [\rho_0, \rho_1]$, where $\left\| f^d \right\|_T$ is a positive real number. $\left\| f - f^d \right\|_1$ is referred as the disagreement, or the tail part (when considered as finite part of an infinite series), and is generally bounded by using a complex analytic-like property and is shown to be small when f^d is chosen appropriately as in the following Remark.

Remark 2.5.4. As discussed in detail in Lemma A.1.10, when $f \in \mathcal{C}$ and if the complex function

$$\hat{f}(\eta, \zeta) := \sum_{N=0}^{\infty} \sum_{j=0}^{\infty} \mathcal{T}_j(F_N(f)) \frac{(\zeta^N + \zeta^{-N})}{2} \frac{(\eta^j + \eta^{-j})}{2} \tag{2.5.9}$$

is analytic in some annular region $\mathcal{D}_0 = \left\{ (\eta, \zeta) \in \mathbb{C}^2 : \frac{1}{r_\zeta} \leq |\zeta| \leq r_\zeta, \frac{1}{r_\eta} \leq |\eta| \leq r_\eta \right\}$ for some $r_\zeta > 1$ and $r_\eta > 1$, and if $\hat{f}^d$ is a discretized approximation of $\hat{f}$ as in A.1.7 of degree $\deg(f^d)$, then

$$\left| F_N(f - f^d) \right| \leq \frac{\|\hat{f}\|_{\mathcal{D}_0}}{r_\zeta^N} \left(\frac{1_{N \leq \deg(f^d)} \alpha_1}{r_\eta^{\deg(f)}} + 1_{N > \deg(f^d)} (r_\zeta - 1) \alpha_2 \right), \tag{2.5.10}$$

for all $N \geq 0$, where α_0, α_1 are specified with r_ζ and r_η, and $\|\hat{f}\|_{\mathcal{D}_0}$ is an upper bound for $\hat{f}$ in $\mathcal{D}_0$. For a given $f \in \mathcal{C}$, this result simplifies tail analysis to finding r_ζ, r_η and $\|\hat{f}\|_{\mathcal{D}_0}$ for which $\hat{f}$ is analytic in $\mathcal{D}_0$ with $|\hat{f}| \leq \|\hat{f}\|_{\mathcal{D}_0}$. When r_ζ, r_η and $\|\hat{f}\|_{\mathcal{D}_0}$ are fixed, the tail

can be made as small as desired by increasing the degree of the approximation f^d.

Remark 2.5.5. If $f \in \mathcal{C}$ (or $\mathcal{S}$), then f^d refers to a discretized approximation of degree $\deg(f^d)$ as defined in A.1.7, which is a finite series in $\mathcal{C}$ (or $\mathcal{S}$) whose all non-zero Fourier modes are polynomials. From this point on, for $f = \psi_\nu^0, \phi_\nu^0, A_1, A_2, A_3$ we fix a positive integer

$$N_1 \ , \ N_1 > N_0 \tag{2.5.11}$$

and set f^d to be the discrete approximation of f of degree N_1. Similarly, for $f = \mathcal{N}^0$ we fix

$$N_2 \ , \ N_2 \geq N_1 \tag{2.5.12}$$

and set f^d to be the discrete approximation of f of degree N_1. A higher degree of approximation for $\mathcal{N}^0$ is enforced due to its small norm.

Theorem 2.4.14 requires bounds on $\|\chi_0\|_1$, $\|\epsilon\|_1$ and $\|\mathcal{R}[\chi]\|_1$, which suggests an investigation on the bounds for operators $K\mathcal{L}^I$, $K\mathcal{E}_\mathcal{L}$ and $J\mathcal{E}_M$. We start with introducing a general form of $\mathcal{L}^I$.

2.6 Bounds on Linear Terms

2.6.1 $\mathcal{L}^I$ and $\mathcal{E}_\mathcal{L}$

Remark 2.6.1. Now, we invent a suitable quasi-inverse of operator $\mathcal{L}$ in the following manner. We rely on the observation that

$$\mathcal{L}_0 E_\nu := E_{\nu\nu} + \frac{q_\nu}{2q} E_\nu + \frac{(1-\rho^4)}{q} F_\nu = \tilde{r} \tag{2.6.1}$$

is explicitly solvable $\tilde{r} \in \mathcal{S}_0$. However, we find for large B the quantity $\mathcal{L}_0^{-1}\mathcal{L} - I$ is not sufficiently small since this quasi-inverse does not capture the first few Fourier-modes of inversion well-enough. Instead, we correct $\mathcal{L}_0^{-1}$ by including an empirically obtained polynomials in ρ for the first few Fourier coefficients. More specifically, we introduce

$$(\mathcal{L} - \mathcal{L}_0)E_\nu = -E_\nu \phi_\nu^0 - F_\nu \psi_\nu^0 + FA_2 + EA_3 =: \mathcal{L}_1 E_\nu \ , \tag{2.6.2}$$

and we construct a quasi-inverse

$$\mathcal{L}^I = \mathcal{L}_0^I + \mathcal{L}_1^I \tag{2.6.3}$$

where $\mathcal{L}_0^I$ and $\mathcal{L}_1^I$ chosen such that $\mathcal{L}^I \sin(n\nu)$ is a polynomial in ρ for all $n \geq 1$(a property which enables explicit calculation of bounds as discussed above), and the linear operators $(\mathcal{L}_0^I\mathcal{L}_0 - \mathcal{I})$ and $(\mathcal{L}_1^I\mathcal{L}_0 + \mathcal{L}^I\mathcal{L}_1)$ are have conveniently small $\| \|_1$-norm, which ensures the smallness of

$$\mathcal{E}_{\mathcal{L}} = \mathcal{L}^I\mathcal{L} - \mathcal{I} = (\mathcal{L}_0^I + \mathcal{L}_1^I)(\mathcal{L}_0 + \mathcal{L}_1) - \mathcal{I} = (\mathcal{L}_0^I\mathcal{L}_0 - \mathcal{I}) + (\mathcal{L}_1^I\mathcal{L}_0 + \mathcal{L}^I\mathcal{L}_1), \tag{2.6.4}$$

Other strategies are possible when B is small, including using a WKB-like inversion. However, our focus is larger B where no analytic methods exist.

Lemma 2.6.2 ($\mathcal{L}_0^{-1}$)**.** Let N, n be positive integers, then;

$$F_N(\mathcal{L}_0 \cos(n\nu)) = -1_{N=n}(N+1) - 1_{n>N}\frac{\left(1-\rho^{2N}\right)}{(1-\rho^{2n})}\rho^{n-N}(1+(-1)^{n-N}) =: L^0_{N,n}, \tag{2.6.5}$$

$$\mathcal{L}_1 : \cos(n\nu) \to \Lambda_{1,n}\cos(n\nu) + \Lambda_{2,n}\sin(n\nu), \tag{2.6.6}$$

where

$$\Lambda_{1,n} = \left(-\phi_\nu^0 + \frac{(1+\rho^{2n})}{(1-\rho^{2n})}\frac{A_2}{n}\right) \ , \ \Lambda_{2,n} = \left(\frac{(1+\rho^{2n})}{(1-\rho^{2n})}\psi_\nu^0 + \frac{A_3}{n}\right) \tag{2.6.7}$$

Let

$$\beta_{N,n} := 1_{n>N}\frac{\left(1-\rho^{2N}\right)}{(1-\rho^{2n})}\frac{\rho^{n-N}}{(n^2-1)}(1+(-1)^{n-N}), \tag{2.6.8}$$

Then

$$\sum_{k=1}^{\infty}\left(-\frac{1_{N=k}}{(N+1)} + \beta_{N,k}\right)L^0_{k,n} = 1_{N=n}, \tag{2.6.9}$$

Proof. Setting $E_\nu = \cos(n\nu)$ and hence $E = \frac{\sin(n\nu)}{n}$, $F = \frac{(1+\rho^{2n})\cos(n\nu)}{(1-\rho^{2n})n}$, $-F_\nu = \frac{(1+\rho^{2n})\sin(n\nu)}{(1-\rho^{2n})}$ and $E_{\nu\nu} = -n\sin(n\nu)$, (2.6.1) and (2.6.2) imply

$$\mathcal{L}_0\cos(n\nu) = -n\sin(n\nu) + \frac{q_\nu}{2q}\cos(n\nu) - \frac{(1+\rho^4)}{q}\frac{(1+\rho^{2n})}{(1-\rho^{2n})}\sin(n\nu),$$
$$\mathcal{L}_1\cos(n\nu) = \left(-\phi_\nu^0 + \frac{(1+\rho^{2n})}{(1-\rho^{2n})}\frac{A_2}{n}\right)\cos(n\nu) + \left(\frac{(1+\rho^{2n})}{(1-\rho^{2n})}\psi_\nu^0 + \frac{A_3}{n}\right)\sin(n\nu),$$

(2.6.5) follows from computing $F_N(\mathcal{L}_0\cos(n\nu))$ through the identity $\frac{1-\rho^4}{q} + i\frac{q_\nu}{2q} = 1 +$

$\sum_{k=1}^{\infty}(\rho^k+(-\rho^k))e^{i\nu}$. (2.6.9) is obtained in the following way: when $N \geq n$, evidently $\sum_{k=1}^{\infty}\left(-\frac{1_{N=k}}{(N+1)}+\beta_{N,k}\right)C^0_{k,n} = 1_{N=n}$, and when $n > N$ it is observed that;

$$\sum_{k=1}^{\infty}\left(-\frac{1_{N=k}}{(N+1)}+\beta_{N,k}\right)L^0_{k,n} = \left(-\frac{1}{(N+1)}+\beta_{N,N}\right)L^0_{N,n}+\beta_{N,n}L^0_{n,n}+\sum_{k=N+1}^{n-1}\beta_{N,k}L^0_{k,n}$$
$$= \frac{(1-\rho^{2n})\,\rho^{n-N}\left(1+(-1)^{n-N}\right)}{(1-\rho^{2n})}\left(\frac{1}{(N+1)}-\frac{1}{(n-1)}-\sum_{k=N+1}^{n-1}\frac{\left(1+(-1)^{k-N}\right)}{(k^2-1)}\right)$$

which follows from the identity $(1+(-1)^{k-N})(1+(-1)^{n-k}) = (1+(-1)^{n-N})(1+(-1)^{k-N})$. If n and N have different parity then evidently $\sum_{k=1}^{\infty}\beta_{N,k}C^0_{k,n} = 0$ and if they have the same parity then $\frac{1}{(N+1)}-\frac{1}{(n-1)}-\sum_{k=N+1}^{n-1}\frac{\left(1+(-1)^{k-N}\right)}{(k^2-1)} = 0$, and the lemma is proved. □

Definition 2.6.3 ($\mathcal{L}^I$ and $\mathcal{E}_{\mathcal{L}}$)**.** Let

$$B > 0\ , \tag{2.6.10}$$

be a fixed positive real number which is large as in the sense of discussed in Remark 2.6.1. Let $(\psi^0_\nu, U^0, \omega^0_0)$ be a fixed quasi-solution with surface tension B, and let

$$N_0 \geq 1 \tag{2.6.11}$$

be a fixed positive integer for which the Fourier-modes higher than N_0 are well captured by $\mathcal{L}_0^{-1}$ in the sense pointed out in Remark 2.6.1. We define a $\mathcal{L}^I_0, \mathcal{L}^I_1 : \mathcal{S} \to \mathcal{C}_0$ as

$$F_N(\mathcal{L}^I_0 \sin(n\nu)) = -\frac{1_{N=n}}{(N+1)}+(1-\rho^{2c_0 n})\beta_{N,n} \text{ for all } N,n \geq 1\ \ , \tag{2.6.12}$$

$$F_N(\mathcal{L}^I_1 \sin(n\nu)) = P_{N,n} \text{ for all } N,n \geq 1, \tag{2.6.13}$$

where c_0 is a fixed positive integer,

$$P_{N,n} = \begin{cases} \text{fixed polynomials in } \rho & \text{when } N,n \leq N_0, \\ 0 & \text{otherwise,} \end{cases} \tag{2.6.14}$$

We define

$$\mathcal{L}^I = \mathcal{L}^I_0 + \mathcal{L}^I_1\ , \tag{2.6.15}$$

$$\mathcal{E}_{\mathcal{L}} = (\mathcal{L}^I_0\mathcal{L}_0 - \mathcal{I}) + \mathcal{L}^I_1\mathcal{L}_0 + \mathcal{L}^I\mathcal{L}_1, \tag{2.6.16}$$

and for convenience we set

$$L^I_{N,n} := F_N(\mathcal{L}^I \sin(-n\nu)) = -\frac{1_{N=n}}{(N+1)} + (1-\rho^{2c_0 n})\beta_{N,n} + P_{N,n}, \tag{2.6.17}$$

for all $N \geq 1$ and $n \geq 1$.

Now, we introduce a number of useful Lemmas.

Lemma 2.6.4. Let $N, n \geq 1$ be integers, then

$$\begin{aligned} F_N(\mathcal{E}_{\mathcal{L}} \cos(n\nu)) = \tilde{P}_{N,n} + \tilde{Q}_{N,n} + \sum_{l=1}^{\infty} F_l(\Lambda_{1,n}) \frac{(L^I_{N,l+n} + L^I_{N,l-n})}{2} \\ + \sum_{l=0}^{\infty} F_l(\Lambda_{2,n}) \frac{(L^I_{N,n+l} + L^I_{N,n-l})}{2} \end{aligned} \tag{2.6.18}$$

where

$$\tilde{Q}_{N,n} := -1_{n>N} \frac{(1-\rho^{2N})\rho^{n-N}(1+(-1)^{n-N})}{(1-\rho^{2n})} \left(\sum_{k=N+1}^{n-1} \frac{\rho^{2c_0 k}(1+(-1)^{k-N})}{(k^2-1)} + \frac{\rho^{2c_0 n}}{n+1} \right) \tag{2.6.19}$$

$$\tilde{P}_{N,n} := \sum_{k=1}^{N_0} P_{N,k} L^0_{k,n} \ , \tag{2.6.20}$$

Proof. Evidently

$$F_N(\mathcal{E}_{\mathcal{L}} \cos(n\nu)) = F_N((\mathcal{L}^I_0 \mathcal{L}_0 - \mathcal{I}) \cos(n\nu)) + F_N(\mathcal{L}^I_1 \mathcal{L}_0 \cos(n\nu)) + F_N(\mathcal{L}^I \mathcal{L}_1 \cos(n\nu))$$

Linearity and (2.6.9) imply

$$\begin{aligned} &F_N((\mathcal{L}^I_0 \mathcal{L}_0 - \mathcal{I}) \cos(n\nu)) = F_N((\mathcal{L}^I_0 \mathcal{L}_0) \cos(n\nu)) - 1_{N=n} \\ &= F_N\left(\mathcal{L}^I_0 \sum_{k=1}^{\infty} F_k(\mathcal{L}_0 \cos(n\nu)) \sin(k\nu) \right) - 1_{N=n} = \sum_{k=1}^{\infty} L^0_{k,n} F_N\left(\mathcal{L}^I_0 \sin(k\nu) \right) - 1_{N=n} \\ &= \sum_{k=1}^{\infty} \left(-\frac{1_{N=k}}{(N+1)} + \beta_{N,k} \right) L^0_{k,n} - 1_{N=n} - \sum_{k=1}^{\infty} \rho^{2c_0 k} \beta_{N,k} L^0_{k,n} = -\sum_{k=1}^{\infty} \rho^{2c_0 k} \beta_{N,k} L^0_{k,n} \end{aligned}$$

and the identity $(1+(-1)^{n-k})(1+(-1)^{k-N}) = (1+(-1)^{n-N})(1+(-1)^{k-N})$ implies

$$\sum_{k=1}^{n} \rho^{2c_0 k} \beta_{N,k} L^0_{k,n} = \sum_{k=N+1}^{n-1} \rho^{2c_0 k} \frac{(1-\rho^{2N})\rho^{k-N}(1+(-1)^{k-N})}{(1-\rho^{2k})(k^2-1)} \frac{(1-\rho^{2k})\rho^{n-k}(1+(-1)^{n-k})}{(1-\rho^{2n})}$$
$$+ \rho^{2c_0 n} \frac{(1-\rho^{2N})\rho^{n-N}(1+(-1)^{n-N})}{(1-\rho^{2n})(n+1)}$$
$$= \frac{(1-\rho^{2N})\rho^{n-N}(1+(-1)^{n-N})}{(1-\rho^{2n})} \left[\sum_{k=N+1}^{n-1} \frac{\rho^{2c_0 k}(1+(-1)^{k-N})}{(k^2-1)} + \frac{\rho^{2c_0 n}}{n+1} \right]$$

Similarly, linearity implies

$$F_N(\mathcal{L}_1^I \mathcal{L}_0 \cos(n\nu)) = F_N\left(\mathcal{L}_1^I \sum_{k=1}^{\infty} F_k(\mathcal{L}_0 \cos(n\nu)) \sin(k\nu)\right) = \sum_{k=1}^{\infty} P_{k,n} L^0_{N,k} = \tilde{P}_{N,n} \quad (2.6.21)$$

and finally

$$F_N(\mathcal{L}^I \mathcal{L}_1 \cos(n\nu)) = F_N(\mathcal{L}^I \Lambda_{1,n} \cos(n\nu)) + F_N(\mathcal{L}^I \Lambda_{2,n} \sin(n\nu))$$
$$= F_N\left(\mathcal{L}^I \sum_{l=0}^{N_1} F_l(\Lambda_{1,n}) \sin(l\nu)\cos(n\nu)\right) + F_N\left(\mathcal{L}^I \sum_{l=0}^{N_1} F_l(\Lambda_{2,n}) \cos(l\nu)\sin(n\nu)\right)$$
$$= \sum_{l=0}^{N_1} F_l(\Lambda_{1,n}) \frac{F_N(\mathcal{L}^I \sin((l+n)\nu)) + F_N(\mathcal{L}^I \sin((l-n)\nu))}{2}$$
$$+ \sum_{l=0}^{N_1} F_l(\Lambda_{2,n}) \frac{F_N(\mathcal{L}^I \sin((n+l)\nu)) + F_N(\mathcal{L}^I \sin((n-l)\nu))}{2}$$

which completes the proof. □

Lemma 2.6.5.

$$\|\epsilon\|_1 \leq \max_{i=1,2} \{\|K_i \mathcal{E}_{\mathcal{L}}\|_1, \|J_i \mathcal{E}_M\|_1\} \quad (2.6.22)$$

Proof. Let $\chi = \langle E_\nu, \vec{v}\rangle$ and $\vec{v} = [w,u]^T$, then

$$\|\epsilon\chi\|_1 \leq \sum_{i=1}^{2} (\|K_i \mathcal{E}_{\mathcal{L}} E_\nu\|_1 + \|J_i \mathcal{E}_M \vec{v}\|_1) \leq \max_{i=1,2} \{\|K_i \mathcal{E}_{\mathcal{L}}\|_1, \|J_i \mathcal{E}_M\|_1\} \|\chi\|_1$$

□

Remark 2.6.6. The bound $\|\epsilon\|_1$ is controlled by $\|K_1 \mathcal{E}_{\mathcal{L}}\|_1, \|K_2 \mathcal{E}_{\mathcal{L}}\|_1, \|J_1 \mathcal{E}_M\|_1$ and $\|J_2 \mathcal{E}_M\|_1$. We are going to obtain bounds for $\|K_2 \mathcal{E}_{\mathcal{L}}\|_1$, $\|K_1 \mathcal{E}_{\mathcal{L}}\|_1$, $\|J_2 \mathcal{E}_{\mathcal{L}}\|_1$ and $\|K_2 \mathcal{E}_{\mathcal{L}}\|_1$ in that order. In cases, the linear operators or the functions are decomposed into their discretized and tail parts, which provides efficient bounds.

Remark 2.6.7. Let $L = K_1, K_2, J_1, J_2$, then it is evident that

$$\|L\mathcal{E}_{\mathcal{L}}\|_1 = \sup_{n\geq 1} \|L\mathcal{E}_{\mathcal{L}} \cos(n\nu)\|_1 \tag{2.6.23}$$

Our strategy is to find a uniform bound on $\|L\mathcal{E}_{\mathcal{L}} \cos(n\nu)\|_1$ when $n \geq N_1 + 1$, and find individual yet more sensitive bounds for $n \leq N_1$.

Definition 2.6.8. We define Λ_i^d, $\mathcal{L}_1^d$ and $\mathcal{E}_{\mathcal{L}}^d$ as:

$$\Lambda_{1,n}^d = -\phi_\nu^{0d} + \frac{(1+\rho^{2n})}{(1-\rho^{2n})}\frac{A_2^d}{n} \ , \ \Lambda_{2,n}^d = \frac{(1+\rho^{2n})}{(1-\rho^{2n})}\psi_\nu^{0d} + \frac{A_3^d}{n}, \tag{2.6.24}$$

$$\mathcal{L}_1^d : \cos(n\nu) \to \Lambda_{1,n}^d \cos(n\nu) + \Lambda_{1,n}^d \sin(n\nu), \tag{2.6.25}$$

$$\mathcal{E}_{\mathcal{L}}^d = \mathcal{L}_1^I \mathcal{L}_0 + \mathcal{L}^I \mathcal{L}_1^d, \tag{2.6.26}$$

Corollary 2.6.9. Let $n \geq 1$ be an integer, then

$$\begin{aligned} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) &= \tilde{P}_{N,n} + \sum_{l=1}^{\infty} F_l(\Lambda_{1,n}^d)\frac{(L_{N,l+n}^I + L_{N,l-n}^I)}{2} \\ &\quad + \sum_{l=0}^{\infty} F_l(\Lambda_{2,n}^d)\frac{(L_{N,n+l}^I + L_{N,n-l}^I)}{2} \end{aligned} \tag{2.6.27}$$

$$\begin{aligned} F_N((\mathcal{E}_{\mathcal{L}} - \mathcal{E}_{\mathcal{L}}^d)\cos(n\nu)) &= \tilde{Q}_{N,n} + \sum_{l=1}^{\infty} F_l(\Lambda_{1,n} - \Lambda_{1,n}^d)\frac{(L_{N,l+n}^I + L_{N,l-n}^I)}{2} \\ &\quad + \sum_{l=0}^{\infty} F_l(\Lambda_{2,n} - \Lambda_{2,n}^d)\frac{(L_{N,n+l}^I + L_{N,n-l}^I)}{2} \ , \end{aligned} \tag{2.6.28}$$

Definition 2.6.10. Let $\gamma = \{\gamma_1, \gamma_2, \cdots\}$ be a sequence of positive real numbers where γ_k is decreasing after some point. We define

$$dec(\gamma) : \gamma \text{ in decreasing order}, \tag{2.6.29}$$

and

$$\tilde{\gamma} := \gamma \wedge dec(\gamma) = \{\max\{\gamma_n, ord(\gamma)_n\}, n = 1, 2, \dots\} \tag{2.6.30}$$

2.6.2 $\|K_2\mathcal{E}_{\mathcal{L}}\|_1$ Case

Lemma 2.6.11. Let

$$K_2 f =: [K_{2,1}f, K_{2,2}f]^T \ , \tag{2.6.31}$$

Let $N, n > 1$ be positive integers, then

$$K_{2,j}\cos(N\nu) = m_{j,1}\frac{(\rho^N + (-\rho)^N)}{N} + m_{j,2}\frac{(\rho^N - (-\rho)^N)}{N} =: v_{j,N} \ , \tag{2.6.32}$$

$$K_{2,j}\mathcal{L}^I \sin(n\nu) = m_{j,1}\tilde{v}_{+,n} + m_{j,2}\tilde{v}_{-,n} = \begin{cases} m_{j,1}\tilde{v}_{+,n} & n \text{ even}, \\ m_{j,2}\tilde{v}_{-,n} & n \text{ odd}, \end{cases} =: \tilde{v}_{j,n} \ , \tag{2.6.33}$$

for $j = 1, 2$ where

$$\begin{aligned} \tilde{v}_{\pm,n} := \frac{(\rho^n \pm (-\rho)^n)}{(n+1)} \sum_{N=1}^{n} \left(-\frac{1_{N=n}}{N} + \frac{1_{n>N}(1-\rho^{2c_0 n})(1-\rho^{2N})(1\pm(-1)^N)}{(1-\rho^{2n})(n-1)N} \right) \\ + \sum_{N=1}^{N_0} P_{N,n}\frac{(\rho^N \pm (-\rho)^N)}{N}, \end{aligned} \tag{2.6.34}$$

Proof. (2.6.32) is immediate from the definition of K_2. Note that

$$\begin{aligned} K_2\mathcal{L}^I \sin(n\nu) &= K_2 \sum_{N=1}^{\infty} F_N(\mathcal{L}^I \sin(n\nu))\cos(N\nu) \\ &= \sum_{N=1}^{n} F_N(\mathcal{L}^I \sin(n\nu)) K_2 \cos(N\nu) = \sum_{N=1}^{n} L^I_{N,n} K_2 \cos(N\nu) \end{aligned}$$

and the identity $(1 \pm (-1)^{n-N})(1 \pm (-1)^N) = (1 + (-1)^n)(1 \pm (-1)^N)$ implies

$$\sum_{N=1}^{n} L^I_{N,n}\frac{\rho^N(1 \pm (-1)^N)}{N} = \tilde{v}_{\pm,n}$$

and (2.6.33) follows. □

Lemma 2.6.12. Let $L : \mathcal{S} \to \mathcal{C}_0$. Assume that γ_L is a sequence of positive real numbers where

$$\gamma_L = \{\gamma_{L,1}, \gamma_{L,2}, \gamma_{L,3}, \cdots\} \ : \ \gamma_{L,k} \geq |L\mathcal{L}^I \sin(k\nu)| \ \forall\, k \geq 1 \ , \ \gamma_{L,k} \text{ is decreasing after a point}, \tag{2.6.35}$$

Let $f \in \mathcal{S}$ and $g \in \mathcal{C}$, and let f^d and g^d be their discretized approximation, respectively. Then

$$\begin{aligned} \left\| L\mathcal{L}^I(f - f^d)\cos(n\nu) \right\|_1 &\leq t_1(f, f^d, \gamma_L, n), \\ \left\| L\mathcal{L}^I(g - g^d)\sin(n\nu) \right\|_1 &\leq t_1(g, g^d, \gamma_L, n) \end{aligned} \tag{2.6.36}$$

where

$$t_1(f,f^d,\gamma,n)=\begin{cases}\frac{\|\hat{f}\|_{\mathcal{D}_0}\alpha_1}{r_\eta^{\deg(f^d)}}\sum_{l=0}^{\deg(f^d)}\frac{(\gamma_{l-n}+\gamma_{l+n})}{2r_\zeta^l} & \\ +\frac{\|\hat{f}\|_{\mathcal{D}_0}\alpha_2}{r_\zeta^{\deg(f^d)}}\frac{(\tilde{\gamma}_{\deg(f^d)+1-n}+\tilde{\gamma}_{\deg(f^d)+1+n})}{2} & 0\le n\le\deg(f^d),\\ \frac{\|\hat{f}\|_{\mathcal{D}_0}\alpha_1}{r_\eta^{\deg(f^d)}}\sum_{l=0}^{\deg(f^d)}\frac{\left(\tilde{\gamma}_{\deg(f^d)+1-l}+\tilde{\gamma}_{\deg(f^d)+1+l}\right)}{2r_\zeta^l} & \\ +\frac{\|\hat{f}\|_{\mathcal{D}_0}\alpha_2}{r_\zeta^{\deg(f^d)}}\frac{(\tilde{\gamma}_1+\tilde{\gamma}_{2\deg(f^d)+2})}{2} & n\ge\deg(f^d)+1,\end{cases}\tag{2.6.37}$$

after the convention $\gamma_0=0$ and $\gamma_{-n}=\gamma_n$ made for all $i=1,2$ and $n\ge 1$.

Proof. Linearity implies

$$\begin{aligned}L\mathcal{L}^I(f-f^d)\cos(n\nu)&=L\mathcal{L}^I\sum_{l=1}^{\infty}F_l(f-f^d)\sin(l\nu)\cos(n\nu)\\&=\sum_{l=1}^{\infty}F_l(f-f^d)\frac{(L\mathcal{L}^I\sin((l+n)\nu)+L\mathcal{L}^I\sin((l-n)\nu))}{2}\end{aligned}$$

Hence, (A.1.24) implies

$$\begin{aligned}&\left\|\sum_{l=1}^{\infty}F_l(f-f^d)\frac{(L\mathcal{L}^I\sin((l+n)\nu)+L\mathcal{L}^I\sin((l-n)\nu))}{2}\right\|_1\\&\le\sum_{l=0}^{\deg(f^d)}+\sum_{l=\deg(f^d)+1}^{\infty}\left|F_l(f-f^d)\right|\frac{(\gamma_{l-n}+\gamma_{l+n})}{2}\\&\le\frac{\|\hat{f}\|_{\mathcal{D}_0}\alpha_1}{r_\eta^{\deg(f^d)}}\sum_{l=0}^{\deg(f^d)}\frac{1}{r_\zeta^l}\frac{(\gamma_{l-n}+\gamma_{l+n})}{2}+\|\hat{f}\|_{\mathcal{D}_0}\alpha_2(r_\zeta-1)\sum_{l=0}^{\deg(f^d)}\frac{1}{r_\zeta^l}\frac{(\gamma_{l-n}+\gamma_{l+n})}{2}\end{aligned}$$

and the result is obtained from the observations $\gamma_{l-n}+\gamma_{l+n}\le\tilde{\gamma}_{l-n}+\tilde{\gamma}_{l+n}\le\tilde{\gamma}_{\deg(f^d)+1-n}+\tilde{\gamma}_{\deg(f^d)+1+n}$ when $n\le\deg(f^d)$ and $l\ge\deg(f^d)+1$, and from $\gamma_{l-n}+\gamma_{l+n}\le\tilde{\gamma}_{n-l}+\tilde{\gamma}_{n+l}\le\tilde{\gamma}_{\deg(f^d)+1-l}+\tilde{\gamma}_{\deg(f^d)+1+l}$ when $n\ge\deg(f^d)+1$ and $l\le\deg(f^d)$, and $\gamma_{l-n}+\gamma_{l+n}\le\tilde{\gamma}_{l-n}+\tilde{\gamma}_{l+n}\le\tilde{\gamma}_1+\tilde{\gamma}_{2\deg(f^d)+2}$ when $n\ge\deg(f^d)+1$ and $l\ge\deg(f^d)+1$. The same exact proof for g case. □

Corollary 2.6.13. Let $\gamma_{\mathcal{I}}$ be as defined in (2.6.35) in the case when $L=\mathcal{I}$ is the identity operator. Then,

$$\sum_{k=1}^{\infty}\left|F(a_j-a_j^d)\right|\le t_1(A_j,A_j^d,\gamma_{\mathcal{I}},0)\ ,\tag{2.6.38}$$

for $j=1,2$.

Lemma 2.6.14. Let $\gamma_{K_{2,j}}$ be as defined in (2.6.35) for $j = 1, 2$. Then

$$\begin{aligned}\left\|K_{2,j}(\mathcal{E}_{\mathcal{L}} - \mathcal{E}^d_{\mathcal{L}})\cos(n\nu)\right\|_1 &\leq \frac{2\rho^{2c_0}}{(1-\rho^2)}\|K_{2,j}\|_1 + t_1(-\phi^0_\nu, -\phi^{0d}_\nu, \gamma_{K_{2,j}}, n) \\ &+ \frac{t_1(A_3, A^d_3, \gamma_{K_{2,j}}, n)}{n} + \frac{(1+\rho^{2n})}{(1-\rho^{2n})}\left(t_1(\psi^0_\nu, \psi^{0d}_\nu, \gamma_{K_{2,j}}, n) + \frac{t_2(A_2, A^d_2, \gamma_{K_{2,j}}, n)}{n}\right)\end{aligned} \tag{2.6.39}$$

for $j = 1, 2$.

Proof. Linearity implies

$$\begin{aligned}&K_{2,j}(\mathcal{E}_{\mathcal{L}} - \mathcal{E}^d_{\mathcal{L}})\cos(n\nu) = K_{2,j}(\mathcal{L}^I_0\mathcal{L}_0 - \mathcal{I})\cos(n\nu) + K_{2,j}\mathcal{L}^I(\mathcal{L}_1 - \mathcal{L}^d_1)\cos(n\nu) \\ &= K_{2,j}(\mathcal{L}^I_0\mathcal{L}_0 - \mathcal{I})\cos(n\nu) + K_{2,j}\mathcal{L}^I(\Lambda_{1,n} - \Lambda^d_{1,n})\cos(n\nu) + K_{2,j}\mathcal{L}^I(\Lambda_{2,n} - \Lambda^d_{2,n})\sin(n\nu)\end{aligned}$$

moreover,

$$K_{2,j}\sum_{N=1}^{\infty} F_N\left((\mathcal{L}^I_0\mathcal{L}_0 - \mathcal{I})\cos(n\nu)\right)\cos(N\nu) = \sum_{N=1}^{\infty}\tilde{Q}_{N,n}K_{2,j}\cos(N\nu)$$

and

$$\begin{aligned}&\sum_{N=1}^{\infty}\left|\tilde{Q}_{N,n}K_{2,j}\cos(N\nu)\right| \leq \|K_{2,j}\|_1\sum_{N=1}^{\infty}\left|\tilde{Q}_{N,n}\right| = \|K_{2,j}\|_1\left|\sum_{k=1}^{\infty}\rho^{2c_0k}\beta_{N,k}L^0_{k,n}\right| \\ &\leq \|K_{2,j}\|_1\rho^{2c_0}\sum_{k=1}^{\infty}\left|\beta_{N,k}L^0_{k,n}\right| = \|K_{2,j}\|_1\rho^{2c_0}\sum_{k=1}^{\infty}\beta_{N,k}L^0_{k,n} = \|K_{2,j}\|_1\rho^{2c_0}\left(1_{N=n} + \frac{L^0_{N,n}}{N+1}\right) \\ &\leq \|K_{2,j}\|_1\rho^{2c_0}\sum_{N=1}^{n}\left(2_{N=n} + (1+(-1)^{n-N})\rho^{n-N}\right) \leq \|K_{2,j}\|_1\rho^{2c_0}\left(2 + \sum_{k=1}^{\infty}2\rho^{2k}\right) \\ &\leq \|K_{2,j}\|_1\frac{2\rho^{2c_0}}{(1-\rho^2)}\end{aligned}$$

and the result is obtained from (2.6.7), triangle inequality and Lemma 2.6.12. □

Lemma 2.6.15. Let $f^d \in \mathcal{S}$ and $g^d \in \mathcal{C}$ be a discretized approximation of some $f \in \mathcal{S}$ and $g \in \mathcal{C}$, respectively. Let $n \geq 0$ be an integer, then

$$\begin{aligned}\left\|K_{2,j}\mathcal{L}^I f^d\cos(n\nu)\right\|_1 &\leq d_1(f^d, \gamma_{K_{2,j}}, \tilde{v}_j, n)\ , \\ \left\|K_{2,j}\mathcal{L}^I g^d\sin(n\nu)\right\|_1 &\leq d_1(g^d, \gamma_{K_{2,j}}, \tilde{v}_j, n)\ ,\end{aligned} \tag{2.6.40}$$

where

$$d_1(f^d,\gamma,\tilde{v},n)=\begin{cases}\left\|\sum_{l=0}^{\deg(f^d)}F_l(f^d)\frac{(\tilde{v}_{l-n}+\tilde{v}_{l+n})}{2}\right\|_T & \text{when } n\le\deg(f^d),\\ \sum_{l=0}^{\deg(f^d)}\|F_l(f^d)\|_T\frac{\left(\tilde{\gamma}_{\deg(f^d)+1-l}+\tilde{\gamma}_{\deg(f^d)+1+l}\right)}{2} & \text{when } n\ge\deg(f^d)+1,\end{cases} \tag{2.6.41}$$

and

$$d_1(g^d,\gamma,\tilde{v},n)=\begin{cases}\left\|\sum_{l=0}^{\deg(g^d)}F_l(g^d)\frac{(\tilde{v}_{n-l}+\tilde{v}_{n+l})}{2}\right\|_T & \text{when } n\le\deg(g^d),\\ \sum_{l=0}^{\deg(g^d)}\|F_l(g^d)\|_T\frac{\left(\tilde{\gamma}_{\deg(g^d)+1-l}+\tilde{\gamma}_{\deg(g^d)+1+l}\right)}{2} & \text{when } n\ge\deg(g^d)+1,\end{cases} \tag{2.6.42}$$

Proof. Linearity implies

$$\begin{aligned}K_{2,j}\mathcal{L}^I f^d\cos(n\nu)&=K_{2,j}\mathcal{L}^I\sum_{l=1}^{\deg(f^d)}F(f^d)\sin(l\nu)\cos(n\nu)\\&=\sum_{l=1}^{\deg(f^d)}F(f^d)\frac{(K_{2,j}\mathcal{L}^I\sin((l+n)\nu)+K_{2,j}\mathcal{L}^I\sin((l-n)\nu))}{2}\\&=\sum_{l=1}^{\deg(f^d)}F(f^d)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2},\end{aligned}$$

and evidently

$$\left|K_{2,j}\mathcal{L}^I f^d\cos(n\nu)\right|=\left|\sum_{l=1}^{\deg(f^d)}F(f^d)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right|\le d_1(f^d,\gamma_{K_{2,j}},\tilde{v}_j,n).$$

Similarly, by interchanging sine and cosine in the previous proof and using the identity $\cos(l\nu)\sin(n\nu)=\frac{\sin((n-l)\nu)+\sin((n+l)\nu)}{2}$ the result for the case g is obtained, and the proof is complete. □

Lemma 2.6.16. Let $\gamma_{K_{2,j}}$ be a sequences of positive real numbers as defined in (2.6.35) for $j=1,2$ and let $n\ge 0$ be an integer. Then, when $n\le N_1$:

$$\begin{aligned}\left\|K_{2,j}\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)\right\|_1\le\frac{1}{(1-\rho^{2c_0n})}\Bigg\|&\sum_{N=1}^{N_0}\tilde{P}_{N,n}(1-\rho^{2c_0n})v_{j,N}\\&+\sum_{l=1}^{N_1}F_l(\Lambda_1^d(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}+\sum_{l=0}^{N_1}F_l(\Lambda_2^d(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}\Bigg\|_T\end{aligned} \tag{2.6.43}$$

and when $n \geq N_1 + 1$:

$$\left\|K_{2,j}\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)\right\|_1 \leq \sum_{k=1}^{N_0} 2\rho^{N_1+1-k}\left\|\sum_{N=1}^{N_0} P_{N,k}v_{j,N}\right\|_T + d_1(-\phi^{0d}_{\nu},\gamma_{K_{2,j}},n) + \frac{d_1(A^d_3,\gamma_{K_{2,j}},n)}{N_1+1} \tag{2.6.44}$$

$$+ \frac{(1+\rho^{2(N_1+1)})}{(1-\rho^{2(N_1+1)})}\left(d_1(\psi^{0d}_{\nu},\gamma_{K_{2,j}},n) + \frac{d_1(A^d_2,\gamma_{K_{2,j}},n)}{N_1+1}\right)$$

Proof. Linearity implies

$$\begin{aligned}
&K_{2,j}\mathcal{E}^d_{\mathcal{L}}\cos(n\nu) = K_{2,j}\mathcal{L}^I_1\mathcal{L}_0\cos(n\nu) + K_{2,j}\mathcal{L}^I\mathcal{L}^d_1\cos(n\nu)\\
&= K_{2,j}\mathcal{L}^I_1\mathcal{L}_0\cos(n\nu) + K_{2,j}\mathcal{L}^I\Lambda^d_1\cos(n\nu)\\
&+ K_{2,j}\mathcal{L}^I\Lambda^d_2\sin(n\nu) = \sum_{N=1}^{\infty} F_N(\mathcal{L}^I_1\mathcal{L}_0\cos(n\nu))K_{2,j}\cos(N\nu)\\
&+ \sum_{l=1}^{N_1} F(\Lambda^d_{1,n})\frac{(K_{2,j}\mathcal{L}^I\sin((l+n)\nu) + K_{2,j}\mathcal{L}^I\sin((l-n)\nu))}{2}\\
&+ \sum_{l=0}^{N_1} F(\Lambda^d_{2,n})\frac{(K_{2,j}\mathcal{L}^I\sin((n+l)\nu) + K_{2,j}\mathcal{L}^I\sin((n-l)\nu))}{2}\\
&= \sum_{N=1}^{N_0}\tilde{P}_{N,n}v_{j,N} + \sum_{l=1}^{N_1} F(\Lambda^d_{1,n})\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2} + \sum_{l=1}^{N_1} F(\Lambda^d_{2,n})\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}
\end{aligned}$$

When $n \leq N_1$;

$$\begin{aligned}
&\left\|K_{2,j}\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)\right\|_1 = \frac{\left\|K_{2,j}\mathcal{E}^d_{\mathcal{L}}(1-\rho^{2c_0n})\cos(n\nu)\right\|_1}{(1-\rho^{2c_0n})}\\
&\leq \frac{1}{(1-\rho^{2c_0n})}\left|\sum_{N=1}^{N_0}\tilde{P}_{N,n}(1-\rho^{2c_0n})v_{j,N} + \sum_{l=1}^{N_1} F(\Lambda^d_{1,n}(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right.\\
&\left.+ \sum_{l=0}^{N_1} F(\Lambda^d_{2,n}(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}\right| \leq \frac{1}{(1-\rho^{2c_0n})}\left\|\sum_{N=1}^{N_0}\tilde{P}_{N,n}(1-\rho^{2c_0n})v_{j,N}\right.\\
&\left.+ \sum_{l=1}^{N_1} F(\Lambda^d_{1,n}(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2} + \sum_{l=0}^{N_1} F(\Lambda^d_{2,n}(1-\rho^{2c_0n}))\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}\right\|_T
\end{aligned}$$

and (2.6.43) follows. When $n \geq N_1 + 1$; $k \leq N_0$ implies $\left|L^0_{k,n}\right| \leq 2\rho^{n-l} \leq 2\rho^{N_1+1-l}$, and hence

$$\left|\sum_{N=1}^{N_0}\tilde{P}_{N,n}v_{j,N}\right| = \left|\sum_{N=1}^{N_0}\sum_{k=1}^{N_0} P_{N,k}L^0_{k,n}v_{j,N}\right| \leq \sum_{k=1}^{N_0}\left|L^0_{k,n}\right|\left|\sum_{N=1}^{N_0} P_{N,k}v_{j,N}\right|$$

$$\leq \sum_{k=1}^{N_0} 2\rho^{N_1+1-k} \left\| \sum_{N=1}^{N_0} P_{N,k} v_{j,N} \right\|_T$$

When $n \geq N_1 + 1$, (2.6.24) and Lemma 2.6.15 imply

$$\left| \sum_{l=1}^{N_1} F(\Lambda_{1,n}^d) \frac{(\tilde{v}_{j,l-n} + \tilde{v}_{j,l+n})}{2} \right| \leq \left| \sum_{l=1}^{N_1} F(-\phi_\nu^{0d}) \frac{(\tilde{v}_{j,l-n} + \tilde{v}_{j,l+n})}{2} \right|$$
$$+ \frac{(1+\rho^{2n})}{(1-\rho^{2n})n} \left| \sum_{l=1}^{N_1} F(A_2^d) \frac{(\tilde{v}_{j,l-n} + \tilde{v}_{j,l+n})}{2} \right| \leq d_1(-\phi_\nu^{0d}, \gamma_{K_{2,j}}, n) + \frac{(1+\rho^{2n})}{(1-\rho^{2n})} \frac{d_1(A_2^d, \gamma_{K_{2,j}}, n)}{n}$$

and similarly

$$\left| \sum_{l=1}^{N_1} F(\Lambda_{1,n}^d) \frac{(\tilde{v}_{j,l-n} + \tilde{v}_{j,l+n})}{2} \right| \leq \frac{(1+\rho^{2n})}{(1-\rho^{2n})} d_1(\psi_\nu^{0d}, \gamma_{K_{2,j}}, n) + \frac{d_1(A_3^d, \gamma_{K_{2,j}}, n)}{n}$$

and the result follows. □

2.6.3 $\|K_1 \mathcal{E}_{\mathcal{L}}\|_1$ Case

Lemma 2.6.17. Let γ_{K_1} be a as in (2.6.35). Let $n \geq 1$ be an integer, then

$$\left\| K_1(\mathcal{E}_{\mathcal{L}} - \mathcal{E}_{\mathcal{L}}^d)\cos(n\nu) \right\|_1 \leq \frac{2\rho^{2c_0}}{(1-\rho^2)} \|K_1\|_1 + t_1(-\phi_\nu^0, -\phi_\nu^{0d}, \gamma_{K_1}, n) + \frac{t_1(A_3, A_3^d, \gamma_{K_1}, n)}{n}$$
$$+ \frac{(1+\rho^{2n})}{(1-\rho^{2n})} \left(t_1(\psi_\nu^0, \psi_\nu^{0d}, \gamma_{K_1}, n) + \frac{t_1(A_2, A_2^d, \gamma_{K_1}, n)}{n} \right) \tag{2.6.45}$$

Proof. The proof is immediate from the proof of Lemma 2.6.14 when $K_{2,j}$ are replaced by K_1. □

Lemma 2.6.18. Let $f^d \in \mathcal{S}$ and $g^d \in \mathcal{C}$ be a discretized approximation of some $f \in \mathcal{S}$ and $g \in \mathcal{C}$, respectively. Let $n \geq 0$ be an integer, then

$$\begin{aligned} \left\| K_1 \mathcal{L}^I f^d \cos(n\nu) \right\|_1 &\leq d_2(f^d, n) \ , \\ \left\| K_1 \mathcal{L}^I g^d \sin(n\nu) \right\|_1 &\leq d_2(g^d, n) \ , \end{aligned} \tag{2.6.46}$$

where

$$d_2(f^d,n)=\begin{cases}\sum_{k=1}^{n+\deg(f^d)}\Big\|\sum_{l=1}^{\deg(f^d)}\Big[\frac{(L^I_{k,l-n}+L^I_{k,l+n})}{2} & \\ +\sum_{j=1}^{2}F_k(a_j^d)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\Big]\Big\|_T & n\le\deg(f^d)\\ +\sum_{j=1}^{2}\left(\left\|\sum_{l=1}^{\deg(f^d)}F_l(f^d)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right\|_T\right)t_1(A_j,A_j^d,\gamma_{\mathcal{I}},0) & \\ d_1(f^d,\gamma_{\mathcal{I}},n)+\sum_{j=1}^{2}\Big(\sum_{k=1}^{N_1}\|F_k(a_j^d)\|_T & \\ +t_1(A_j,A_j^d,\gamma_{\mathcal{I}},0)\Big)d_1(f^d,\gamma_{K_{2,j}},n) & n>\deg(f^d)\end{cases}\tag{2.6.47}$$

and

$$d_2(g^d,n)=\begin{cases}\sum_{k=0}^{n+\deg(g^d)}\Big\|\sum_{l=0}^{\deg(g^d)}\Big[\frac{(L^I_{k,n-l}+L^I_{k,n+l})}{2} & \\ +\sum_{j=1}^{2}F_k(a_j^d)\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}\Big]\Big\|_T & n\le\deg(g^d)\\ +\sum_{j=1}^{2}\left(\left\|\sum_{l=0}^{\deg(g^d)}F_l(g^d)\frac{(\tilde{v}_{j,n-l}+\tilde{v}_{j,n+l})}{2}\right\|_T\right)t_1(A_j,A_j^d,\gamma_{\mathcal{I}},0) & \\ d_1(g^d,\gamma_{\mathcal{I}},n)+\sum_{j=1}^{2}\Big(\sum_{k=1}^{N_1}\|F_k(a_j^d)\|_T & \\ +t_1(A_j,A_j^d,\gamma_{\mathcal{I}},0)\Big)d_1(g^d,\gamma_{K_{2,j}},n) & n>\deg(g^d)\end{cases}\tag{2.6.48}$$

Proof. It is noted that

$$\begin{aligned}K_1\mathcal{L}^I f^d\cos(n\nu)&=K_1\sum_{l=1}^{\deg(f^d)}F_l(f^d)\mathcal{L}^I\sin(l\nu)\cos(n\nu)\\&=K_1\sum_{l=1}^{\deg(f^d)}F_l(f^d)\mathcal{L}^I\left(\frac{\sin((l-n)\nu)+\sin((l+n)\nu)}{2}\right)\\&=\sum_{N=1}^{\deg(f^d)}F_l(f^d)\frac{K_1\mathcal{L}^I\sin((l-n)\nu)+K_1\mathcal{L}^I\sin((l+n)\nu)}{2}\\&=\sum_{N=1}^{\deg(f^d)}F_l(f^d)\left(\frac{\mathcal{L}^I\sin((l-n)\nu)+\mathcal{L}^I\sin((l+n)\nu)}{2}+\sum_{j=1}^{2}a_j\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\\&=\sum_{k=1}^{\infty}\cos(k\nu)\left(\sum_{l=1}^{\deg(f^d)}F_l(f^d)\left(\frac{L^I_{k,n-l}+L^I_{k,n+l}}{2}+\sum_{j=1}^{2}F_k(a_j)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\right),\end{aligned}$$

when $n \leq \deg(f^d)$,

$$\begin{aligned}
&\left\|K_1\mathcal{L}^I f^d \cos(n\nu)\right\|_1 = \sum_{k=1}^{\infty}\left|\sum_{l=1}^{\deg(f^d)} F_l(f^d)\left(\frac{L^I_{k,l-n}+L^I_{k,l+n}}{2}+\sum_{j=1}^{2}F_k(a_j)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\right| \\
&\leq \sum_{k=1}^{n+\deg(f^d)}\left|\sum_{l=1}^{\deg(f^d)} F_l(f^d)\left(\frac{L^I_{k,l-n}+L^I_{k,l+n}}{2}+\sum_{j=1}^{2}F_k(a^d_j)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\right| \\
&+\sum_{k=1}^{\infty}\left|\sum_{l=1}^{\deg(f^d)} F_l(f^d)\left(\frac{L^I_{k,l-n}+L^I_{k,l+n}}{2}+\sum_{j=1}^{2}F_k(a_j-a^d_j)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\right| \\
&\leq \sum_{k=1}^{n+\deg(f^d)}\left\|\sum_{l=1}^{\deg(f^d)} F_l(f^d)\left(\frac{L^I_{k,l-n}+L^I_{k,l+n}}{2}+\sum_{j=1}^{2}F_k(a^d_j)\frac{(\tilde{v}_{j,l-n}+\tilde{v}_{j,l+n})}{2}\right)\right\|_T \\
&+\sum_{l=1}^{\deg(f^d)}|F_l(f^d)|\left(\frac{\gamma_{\mathcal{I},l-n}+\gamma_{\mathcal{I},l+n}}{2}+\sum_{j=1}^{2}\frac{(\gamma_{K_{2,j},l-n}+\gamma_{K_{2,j},l+n})}{2}\sum_{k=1}^{\infty}|F_k(a_j-a^d_j)|\right) \\
&\leq d_2(f^d,n),
\end{aligned}$$

and similarly when $n \geq \deg(f^d)+1$,

$$\begin{aligned}
\left\|K_1\mathcal{L}^I f^d \cos(n\nu)\right\|_1 \leq \sum_{l=1}^{\deg(f^d)}|F_l(f^d)|\Bigg(&\frac{\gamma_{\mathcal{I},l-n}+\gamma_{\mathcal{I},l+n}}{2} \\
&+\sum_{j=1}^{2}\frac{(\gamma_{K_{2,j},l-n}+\gamma_{K_{2,j},l+n})}{2}\sum_{k=1}^{\infty}(|F_k(a^d_j)|+|F_k(a_j-a^d_j)|)\Bigg) \leq d_2(f^d,n),
\end{aligned}$$

and the result follows. g case is the same. □

Corollary 2.6.19. Let $n \geq 0$, then

$$\gamma_{K_1,n} \leq d_2(1,n) \tag{2.6.49}$$

Proof. This is immediate from the previous lemma by setting $g^d = 1$. □

Lemma 2.6.20. Let $\gamma_{\mathcal{I}}$ be as defined in Corollary 2.6.13 and net $n \geq 0$ be an integer.

Then, when $n \leq N_1$:

$$\left\|K_1\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)\right\|_1 \leq \frac{\sum_{k=1}^{n+N_1}\left\|\sum_{N=1}^{n+N_1}F_N(\mathcal{E}^d_{\mathcal{L}}(1-\rho^{2c_0n})\cos(n\nu))(1_{N=k}+\sum_{j=1}^{2}v_{j,N}F_k(a^d_j))\right\|_T}{(1-\rho^{2c_0n})} + \sum_{j=1}^{2}\left(\frac{\left\|\sum_{N=1}^{n+N_1}v_{j,N}F_N(\mathcal{E}^d_{\mathcal{L}}(1-\rho^{2c_0n})\cos(n\nu))\right\|_T}{(1-\rho^{2c_0n})}\right)t_1(A_j,A^d_j,\gamma_{\mathcal{I}},0)\ , \tag{2.6.50}$$

and when $n \geq N_1+1$:

$$\begin{aligned}\left\|K_1\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)\right\|_1 &\leq \sum_{l=1}^{N_0}2\rho^{N_1+1-l}\left(\sum_{k=1}^{N_1}\left\|\sum_{N=1}^{N_0}P_{N,l}\left(1_{N=k}+\sum_{j=1}^{2}F_k(a^d_j)v_{j,N}\right)\right\|_T\right. \\ &\left.+\sum_{j=1}^{2}t_1(A_j,A^d_j,\gamma_{\mathcal{I}},0)\sum_{N=1}^{N_0}\|P_{N,l}v_{j,N}\|_T\right)+d_2(-\phi^{0d}_\nu,n)+\frac{d_2(A^d_3,n)}{N_1+1} \\ &+\frac{(1+\rho^{2(N_1+1)})}{(1-\rho^{2(N_1+1)})}\left(d_2(\psi^{0d}_\nu,n)+\frac{d_2(A^d_2,n)}{N_1+1}\right),\end{aligned} \tag{2.6.51}$$

Remark 2.6.21. Note that for each $N, n \geq 1$, the $(1-\rho^{2c_0n})$ factor makes the Fourier coefficients

$$\begin{aligned}&F_N(\mathcal{E}^d_{\mathcal{L}}(1-\rho^{2c_0n})\cos(n\nu)) = \tilde{P}_{N,n}(1-\rho^{2c_0n}) \\ &+\sum_{l=0}^{\infty}\Big[\left(-F_l(\phi^{0d}_\nu)+\frac{(1+\rho^{2n})}{(1-\rho^{2n})}\frac{F_l(A^d_2)}{n}\right)\frac{(L^I_{N,l+n}+L^I_{N,l-n})}{2} \\ &+\left(\frac{(1+\rho^{2n})}{(1-\rho^{2n})}F_l(\psi^{0d}_\nu)+\frac{F_l(A^d_3)}{n}\right)\frac{(L^I_{N,n+l}+L^I_{N,n-l})}{2}\Big](1-\rho^{2c_0n}),\end{aligned} \tag{2.6.52}$$

a polynomial in ρ, which enables the evaluation of $\|\ \|_T$-norms that are introduced in the previous Lemma.

Proof of Lemma 2.6.20. Let

$$K^d_1 f := f + \mathbf{a}^T_d M^I \mathbf{m}[f] \quad \text{where} \quad \mathbf{a}_d = [a^d_1, a^d_2]^T\ , \tag{2.6.53}$$

When $n \leq N_1$; it is noted that $K_1\mathcal{E}^d_{\mathcal{L}}\cos(n\nu) = K^d_1\mathcal{E}^d_{\mathcal{L}}\cos(n\nu) + (K_1-K^d_1)\mathcal{E}^d_{\mathcal{L}}\cos(n\nu)$, and

$$K^d_1\mathcal{E}^d_{\mathcal{L}}\cos(n\nu) = K^d_1\sum_{N=1}^{\infty}F_N(\mathcal{E}^d_{\mathcal{L}}\cos(n\nu))\cos(N\nu) = \sum_{N=1}^{n+N_1}F_N(\mathcal{E}^d_{\mathcal{L}}\cos(n\nu))K^d_1\cos(N\nu)$$

$$\sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \left(\cos(N\nu) + \sum_{j=1}^{2} a_j^d v_{j,N} \right)$$
$$= \sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \sum_{k=1}^{n+N_1} \cos(k\nu) \left(1_{N=k} + \sum_{j=1}^{2} F_k(a_j^d) v_{j,N} \right)$$
$$= \sum_{k=1}^{n+N_1} \cos(k\nu) \left(\sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \left(1_{N=k} + \sum_{j=1}^{2} F_k(a_j^d) v_{j,N} \right) \right)$$

and thus

$$\left\| K_1^d \mathcal{E}_{\mathcal{L}}^d \cos(n\nu) \right\|_1 = \frac{1}{(1-\rho^{2c_0 n})} \left\| K_1^d \mathcal{E}_{\mathcal{L}}^d (1-\rho^{2c_0 n}) \cos(n\nu) \right\|_1 \tag{2.6.54}$$
$$\leq \frac{1}{(1-\rho^{2c_0 n})} \sum_{k=1}^{n+N_1} \left| \sum_{N=1}^{n+N_1} \left(1_{N=k} + \sum_{j=1}^{2} F_k(a_j) v_{j,N} \right) F_N(\mathcal{E}_{\mathcal{L}}^d (1-\rho^{2c_0 n}) \cos(n\nu)) \right|$$

Similarly

$$(K_1 - K_1^d) \mathcal{E}_{\mathcal{L}}^d \cos(n\nu) = \sum_{k=1}^{n+N_1} \cos(k\nu) \left(\sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \left(\sum_{j=1}^{2} F_k(a_j - a_j^d) v_{j,N} \right) \right)$$

and thus

$$\left\| (K_1 - K_1^d) \mathcal{E}_{\mathcal{L}}^d \cos(n\nu) \right\|_1 = \sum_{k=1}^{\infty} \left| \sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \left(\sum_{j=1}^{2} F_k(a_j - a_j^d) v_{j,n} \right) \right|$$
$$\leq \sum_{j=1}^{2} \sum_{k=1}^{\infty} \left| F_k(a_j - a_j^d) \right| \left| \sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d \cos(n\nu)) \right|$$
$$\leq \sum_{j=1}^{2} \sum_{k=1}^{\infty} \left| F_k(a_j - a_j^d) \right| \frac{\left| \sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d (1-\rho^{2c_0 n}) \cos(n\nu)) \right|}{(1-\rho^{2c_0 n})}$$
$$\leq \sum_{j=1}^{2} \left(\frac{\left\| \sum_{N=1}^{n+N_1} F_N(\mathcal{E}_{\mathcal{L}}^d (1-\rho^{2c_0 n}) \cos(n\nu)) \right\|_T}{(1-\rho^{2c_0 n})} \right) \sum_{k=1}^{\infty} \left| F_k(a_j - a_j^d) \right|$$

and (2.6.50) follows from triangle inequality, taking $\|\|_T$-norm of (2.6.54), and from Corollary 2.6.13. When $n \geq N_1 + 1$; it is noted that

$$K_1 \mathcal{E}_{\mathcal{L}}^d \cos(n\nu) = K_1 \mathcal{L}_1^I \mathcal{L}_0 \cos(n\nu) + K_1 \mathcal{L}^I \mathcal{L}_1^d \cos(n\nu)$$
$$= K_1 \mathcal{L}^I \mathcal{L}_0 \cos(n\nu) + K_1 \mathcal{L}^I \Lambda_{1,n}^d \cos(n\nu) + K_1 \mathcal{L}^I \Lambda_{2,n}^d \sin(n\nu)$$

and that

$$K_1\mathcal{L}_1^I\mathcal{L}_0\cos(n\nu) = K_1\sum_{N=1}^{\infty} F_N(\mathcal{L}_1^I\mathcal{L}_0\cos(n\nu))\cos(N\nu) = \sum_{N=1}^{N_0}\tilde{P}_{N,n}K_1\cos(N\nu)$$
$$= \sum_{k=1}^{\infty}\cos(k\nu)\left(\sum_{N=1}^{N_0}\tilde{P}_{N,n}\left(1_{N=k}+\sum_{j=1}^{2}F_k(a_j)v_{j,N}\right)\right)$$

and hence

$$\sum_{k=1}^{\infty}\left|\sum_{N=1}^{N_0}\tilde{P}_{N,n}\left(1_{N=k}+\sum_{j=1}^{2}F_k(a_j)v_{j,N}\right)\right|$$
$$= \sum_{k=1}^{N_1}\left|\sum_{l=1}^{N_0}L_{l,n}^0\left[\sum_{N=1}^{N_0}P_{N,l}\left(1_{N=k}+\sum_{j=1}^{2}(F_k(a_j^d)+F_k(a_j-a_j^d))v_{j,N}\right)\right]\right|$$
$$\leq \sum_{l=1}^{N_0}\left|L_{l,n}^0\right|\sum_{k=1}^{N_1}\left\|\sum_{N=1}^{N_0}P_{N,l}\left(1_{N=k}+\sum_{j=1}^{2}F_k(a_j^d)v_{j,N}\right)\right\|_T$$
$$+\sum_{l=1}^{N_0}\left|L_{l,n}^0\right|\sum_{j=1}^{2}\sum_{N=1}^{N_0}\|P_{N,l}v_{j,N}\|_T\sum_{k=1}^{\infty}\left|F_k(a_j-a_j^d)\right|$$
$$\leq \sum_{l=1}^{N_0}2\rho^{N_1+1-l}\left(\sum_{k=1}^{N_1}\left\|\sum_{N=1}^{N_0}P_{N,l}(1_{N=k}+\sum_{j=1}^{2}F_k(a_j^d)v_{j,N})\right\|_T+\sum_{j=1}^{2}t_1(A_j,\gamma_0)\sum_{N=1}^{N_0}\|P_{N,l}v_{j,N}\|_T\right)$$

since $\left|L_{l,n}^0\right| \leq 2\rho^{n-l} \leq 2\rho^{N_1+1-l}$ when $n \geq N_1+1$. The result follows from Lemma 2.6.18 by choosing $f = -\phi_\nu^0, \psi_\nu^0, A_2, A_3$. □

2.6.4 $\|J_2\mathcal{E}_M\|_1$ Case

Lemma 2.6.22. Let γ_1 and γ_2 be as in defined in (2.6.35), let

$$J_2\mathcal{E}_M = [J_{2,1}\mathcal{E}_M\ ,\ J_{2,2}\mathcal{E}_M]\ ,\ J_2\mathcal{E}_M^d = \left[J_{2,1}\mathcal{E}_M^d\ ,\ J_{2,2}\mathcal{E}_M^d\right] \tag{2.6.55}$$

and let

$$\begin{bmatrix} m_{1,1}^d & m_{1,2}^d \\ m_{2,1}^d & m_{2,2}^d \end{bmatrix} = I_2 - M^I\begin{bmatrix} 1-\sum_{n=1}^{N_1}\frac{(\rho^n+(-\rho)^n)}{n}F_n(a_2^d) & -\sum_{n=1}^{N_1}\frac{(\rho^n+(-\rho)^n)}{n}F_n(a_1^d) \\ -\sum_{n=1}^{N_1}\frac{(\rho^n-(-\rho)^n)}{n}F_n(a_2^d) & -\sum_{n=1}^{N_1}\frac{(\rho^n-(-\rho)^n)}{n}F_n(a_1^d) \end{bmatrix} \tag{2.6.56}$$

where $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then

$$\|J_{2,i}\mathcal{E}_M\|_1 \leq \left\|m_{1,i}^d\right\|_T + \left\|m_{2,i}^d\right\|_T + \sum_{j=1}^{2} t_1(A_j, A_j^d, \gamma_{\mathcal{I}}, 0) \tag{2.6.57}$$

for $i = 1, 2$.

Proof. It is evident that $J_{2,i}\mathcal{E}_M^d = [m_{1,i}^d, m_{2,i}^d]^T$ and that $J_{2,i}(\mathcal{E}_M - \mathcal{E}_M^d) = M^I \mathbf{m}\mathcal{L}^I(A_i - A_i^d) = K_2\mathcal{L}^I(A_i - A_i^d)$. Hence, evidently $\left\|J_{2,i}\mathcal{E}_M^d\right\|_1 = \|m_{1,i}^d\|_T + \|m_{2,i}^d\|_T$, and

$$\begin{aligned}
&\left\|K_2\mathcal{L}^I(A_i - A_i^d)\right\|_1 = \sum_{j=1}^{2}\left|K_{2,j}\mathcal{L}^I(A_i - A_i^d)\right| = \sum_{j=1}^{2}\left|K_{2,j}\mathcal{L}^I\sum_{n=1}^{\infty}F_n(A_i - A_i^d)\sin(n\nu)\right| \\
&= \sum_{j=1}^{2}\left|\sum_{n=1}^{\infty}F_n(A_i - A_i^d)\tilde{v}_{j,n}\right| \leq \sum_{j=1}^{2}\sum_{n=1}^{\infty}\left|F_n(A_i - A_i^d)\right|\gamma_{j,n} \leq \sum_{j=1}^{2} t_1(A_j, A_j^d, \gamma_{\mathcal{I}}, 0)
\end{aligned}$$

and the result follows. □

Corollary 2.6.23.

$$\|J_2\mathcal{E}_M\|_1 \leq \max_{i=1,2}\left\{\left\|m_{1,i}^d\right\|_T + \left\|m_{2,i}^d\right\|_T + \sum_{j=1}^{2} t_1(A_j, A_j^d, \gamma_{\mathcal{I}}, 0)\right\} \tag{2.6.58}$$

2.6.5 $\|J_1\mathcal{E}_M\|_1$ Case

Lemma 2.6.24.

$$\|J_1\mathcal{E}_M\|_1 \leq \sum_{j=1}^{2}\left(t_1(A_j, \gamma_0) + \sum_{n=1}^{N_1}\|F_n(a_j^d)\|_T\right)\|J_2\mathcal{E}_M\|_1 \tag{2.6.59}$$

Proof. This follow from $\|J_1\mathcal{E}_M\|_1 = \|\mathbf{a}J_2\mathcal{E}_M\|_1 \leq \|\mathbf{a}\|_1\|J_2\mathcal{E}_M\|_1$, and from

$$\begin{aligned}
\|\mathbf{a}\|_1 &= \sum_{j=1}^{2}\|a_j\|_1 \leq \sum_{j=1}^{2}\sum_{n=1}^{\infty}|F_n(a_j)| \leq \sum_{j=1}^{2}\sum_{n=1}^{N_1}\left|F_n(a_j^d)\right| + \sum_{j=1}^{2}\sum_{n=1}^{\infty}\left|F_n(a_j - a_j^d)\right| \\
&\leq \sum_{j=1}^{2}\left(t_1(A_j, A_j^d, \gamma_{\mathcal{I}}, 0) + \sum_{n=1}^{N_1}\|F_n(a_j^d)\|_T\right)
\end{aligned}$$

□

Lemma 2.6.25. Let

$$\gamma_{\mathcal{I}} := \{\gamma_{\mathcal{I},n}\}_{n=1}^{\infty} \text{ where } \gamma_{\mathcal{I},n} := \begin{cases} \sum_{N=1}^{n} \|L_{N,n}^{I}\|_T & \text{if } 1 \le n \le N_0, \\ \frac{1}{n+1} + \frac{2\rho^2}{(n^2-1)(1-\rho^2)} & \text{if } n > N_0, \end{cases} \tag{2.6.60}$$

Then

$$\left\| \mathcal{L}^I \sin(n\nu) \right\|_1 \le \gamma_{\mathcal{I},n} \text{ for all } n \ge 1, \tag{2.6.61}$$

Proof. It is noted that $\left\| \mathcal{L}^I \sin(n\nu) \right\|_1 = \left\| \sum_{N=1}^{\infty} F_N(\mathcal{L}^I \sin(n\nu)) \cos(N\nu) \right\|_1 = \sum_{N=1}^{n} \left| L_{N,n}^{I} \right|$ and hence, when $n > N_0$ one has

$$\begin{aligned} \sum_{N=1}^{n} \left| L_{N,n}^{I} \right| &= \sum_{N=1}^{n} \left| -\frac{1_{N=n}}{(N+1)} + (1-\rho^{2c_0 n}) 1_{n>N} \frac{\left(1-\rho^{2N}\right)}{(1-\rho^{2n})} \frac{\rho^{n-N}}{(n^2-1)} (1+(-1)^{n-N}) \right| \\ &\le \sum_{N=1}^{n} \frac{1_{N=n}}{(N+1)} + 1_{n>N} \frac{\rho^{n-N}(1+(-1)^{n-N})}{(n^2-1)} \le \frac{1}{n+1} + \frac{1}{(n^2-1)} \sum_{N=1}^{\infty} 2\rho^{2N} \end{aligned}$$

and when $n \le N_0$ one immediately obtains $\sum_{N=1}^{n} \left| L_{N,n}^{I} \right| \le \sum_{N=1}^{n} \|L_{N,n}^{I}\|_T$, and the result follows. □

Lemma 2.6.26. Let

$$\gamma_{K_{2,j}} := \left\{ \gamma_{K_{2,j},n} \right\}_{n=1}^{\infty} \text{ where } \gamma_{K_{2,j},n} := \begin{cases} \|\tilde{v}_{j,n}\|_T, & 1 \le n \le N_0 \\ 2\rho^n \frac{(1+2\ln n)}{(n^2-1)} \max_{k=1,2} \|m_{j,k}\|_T, & n \ge N_0 + 1 \end{cases} \tag{2.6.62}$$

Then

$$\left\| K_{2,j} \mathcal{L}^I \sin(n\nu) \right\|_1 \le \gamma_{K_{2,j},n} \text{ for all } n \ge 1 \text{ and } j = 1, 2 \tag{2.6.63}$$

Proof. Since $K_{2,j}\mathcal{L}^I \sin(n\nu) = \tilde{v}_{j,n}$, when $n \le N_0$ it is evident that $\|K_{2,j}\mathcal{L}^I \sin(n\nu)\|_1 \le \|\tilde{v}_{j,n}\|_T$, and when $n \ge N_0 + 1$, one has

$$|\tilde{v}_{j,n}| = \begin{cases} |m_{j,1}|\,|\tilde{v}_{+,n}| & n \text{ even} \\ |m_{j,2}|\,|\tilde{v}_{-,n}| & n \text{ odd} \end{cases} \le \max\{|\tilde{v}_{+,n}|, |\tilde{v}_{-,n}|\} \max_{k=1,2} \sum_{k=1}^{2} \|m_{j,k}\|_T \tag{2.6.64}$$

and by taking absolute value of (2.6.34) and the identities $1 \pm (-1)^n \le 2$, $\frac{(1-\rho^{2N})(1\pm(-1)^N)}{(1-\rho^{2n})} \le 2$, $1 - \rho^{2c_0 n} \le 1$, and $\sum_{N=1}^{n-1} \frac{(1+(-1)^N)}{N} \le \sum_{N=1}^{n-1} \frac{2}{N} \le 2\ln(n)$ one obtains

$$|\tilde{v}_{\pm,n}| \le 2\rho^n \frac{(1+2\ln n)}{(n^2-1)} \text{ for } n > N_0, \tag{2.6.65}$$

and the result follows. □

2.6.6 $\|\chi_0\|_1$ and $\|K\mathcal{L}^I \circ (f)\|$ Cases

Lemma 2.6.27. Let $f^d \in \mathcal{S}$ and $g^d \in \mathcal{C}$ be a discretized approximation of some $f \in \mathcal{S}$ and $g \in \mathcal{C}$, respectively. Let $n \geq 0$ be an integer, then

$$\begin{aligned}
\|K\mathcal{L}^I f\cos(n\nu)\|_1 &\leq d_2(f^d, n) + t_1(f, f^d, \gamma_{K_1}, n) + \sum_{j=1}^{2} d_1(f^d, \gamma_{K_{2,j}}, \tilde{v}_j, n) + t_1(f, f^d, \gamma_{K_{2,j}}, n) \\
\|K\mathcal{L}^I g\sin(n\nu)\|_1 &\leq d_2(g^d, n) + t_1(f, g^d, \gamma_{K_1}, n) + \sum_{j=1}^{2} d_1(g^d, \gamma_{K_{2,j}}, \tilde{v}_j, n) + t_1(g, g^d, \gamma_{K_{2,j}}, n)
\end{aligned} \tag{2.6.66}$$

Proof. It is observed that $\|K\mathcal{L}^I f\|_1 = \|K_1\mathcal{L}^I f\|_1 + \sum_{j=1}^{2} |K_{2,j}\mathcal{L}^I f|$ The result follows from Lemma 2.6.12, Lemma 2.6.15 and Lemma 2.6.18. □

Corollary 2.6.28.

$$\|\chi_0\|_1 \leq d_2(\mathcal{N}^{0d}, 0) + t_1(\mathcal{N}^0, \mathcal{N}^{0d}, \gamma_{K_1}, 0) + \sum_{j=1}^{2}[d_1(\mathcal{N}^{0d}, \gamma_{K_{2,j}}, \tilde{v}_j, 0) + t_1(\mathcal{N}^0, \mathcal{N}^{0d}, \gamma_{K_{2,j}}, 0)] \tag{2.6.67}$$

Proof. This is immediate by setting $f = \mathcal{N}^0$ and $n = 0$ in the previous Lemma. □

Corollary 2.6.29. Let $f^d \in \mathcal{S}$ and $g^d \in \mathcal{C}$ be a discretized approximation of some $f \in \mathcal{S}$ and $g \in \mathcal{C}$, respectively. Let

$$\begin{aligned}
\mathcal{L}^I \circ (f) : \mathcal{C} \to \mathcal{S}\ ,\ \mathcal{L}^I \circ (f)[h] = \mathcal{L}^I f h\ , \\
\mathcal{L}^I \circ (g) : \mathcal{S} \to \mathcal{S}\ ,\ \mathcal{L}^I \circ (g)[h] = \mathcal{L}^I g h\ ,
\end{aligned} \tag{2.6.68}$$

Then

$$\begin{aligned}\left\|K\mathcal{L}^I\circ(f)\right\|_1 &\le \max_{0\le n\le \deg(f^d)+1}\Big\{d_2(f^d,n)+t_1(f,f^d,\gamma_{K_1},n)\\ &\quad+\sum_{j=1}^{2}\Big[d_1(f^d,\gamma_{K_{2,j}},\tilde{v}_j,n)+t_1(f,f^d,\gamma_{K_{2,j}},n)\Big]\Big\},\\ \left\|K\mathcal{L}^I\circ(g)\right\|_1 &\le \max_{1\le n\le \deg(f^d)+1}\Big\{d_2(g^d,n)+t_1(f,g^d,\gamma_{K_1},n)\\ &\quad+\sum_{j=1}^{2}\Big[d_1(g^d,\gamma_{K_{2,j}},\tilde{v}_j,n)+t_1(g,g^d,\gamma_{K_{2,j}},n)\Big]\Big\},\end{aligned}\tag{2.6.69}$$

Proof. This is immediate from $\left\|K\mathcal{L}^I\circ(f)\right\|_1=\sup_{n\ge0}\left\|K\mathcal{L}^I f\cos(n\nu)\right\|_1$ and from the previous lemma. g case is similar. □

2.7 Bounds on nonlinear terms $\mathcal{R}$

Recall $\mathcal{R}=\langle K_1\mathcal{L}^I R, K_2\mathcal{L}^I R\rangle$, where $R=R_0+R_1$, with R_0 are all quadratic terms and R_1 has the remaining nonlinear terms, given by (2.4.49)-(2.4.50). Note R_0, R_1 may be rewritten as

$$R_0=-E_\nu F_\nu+\Im\left[(b+ia)R_{0,1}\right]+\frac{1}{2}\left[aU_0+c(U_0-1)\right](w+F)^2+u(a+c)(w+F)\ , \tag{2.7.1}$$

$$R_1=\Im\left\{(b+ia)R_{1,1}\right\}+\left[U_0a+(U_0-1)c\right]e_3[w+F]+uce_2[w+F]\ , \tag{2.7.2}$$

where for $\chi=\langle\psi_\nu,w,u\rangle$,

$$R_{0,1}\left[\chi\right]=\frac{U_0}{2}\left(w+F+iE\right)^2+\left(U_0(w+F)+u\right)\left[w+F+iE\right]\ , \tag{2.7.3}$$

$$\begin{aligned}R_{1,1}[\chi]&=Ue_2(w+F)e^{w+F+iE}-U_0e_2(w+F)+U_0e_3[w+F+iE]\\ &\quad+\left((1+w+F)U-U_0\right)e_2[w+F+iE]+u(W+F)(w+F+iE),\end{aligned}\tag{2.7.4}$$

Remark 2.7.1. Recall we are using the norm

$$\|\chi\|_1=\|\psi_\nu\|_1+c_w|w|+c_u|u| \tag{2.7.5}$$

Definition 2.7.2. Define

$$C_0 = \max\left\{ \frac{1}{c_w} \ , \ \sup_{\rho\in[\rho_0,\rho_1]} \sup_{k\geq 1} \frac{(1+\rho^{2k})}{k(1-\rho^{2k})} \right\} \tag{2.7.6}$$

$$C_1 = \max\left\{ C_0, \frac{2}{3U_0 c_u} \right\} \ , \ C_2 = \min\left\{ C_0, \frac{2}{3U_0 c_u} \right\} \tag{2.7.7}$$

Lemma 2.7.3. Let $\chi = \langle E_\nu, w, u\rangle$, where $E_\nu \in \mathcal{C}_0$. Then

$$\|F_\nu\|_1 \leq C_0\|E_\nu\|_1 \leq C_0\left(\|\chi\|_1 - c_w|w| - c_u|u|\right) \ , \tag{2.7.8}$$

$$\|w+F\|_1 \leq |w| + C_0\|E_\nu\|_1 \leq C_0\left(\|\chi\|_1 - c_u|u|\right) \ , \tag{2.7.9}$$

$$\|w+F+iE\|_1 \leq |w| + C_0\|E_\nu\|_1 \leq C_0\left(\|\chi\|_1 - c_u|u|\right) \ , \tag{2.7.10}$$

Proof. We note that on $\zeta = e^{i\nu}$, $\omega = F + iE$ has the following Fourier representation

$$\omega = \sum_{k=1}^{\infty} \omega_k \left(e^{ik\nu} - \rho^{2k} e^{-ik\nu} \right) \tag{2.7.11}$$

implying $\|F+iE\|_1 = \sum_{k=1}^{\infty} |\omega_k|(1+\rho^{2k})$,

$$E_\nu = \sum_{k=1}^{\infty} k\omega_k(1-\rho^{2k})\cos(knu) \ = \frac{1}{2}\sum_{k=-\infty,k\neq 0}^{\infty} |k|\omega_{|k|}(1-\rho^{|2k|})e^{ik\nu} \ , \tag{2.7.12}$$

$$F_\nu = \sum_{k=1}^{\infty} k\omega_k(1+\rho^{2k})\sin(knu) \ = \frac{1}{2i}\sum_{k=-\infty,k\neq 0}^{\infty} |k|\mathrm{sgn}k\omega_{|k|}(1-\rho^{|2k|})e^{ik\nu} \ , \tag{2.7.13}$$

implying

$$\|E_\nu\|_1 = \sum_{k=1}^{\infty} k|\omega_k|\left(1-\rho^{2k}\right) \ , \ \|F_\nu\|_1 = \sum_{k=1}^{\infty} k|\omega_k|\left(1+\rho^{2k}\right) \ , \tag{2.7.14}$$

and therefore it easily follows that

$$\|F_\nu\|_1 \leq C_0\|E_\nu\|_1 \leq C_0\left(\|\chi\|_1 - c_w|w| - c_u|u|\right) \tag{2.7.15}$$

Furthermore, we have

$$\|w+F+iE\|_1 = |w| + \sum_{k=1}^{\infty} k|\omega_k|\left(1+\rho^{2k}\right) \leq |w| + C_0\|\psi_\nu\|_1 \leq C_0\left(\|\chi\|_1 - c_u|u|\right) \tag{2.7.16}$$

Clearly the same argument as above applies to $\|w+F\|_1$. □

Lemma 2.7.4.

$$\|E_\nu F_\nu\|_1 \leq C_0 \|E_\nu\|^2 \leq C_0 \left(\|\chi\|_1 - c_w|w| - c_u|u|\right)^2 \ , \tag{2.7.17}$$

$$\left\|\frac{1}{2}(w+F)^2\right\|_1 \leq \frac{1}{2}\|w+F\|_1^2 \leq \frac{1}{2}C_0^2 \left(\|\chi\|_1 - c_u|u|\right)^2 \ , \tag{2.7.18}$$

$$\|u(w+F)\|_1 \leq |u|\,\|w+F\|_1 \leq \frac{C_0}{c_u}\left(\|\chi\|_1 - \|\psi_\nu\|_1 - c_w|w|\right)\left(\|\chi\|_1 - c_u|u|\right) \ , \tag{2.7.19}$$

Proof. We note from Banach Algebra property of l^1 norm in the Fourier space and Lemma and definition of $\|\chi\|_1$ 2.7.3,

$$\|E_\nu F_\nu\|_1 \leq \|E_\nu\|_1 \|F_\nu\|_1 \leq C_0 \|E_\nu\|_1^2 \leq C_0 \left(\|\chi\|_1 - c_w|w| - c_u|u|\right)^2 \tag{2.7.20}$$

Also,

$$\left\|\frac{1}{2}(w+F)^2\right\|_1 \leq \frac{1}{2}\|w+F\|_1^2 \leq \frac{C_0^2}{2}\left(\|\chi\|_1 - c_u|u|\right)^2 \ , \tag{2.7.21}$$

$$\begin{aligned} \|u(w+F)\|_1 &\leq |u|\|w+F\|_1 \leq C_0|u|\left(\|\chi\|_1 - c_u|u|\right) \\ &\leq \frac{C_0}{c_u}\left(\|\chi\|_1 - \|\psi_\nu\|_1 - c_u|u|\right)\left(\|\chi\|_1 - c_u|u|\right) \ , \end{aligned} \tag{2.7.22}$$

□

Lemma 2.7.5. We also have

$$\left\|E_\nu^{(1)}F_\nu^{(1)} - E_\nu^{(2)}F_\nu^{(2)}\right\|_1 \leq C_0 \left(\|E_\nu^{(1)}\|_1 + \|E_\nu^{(2)}\|_1\right)\|E_\nu^{(1)} - E_\nu^{(2)}\|_1 \ , \tag{2.7.23}$$

$$\frac{1}{2}\left\|\left(w^{(1)}+F^{(1)}\right)^2 - \left(w^{(2)}+F^{(2)}\right)^2\right\|_1 \leq \frac{1}{2}C_0^2\left(\|\chi_1\|_1 + \|\chi_2\|_1\right)\|\chi_1 - \chi_2\|_1 \ , \tag{2.7.24}$$

$$\|u^{(1)}(w^{(1)}+F^{(1)}) - u^{(2)}(w^{(2)}+F^{(2)})\|_1 \leq \frac{C_0}{c_u}\left(\|\chi_1\|_1 + \|\chi_2\|_1\right)\|\chi_1 - \chi_2\|_1 \tag{2.7.25}$$

Proof. We note that

$$E_\nu^{(1)}F_\nu^{(1)} - E_\nu^{(2)}F_\nu^{(2)} = E_\nu^{(1)}\left(F_\nu^{(1)} - F_\nu^{(2)}\right) + F_\nu^{(2)}\left(E_\nu^{(1)} - E_\nu^{(2)}\right) \tag{2.7.26}$$

The rest of (2.7.23) from using Banach Algebra properties and using Lemmar 2.7.3. We

note that

$$\begin{aligned}&\frac{1}{2}\left[(w^{(1)}+F^{(1)})^2-(w^{(2)}+F^{(2)})^2\right]\\&=\frac{1}{2}\left(w^{(1)}+F^{(1)}+w^{(2)}+F^{(2)}\right)\left(w^{(1)}+F^{(1)}-w^{(2)}-F^{(2)}\right)\end{aligned}\tag{2.7.27}$$

The rest of (2.7.24) follows from using Banach algebra properties and Lemma 2.7.3 We also note that

$$\begin{aligned}u^{(1)}\left(w^{(1)}+F^{(1)}\right)-u^{(2)}\left(w^{(2)}+F^{(2)}\right)&=\left(u^{(1)}-u^{(2)}\right)\left(w^{(1)}+F^{(1)}\right)\\&\quad+u^{(2)}\left(w^{(1)}+F^{(1)}-w^{(2)}-F^{(2)}\right)\end{aligned}$$

This implies that

$$\begin{aligned}\|u^{(1)}(w^{(1)}+F^{(1)})-u^{(2)}(w^{(2)}+F^{(2)})\|_1&\leq C_0\left|u^{(1)}-u^{(2)}\right|\|\chi_1\|_1+C_0|u^{(2)}|\|\chi_1-\chi_2\|_1\\&\leq\frac{C_0}{c_u}\left(\|\chi_1\|+\|\chi_2\|\right)\|\chi_1-\chi_2\|_1\end{aligned}$$

and therefore (2.7.25) follows. □

Lemma 2.7.6.

$$\begin{aligned}\|R_{0,1}[\chi]\|_1&\leq C_0\left(\|\chi\|_1-c_u|u|\right)\left\{\frac{3}{2}C_0U_0\left(\|\chi\|_1-c_u|u|\right)+\frac{1}{c_u}\left(\|\chi|_1-\|\psi_\nu\|_1-c_w|w|\right)\right\}\\&\leq C_0\left(\frac{3}{2}U_0(C_0-C_2)+\frac{1}{c_u}\right)\|\chi\|_1\left(\|\chi\|_1-c_u|u|\right)\ ,\end{aligned}\tag{2.7.28}$$

Proof. We note that

$$\begin{aligned}&\left\|\frac{U_0}{2}\left(w+F+iE\right)^2+\left(U_0(w+F)+u\right)\left[w+F+iE\right]\right\|_1\\&\leq\|w+F+iE\|_1\left(\frac{U_0}{2}\|w+F+iE\|_1+U_0\|w+F\|_1+|u|\right)\\&\leq C_0\left(\|\chi\|_1-c_u|u|\right)\left\{\frac{3}{2}U_0C_0\left(\|\chi\|_1-c_u|u|\right)+\frac{1}{c_u}\left(\|\chi\|_1-\|\psi_\nu\|_1-c_w|w|\right)\right\}\end{aligned}$$

Combining negative terms and determining its minimum, the the Lemma follows after noting definition of C_2. □

Lemma 2.7.7. For $\|\chi_1\|_1, \|\chi_2\|_1 \leq \delta$,

$$\|R_{0,1}[\chi_1] - R_{0,1}[\chi_2]\|_1 \leq 3U_0C_0C_1\delta\|\chi_1 - \chi_2\|_1 \ , \tag{2.7.29}$$

Proof. We notice that

$$\begin{aligned} R_{0,1}[\chi_1] - R_{0,1}[\chi_2] &= (w_1 + F_1 + iE_1)\left[\frac{U_0}{2}\left(w_1 - w_2 + F_1 - F_2 + i(E_1 - E_2)\right) + \right. \\ &\left. U_0(w_1 - w_2 + F_1 - F_2) + u_1 - u_2\right] \\ &+ \left[\frac{U_0}{2}(w_2 + F_2 + iE_2) + \left(U_0(w_2 + F_2) + u_2\right)\right](w_1 - w_2 + F_1 - F_2 + iE_1 - iE_2) \end{aligned}$$

Using Banach algebra properties, lemma 2.7.3 and definition of $\|\chi\|_1$, C_1, we obtain

$$\|R_{0,1}[\chi_1] - R_{0,1}[\chi_2]\|_1 = 3C_0C_1\delta U_0\|\chi_1 - \chi_2\|_1 \tag{2.7.30}$$

□

Definition 2.7.8. Let $f = 1, b + ia, U^0a + (U^0 - 1)c, a + c$, then, we define

$$\begin{aligned} C_{K\mathcal{L}^I\circ f} := \max_{0\leq n\leq \deg(f^d)+1} &\left\{ d_2(f^d, n) + t_1(f, f^d, \gamma_{K_1}, n) \right. \\ &\left. + \sum_{j=1}^{2}\left[d_1(f^d, \gamma_{K_{2,j}}, \tilde{v}_j, n) + t_1(f, f^d, \gamma_{K_{2,j}}, n)\right]\right\} \end{aligned} \tag{2.7.31}$$

and in the case $f = 1$, we use the notation $C_{K\mathcal{L}^I\circ f} = C_{K\mathcal{L}^I}$.

Corollary 2.7.9.

$$\left\|K\mathcal{L}^I \circ f\right\|_1 \leq C_{K\mathcal{L}^I\circ f}, \tag{2.7.32}$$

Proof. This is immediate from Corollary 2.6.29. □

Lemma 2.7.10. For any $\chi = \langle E_\nu, w, u\rangle$,

$$\|K\mathcal{L}^I R_0[\chi]\|_1 \leq k_{2,0}\|\chi\|_1^2 \tag{2.7.33}$$

where

$$k_{2,0} = C_0C_{K\mathcal{L}^I} + \left(\frac{3}{2}U_0C_0(C_0 - C_2) + \frac{C_0}{c_u}\right)C_{K\mathcal{L}^I\circ(b+ia)} \tag{2.7.34}$$

$$+\frac{1}{2}C_0^2 C_{K\mathcal{L}^I\circ[aU_0+c(U_0-1)]}+\frac{C_0}{c_u}C_{K\mathcal{L}^I\circ(c+a)}\ ,$$

Furthermore, for $\|\chi_1\|_1, \|\chi_2\|_1 \leq \delta$,

$$\|K\mathcal{L}^I R_0(\chi_1) - K\mathcal{L}^I R_0(\chi_2)\|_1 \leq 2k_{2,0}\delta\|\chi_1-\chi_2\|_1\ , \tag{2.7.35}$$

Proof. We note from expression for R_0 that

$$\begin{aligned}\|K\mathcal{L}^I R_0\|_1 &\leq \|K\mathcal{L}^I\|_1\|E_\nu F_\nu\|_1 + \|K\mathcal{L}^I\circ(b+ia)\|_1\|R_{0,1}\|_1\\ &+\|K\mathcal{L}^I\circ(aU_0+c(U_0-1)\|_1\ \frac{1}{2}\|(w+F)^2\|_1 + \|K\mathcal{L}^I\circ(a+c)\|_1\|u(w+F)\|_1\end{aligned}$$

The rest follows from previous estimates in Lemmas 2.7.4, 2.7.5 and 2.7.6 . □

Lemma 2.7.11. For $j \geq 2$, for $\|\chi\| \leq \delta \leq \frac{1}{100C_0}$, $e_j[w+F]$ and $e_j[w+F+iE]$ satisfy

$$\begin{aligned}\|e_j[w+F]\|_1 &\leq \frac{C_0^j\|\chi\|_1^j}{j!}\left(1+100^j j! e_{j+1}[1/100]\right)\ ,\\ \|e_j[w+F+iE]\|_1 &\leq \frac{C_0^j\|\chi\|_1^j}{j!}\left(1+100 j! e_{j+1}[1/100]\right)\ ,\end{aligned} \tag{2.7.36}$$

$$\begin{aligned}\|e_j[w_1+F_2]-e_j[w_2+F_2]\|_1 &\leq e_{j-1}\left[C_0\delta\right]\|\chi_1-\chi_2\|_1\ ,\\ \|e_j[w_1+F_1+iE_1]-e_j[w_2+F_2+iE_2]\|_1 &\leq C_0 e_{j-1}\left[C_0\delta\right]\|\chi_1-\chi_2\|_1\ ,\end{aligned} \tag{2.7.37}$$

Proof. The results follow from the series representation

$$e_j[y] = \sum_{m=j}^{\infty}\frac{y^m}{m!} \tag{2.7.38}$$

after using $\|w+F\|_1 \leq C_0\|\chi\|_1$ and Banach Algebra properties. In the case of $e_j[y_1]-e_j[y_2]$, we use

$$e_j[y_1]-e_j[y_2] = (y_1-y_2)\sum_{m=j}^{\infty}\frac{1}{m!}\sum_{k=0}^{m-1}y_1^{m-1-k}y_2^k \tag{2.7.39}$$

Therefore, for $\|y_1\|_1, \|y_2\|_1 \leq C_0\delta$, we obtain

$$\|e_j[y_1]-e_j[y_2]\|_1 \leq \|y_1-y_2\|_1\sum_{m=j}^{\infty}\frac{1}{(m-1)!}(C_0\delta)^{m-1} \leq \|y_1-y_2\|_1 e_{j-1}\left[C_0\delta\right] \tag{2.7.40}$$

□

Lemma 2.7.12. For $\|\chi\|_1 \leq \delta \leq \frac{1}{100C_0}$,

$$\begin{aligned} \|e_3[w+F]\|_1 &\leq 1.004\frac{C_0^3}{6}\|\chi\|_1^3 \ , \\ \|ue_2[w+F]\|_1 &\leq 1.004\frac{C_0^2}{2c_u}\|\chi\|_1^3 \ , \end{aligned} \tag{2.7.41}$$

Furthermore, if $\|\chi_1\|_1, \|\chi_2\|_1 \leq \delta \leq \frac{1}{100C_0}$,

$$\begin{aligned} \|e_3[w_1+F_1] - e_3[w_2+F_2]\|_1 &\leq 1.004\frac{C_0^3\delta^2}{2}\|\chi_1-\chi_2\|_1 \ , \\ \|u_1e_2[w_1+F_1] - u_2e_2[w_2+F_2]\|_1 &\leq 1.004\frac{C_0^2\delta^2}{2c_u}\left(1+\frac{2}{C_0}\right)\|\chi_1-\chi_2\|_1 \ , \end{aligned} \tag{2.7.42}$$

Proof. It is clear from Banach Algebra properties and 2.7.3 and 2.7.11 that

$$\|e_3[w+F]\|_1 \leq e_3\left[C_0\|\chi\|_1\right] \leq 1.004\frac{\|\chi\|_1^3}{3!} \tag{2.7.43}$$

Also using lemma 2.7.11

$$\begin{aligned} \|e_3[w_1+F_1] - e_3[w_2+F_2]\|_1 &\leq e_2[C_0\delta]\|\chi_1-\chi_2\|_1 \leq 1.004\frac{C_0^2\delta^2}{2!}\|\chi_1-\chi_2\|_1 \ , \\ \|ue_2[w+F]\|_1 &\leq |u|e_2[C_0\|\chi\|_1] \leq 1.004\frac{C_0^2}{2c_u}\|\chi\|_1^3 \ , \end{aligned}$$

and

$$\begin{aligned} \|u_1e_2[w_1+F_1] - u_2e_2[w_2+F_2]\|_1 &\leq \frac{1}{c_u}\left(e_2[C_0\delta] + \delta e_1[C_0\delta]\right)\|\chi_1-\chi_2\|_1 \\ &\leq 1.004\frac{C_0^2\delta^2}{2c_u}\left(1+\frac{2}{C_0}\right)\|\chi_1-\chi_2\|_1 \end{aligned}$$

□

Lemma 2.7.13. For $\|\chi\| \leq \delta \leq \frac{1}{100C_0}$, $R_{1,1}$ defined in (2.7.4) satisfies

$$\|R_{1,1}[\chi]\| \leq C_{R_{1,1}}\|\chi\|_1^3 \ , \tag{2.7.44}$$

where

$$C_{R_{1,1}} = 1.084\frac{C_0^2}{2}\left(\frac{7}{3}U_0C_0 + \frac{5}{c_u}\right) \tag{2.7.45}$$

Proof. From (2.7.4), we may rewrite

$$R_{1,1} = U_0 e_2[w+F]e_1[w+F+iE] + ue_2[w+F]e^{w+F+iE} + U_0 e_3[w+F+iE] \\ + ue_2[w+F+iE] + (u+U_0)(w+F)e_2[w+F+iE] + u(W+F)(w+F+iE)\ , \tag{2.7.46}$$

We note that

$$U_0\|e_2[w+F]e_1[w+F+iE]\|_1 \le U_0 e_2\left[C_0\|\chi\|_1\right] e_1\left[C_0\|\chi\|_1\right] \le \frac{1.084}{2} C_0^3\|\chi\|_1^3\ , \tag{2.7.47}$$

where we used the fact that for $C_0\|\chi\|_1 \le \frac{1}{100}$ for which

$$\frac{2}{C_0^2\|\chi\|_1^3} e_2\left[C_0\|\chi\|_1\right] e_1\left[C_0\|\chi\|_1\right] \le 1.084$$

Also,

$$\|ue_2[w+F]e^{w+F+iE}\|_1 \le \frac{1}{2c_u} e_2\left[C_0\|\chi\|_1\right] e^{C_0\|\chi\|_1} \le 1.014 \frac{C_0^2}{2c_u}\|\chi\|_1^3\ , \tag{2.7.48}$$

since

$$\frac{2}{C_0^2\|\chi\|_1^2} e_2\left[C_0\|\chi\|_1\right] \exp\left[C_0\|\chi\|_1\right] \le 1.014$$

Furthermore

$$U_0\|e_3[w+F+iE]\|_1 \le U_0 e_3\left[C_0\|\chi\|_1\right] \le \frac{1.0026}{6} U_0 C_0^3\|\chi\|_1^3 \tag{2.7.49}$$

since

$$\frac{6}{C_0^3\|\chi\|_1^3} e_3\left[C_0\|\chi\|_1\right] \le 1.0026$$

Also, the same reasoning shows

$$\|ue_2[w+F+iE]\|_1 \le |u| e_2\left[C_0\|\chi\|_1\right] \le \frac{1.0034 C_0^2}{2c_u}\|\chi\|_1^3\ , \tag{2.7.50}$$

$$\|(w+F)(u+U_0)e_2[w+F+iE]\| \le 1.0034\left(U_0 + \frac{\delta}{c_u}\right)\frac{C_0^3}{2}\|\chi\|_1^3\ , \\ \|u(w+F)(w+F+iE)\| \le \frac{C_0^2}{c_u}\|\chi\|_1^3\ ,$$

Combining all the estimates, we get the stated lemma. □

Lemma 2.7.14. For $\|\chi_1\|_1, \|\chi_2\|_1 \leq \delta \leq \frac{1}{100C_0}$, $R_{1,1}$ defined in (2.7.4) satisfies

$$\|R_{1,1}[\chi_1] - R_{1,1}[\chi_2]\|_1 \leq 3\delta^2 C_{R_{1,1}} \|\chi_1 - \chi_2\|_1 \ , \tag{2.7.51}$$

where $C_{R_{1,1}}$ is as defined in the last Lemma.

Proof. We note

$$\begin{aligned}
&U_0 \left\| e_2[w_1 + F_1] \left\{ e_1[w_1 + F_1 + iE_1] - e_1[w_2 + F_2 + iE_2] \right\} \right\|_1 \\
&+ U_0 \| e_1[w_2 + F_2 + iE_2] \left\{ e_2[w_2 + F_2] - e_2[w_1 + F_1] \right\} \|_1 \\
&\leq U_0 C_0 \|\chi_1 - \chi_2\|_1 \exp\left[C_0\delta\right] e_2\left[C_0\delta\right] \leq 1.084 \frac{U_0 C_0^3}{2} \delta^2 \|\chi_1 - \chi_2\|_1
\end{aligned}$$

Also, since

$$\begin{aligned}
&u_1 e_2[w_1 + F_1] e^{w_1 + F_1 + iE_1} - u_2 e_2[w_2 + F_2] e^{w_2 + F_2 + iE_2} = \\
&u_1 e_2[w_1 + F_1] \left[e^{w_1 + F_1 + iE_1} - e^{w_2 + F_2 + iE_2} \right] + u_1 e^{w_2 + F_2 + iE_2} \left[e_2[w_1 + F_1] - e_2[w_2 + F_2] \right] \\
&+ e^{w_2 + F_2 + iE_2} e_2[w_2 + F_2] \left(u_1 - u_2 \right) \ ,
\end{aligned}$$

and so,

$$\begin{aligned}
&\left\| u_1 e_2[w_1 + F_1] e^{w_1 + F_1 + iE_1} - u_2 e_2[w_2 + F_2] e^{w_2 + F_2 + iE_2} \right\|_1 \\
&\leq |u_1| e_2\left[C_0\delta\right] e^{C_0\delta} \|\chi_1 - \chi_2\|_1 + |u_1| e^{C_0\delta} e_1[C_0\delta] \|\chi_1 - \chi_2\|_1 + e^{C_0\delta} e_2[C_0\delta] \left| u_1 - u_2 \right| \\
&\leq \frac{1}{c_u} e^{C_0\delta} \left((1+\delta) e_2[C_0\delta] + \delta e_1[C_0\delta] \right) \|\chi_1 - \chi_2\|_1 \leq 1.03 \frac{C_0^2 \delta^2}{2c_u} \left(1 + \frac{2}{C_0} \right) \|\chi_1 - \chi_2\|_1
\end{aligned}$$

Furthermore

$$\begin{aligned}
U_0 \left\| e_3[w_1 + F_1 + iE_1] - e_3[w_2 + F_2 + iE_2] \right\|_1 &\leq U_0 C_0 e_2\left[C_0\delta\right] \|\chi_1 - \chi_2\|_1 \\
&\leq \frac{1.004 U_0 C_0^2 \delta^2}{2} \|\chi_1 - \chi_2\|_1 \\
\|u_1 e_2[w_1 + F_1 + iE_1] - u_2 e_2[w_2 + F_2 + iE_2]\|_1 &\leq \frac{1}{c_u} \left(e_2\left[C_0\delta\right] + \delta e_1[C_0\delta] \right) \|\chi_1 - \chi_2\|_1 \\
&\leq \frac{1.004 C_0^2 \delta^2}{2c_u} \left(1 + \frac{2}{C_0} \right) \|\chi_1 - \chi_2\|_1
\end{aligned}$$

Also,

$$
\begin{aligned}
&(w_1+F_1)(u_1+U_0)e_2[w_1+F_1+iE_1]-(w_2+F_2)(u_2+U_0)e_2[w_2+F_2+iE_2]\\
&=(w_1+F_1)(u_1+U_0)\left(e_2[w_1+F_1+iE_1]-e_2[w_2+F_2+iE_2]\right)\\
&+(u_1+U_0)e_2[w_2+F_2+iE_2]\left(w_1+F_1-w_2-F_2\right)\\
&+(w_2+F_2)e_2[w_2+F_2+iE_2](u_1-u_2)
\end{aligned}
$$

and so we obtain

$$
\begin{aligned}
&\|(w_1+F_1)(u_1+U_0)e_2[w_1+F_1+iE_1]-(w_2+F_2)(u_2+U_0)e_2[w_2+F_2+iE_2]\|_1\\
&\leq\left\{\left[\frac{\delta}{c_u}+C_0U_0\right](C_0\delta e_1[C_0\delta]+e_2[C_0\delta])+\frac{C_0}{c_u}\delta e_2[C_0\delta]\right\}\|\chi_1-\chi_2\|_1\\
&\leq\frac{3.014C_0^2\delta^2}{2}\left(C_0U_0+\frac{0.01}{C_0c_u}+\frac{0.004}{c_u}\right)
\end{aligned}
$$

Furthermore,

$$
\|u_1(w_1+F_1)(w_1+F_1+iE_1)-u_2(w_2+F_2)(w_1+F_1+iE_1)\|_1\leq\frac{C_0^2}{c_u}\delta^2\left(1+\frac{2}{C_0}\right)\|\chi_1-\chi_2\|_1
$$

and therefore the result follows by adding all of the above and using $C_0>1$ and noting that $3\delta^2C_{R_{1,1}}$ is an upper bound of the sum. □

Lemma 2.7.15. For $\|\chi\|_1\leq\frac{1}{100C_0}$,

$$
\|\mathcal{K}\mathcal{L}^IR_1[\chi]\|_1\leq k_{2,1}\|\chi\|_1^3\ , \tag{2.7.52}
$$

where

$$
k_{2,1}=1.084\left(C_{R_{1,1}}C_{K\mathcal{L}^I\circ(b+ia)}+\frac{1}{6}C_0^3C_{K\mathcal{L}^I\circ(aU_0+c(U_0-1))}+\frac{C_0^2}{2c_u}C_{K\mathcal{L}^I\circ c}\right)
$$

Furthermore for $\|\chi_1\|_1,\|\chi_2\|_1\leq\delta\leq\frac{1}{100}$ we have

$$
\|\mathcal{K}\mathcal{L}^I\left(R_1[\chi_1]-R_1[\chi_2]\right)\|_1\leq3k_{2,1}\delta^2\|\chi_1-\chi_2\|_1\ , \tag{2.7.53}
$$

Proof. The proof follows by combining previous lemmas and noting the expression for R_1 in (2.7.2). □

Proposition 2.7.16. For $\|\chi\|_1 \leq \delta \leq \frac{1}{100C_0}$,

$$\|\mathcal{R}[\chi]\|_1 \leq (k_{2,0} + k_{2,1}\delta) \|\chi\|_1^2 \tag{2.7.54}$$

Furthermore, for $\|\chi_1\|_1, \|\chi_2\|_1 \leq \delta \leq \frac{1}{100C_0}$,

$$\|\mathcal{R}[\chi_1] - \mathcal{R}[\chi_2]\|_1 \leq \left(2\delta k_{2,0} + 3k_{2,1}\delta^2\right) \|\chi_1 - \chi_2\|_1 \tag{2.7.55}$$

Proof. We combine the results for $\mathcal{R}_0 = K\mathcal{L}^I R_0$ and $\mathcal{R}_1 = K\mathcal{L}^I R_1$ as in previous lemmas to obtain this result. □

Remark 2.7.17. We can choose $k_2 = k_{2,0} + \frac{3}{2}\delta k_{2,1}\delta$ for the purpose of Theorem.

2.8 Solution for $B = 0.3$

In this section, we present a quasi solution and a quasi linear inverse for the case when the surface tension parameter

$$B = 0.3 \tag{2.8.1}$$

for which the conditions in main result 2.4.17 are satisfied and the existence and uniqueness of a solution to Hele-Shaw problem follows for the bubble size parameter ρ over the range

$$0.5 \leq \rho \leq 0.7 \ , \tag{2.8.2}$$

Definition 2.8.1 (Quasi Solution)**.** We define $\hat{Z}^0(\eta, \zeta)$ as

$$\hat{Z}^0(\eta, \zeta) := \frac{2}{\pi} \ln\left(\frac{\zeta - \rho}{\zeta + \rho}\right) + \frac{4\rho}{\pi}\left(\hat{z}^0(\eta, \zeta) + \hat{z}^0\left(\eta, \frac{\rho^2}{\zeta}\right)\right), \tag{2.8.3}$$

where

$$10\rho - 6 = \frac{1}{2}\left(\eta + \frac{1}{\eta}\right) \tag{2.8.4}$$

and

$$\hat{z}^0(\eta, \zeta) = \mathtt{p}_0 + \sum_{n=1}^{12} \frac{\mathtt{p}_n}{n^6}\zeta^n + 10^{-2}\sum_{j=1}^{4} \frac{n_j}{d_j - \zeta}, \tag{2.8.5}$$

with

$$\mathtt{p}_n = \sum_{k=0}^{5} \mathtt{p}_{n,k} \frac{1}{2}\left(\eta^k + \frac{1}{\eta^k}\right),\ n_j = \sum_{k=0}^{5} n_{j,k} \frac{1}{2}\left(\eta^k + \frac{1}{\eta^k}\right),\ d_j = \sum_{k=0}^{5} d_{j,k} \frac{1}{2}\left(\eta^k + \frac{1}{\eta^k}\right), \tag{2.8.6}$$

for $n = 0, \ldots, 12$ and $j = 1, \ldots, 4$ where

$$\begin{aligned}
[\mathtt{p}_{0,k}] &= \left[-\tfrac{10}{909}, -\tfrac{5}{1194}, -\tfrac{1}{1646}, -\tfrac{1}{57166}, \tfrac{1}{96598}, \tfrac{1}{251808}\right], & [\mathtt{p}_{1,k}] &= \left[\tfrac{75}{2362}, \tfrac{11}{1252}, \tfrac{1}{2039}, -\tfrac{1}{5957}, -\tfrac{1}{17184}, -\tfrac{1}{65798}\right], \\
[\mathtt{p}_{2,k}] &= \left[\tfrac{42}{209}, \tfrac{123}{1156}, \tfrac{32}{1067}, \tfrac{9}{1423}, \tfrac{5}{3696}, \tfrac{1}{2971}\right], & [\mathtt{p}_{3,k}] &= \left[\tfrac{107}{108}, \tfrac{311}{871}, \tfrac{19}{718}, -\tfrac{5}{388}, -\tfrac{15}{2768}, -\tfrac{3}{1990}\right], \\
[\mathtt{p}_{4,k}] &= \left[\tfrac{102}{71}, \tfrac{149}{170}, \tfrac{5}{17}, \tfrac{39}{574}, \tfrac{25}{1862}, \tfrac{5}{1762}\right], & [\mathtt{p}_{5,k}] &= \left[\tfrac{13}{7}, \tfrac{115}{132}, \tfrac{67}{468}, -\tfrac{2}{163}, -\tfrac{39}{3356}, -\tfrac{15}{4172}\right], \\
[\mathtt{p}_{6,k}] &= \left[\tfrac{125}{64}, \tfrac{77}{59}, \tfrac{212}{425}, \tfrac{25}{196}, \tfrac{27}{1084}, \tfrac{10}{2273}\right], & [\mathtt{p}_{7,k}] &= \left[\tfrac{97}{70}, \tfrac{525}{668}, \tfrac{26}{129}, \tfrac{15}{922}, -\tfrac{2}{357}, -\tfrac{7}{2696}\right], \\
[\mathtt{p}_{8,k}] &= \left[\tfrac{103}{82}, \tfrac{113}{126}, \tfrac{121}{316}, \tfrac{47}{428}, \tfrac{7}{304}, \tfrac{4}{1033}\right] & [\mathtt{p}_{9,k}] &= \left[\tfrac{249}{448}, \tfrac{43}{120}, \tfrac{45}{374}, \tfrac{11}{520}, \tfrac{3}{4792}, -\tfrac{3}{4004}\right], \\
[\mathtt{p}_{10,k}] &= \left[\tfrac{209}{453}, \tfrac{161}{465}, \tfrac{77}{474}, \tfrac{27}{518}, \tfrac{11}{904}, \tfrac{5}{2278}\right], & [\mathtt{p}_{11,k}] &= \left[\tfrac{24}{223}, \tfrac{34}{449}, \tfrac{5}{164}, \tfrac{13}{1743}, \tfrac{2}{1999}, -\tfrac{1}{92724}\right], \\
[\mathtt{p}_{12,k}] &= \left[\tfrac{38}{451}, \tfrac{41}{624}, \tfrac{20}{599}, \tfrac{13}{1096}, \tfrac{7}{2260}, \tfrac{2}{3191}\right],
\end{aligned} \tag{2.8.7}$$

and

$$\begin{aligned}
\{n_{1,k}\} &= [-\tfrac{23}{172}, \tfrac{13}{1715}, \tfrac{4}{1199}, -\tfrac{1}{47386}, \tfrac{1}{53736}, \tfrac{1}{56502}], & [d_{1,k}] &= [-\tfrac{93}{52}, \tfrac{31}{206}, -\tfrac{9}{766}, \tfrac{5}{4369}, -\tfrac{1}{14720}, \tfrac{1}{66322}], \\
\{n_{2,k}\} &= [-\tfrac{49}{89}, -\tfrac{9}{83}, -\tfrac{9}{2404}, \tfrac{6}{7261}, \tfrac{1}{8678}, \tfrac{1}{76114}] & [d_{2,k}] &= [\tfrac{114}{65}, -\tfrac{133}{886}, \tfrac{21}{1640}, -\tfrac{3}{2905}, \tfrac{1}{11418}, -\tfrac{1}{141686}], \\
\{n_{3,k}\} &= [-\tfrac{31}{45}, \tfrac{28}{197}, \tfrac{13}{239}, \tfrac{2}{1225}, -\tfrac{7}{4028}, -\tfrac{1}{1450}] & [d_{3,k}] &= [-\tfrac{215}{98}, \tfrac{61}{321}, -\tfrac{11}{1554}, \tfrac{7}{1730}, \tfrac{7}{9742}, \tfrac{3}{12004}], \\
\{n_{4,k}\} &= [-\tfrac{181}{78}, -\tfrac{377}{942}, -\tfrac{3}{298}, \tfrac{4}{1877}, \tfrac{3}{11980}, \tfrac{1}{24682}] & [d_{4,k}] &= [\tfrac{61}{29}, -\tfrac{31}{170}, \tfrac{19}{1214}, -\tfrac{2}{1577}, \tfrac{1}{9492}, -\tfrac{1}{125458}],
\end{aligned} \tag{2.8.8}$$

We define $\hat{\omega}^0$ and U^0 as

$$\hat{\omega}^0(\eta, \zeta) = \log\left(\frac{\pi}{4\rho}(\zeta^2 - \rho^2) Z_\zeta^0(\eta, \zeta)\right), \tag{2.8.9}$$

$$U^0(\eta) := \sum_{k=0}^{5} U_k^0 \frac{1}{2}\left(\eta^k + \frac{1}{\eta^k}\right),\ \left[U_k^0\right] = \left[\frac{68}{49}, -\frac{53}{880}, \frac{4}{1529}, \frac{3}{11948}, \frac{1}{70618}, \frac{1}{144504}\right] \tag{2.8.10}$$

Definition 2.8.2 (Quasi Linear Inverse)**.** We set

$$N_0 = 9, \tag{2.8.11}$$

We define

$$P = P_0 + \tau P_1 + \tau^2 P_2 \tag{2.8.12}$$

where

$$P_0 = \begin{bmatrix} \frac{1}{7} & \frac{53}{808} & \frac{49}{529} & \frac{5}{1073} & \frac{7}{745} & \frac{1}{4807} & \frac{13}{7645} & -\frac{1}{18644} & \frac{1}{2628} \\ -\frac{72}{245} & \frac{11}{177} & \frac{101}{1262} & \frac{19}{233} & \frac{18}{2161} & \frac{6}{551} & \frac{8}{7657} & \frac{2}{969} & \frac{1}{7154} \\ \frac{26}{1087} & -\frac{3}{269} & -\frac{1}{1300} & \frac{16}{523} & \frac{14}{253} & \frac{19}{3657} & \frac{9}{976} & \frac{2}{2541} & \frac{3}{1537} \\ -\frac{35}{222} & \frac{17}{777} & \frac{3}{118} & \frac{3}{1700} & \frac{3}{170} & \frac{12}{301} & \frac{20}{5221} & \frac{11}{1469} & \frac{4}{5575} \\ -\frac{2}{793} & -\frac{6}{365} & -\frac{12}{2929} & \frac{11}{1071} & \frac{3}{4445} & \frac{11}{1057} & \frac{19}{639} & \frac{13}{4842} & \frac{1}{163} \\ -\frac{11}{180} & \frac{7}{736} & \frac{7}{2284} & -\frac{10}{2569} & \frac{13}{2120} & \frac{2}{3897} & \frac{4}{587} & \frac{13}{569} & \frac{5}{2456} \\ -\frac{6}{1223} & -\frac{11}{1675} & -\frac{1}{2928} & \frac{1}{4594} & -\frac{10}{2929} & \frac{3}{806} & \frac{2}{4657} & \frac{4}{865} & \frac{8}{441} \\ -\frac{77}{3312} & \frac{14}{3933} & \frac{2}{3689} & -\frac{4}{3739} & -\frac{1}{7775} & -\frac{4}{1413} & \frac{9}{3631} & \frac{1}{5437} & \frac{3}{877} \\ -\frac{9}{2933} & -\frac{3}{1285} & \frac{1}{41433} & -\frac{2}{7435} & -\frac{2}{1827} & -\frac{1}{2786} & -\frac{3}{1307} & \frac{2}{1193} & \frac{1}{5165} \end{bmatrix},$$

$$P_1 = \begin{bmatrix} \frac{25}{314} & -\frac{70}{2311} & \frac{5}{344} & \frac{2}{655} & \frac{15}{3863} & \frac{2}{4157} & \frac{1}{819} & \frac{1}{16126} & \frac{2}{4873} \\ -\frac{35}{241} & \frac{19}{826} & \frac{2}{99} & \frac{22}{907} & \frac{10}{1733} & \frac{6}{971} & \frac{2}{1663} & \frac{5}{2689} & \frac{1}{3515} \\ \frac{25}{659} & -\frac{23}{749} & \frac{4}{3393} & \frac{4}{691} & \frac{7}{439} & \frac{5}{1424} & \frac{10}{1849} & \frac{2}{2347} & \frac{4}{2177} \\ -\frac{39}{305} & \frac{5}{224} & \frac{11}{740} & \frac{14}{4201} & \frac{11}{2452} & \frac{17}{1366} & \frac{1}{373} & \frac{7}{1504} & \frac{3}{3847} \\ -\frac{8}{1017} & -\frac{27}{1027} & \frac{2}{297} & \frac{4}{619} & \frac{1}{1342} & \frac{5}{1898} & \frac{8}{837} & \frac{5}{2714} & \frac{9}{2276} \\ -\frac{35}{491} & \frac{9}{832} & \frac{2}{903} & \frac{4}{1481} & \frac{3}{740} & \frac{1}{2822} & \frac{3}{1670} & \frac{9}{1204} & \frac{3}{2083} \\ -\frac{11}{728} & -\frac{53}{4001} & \frac{1}{211} & \frac{2}{4671} & \frac{3}{4606} & \frac{7}{2776} & \frac{1}{4613} & \frac{3}{2525} & \frac{1}{165} \\ -\frac{53}{1360} & \frac{5}{1151} & \frac{1}{1382} & \frac{1}{619} & \frac{1}{39265} & \frac{1}{8429} & \frac{2}{1169} & \frac{1}{10920} & \frac{2}{2031} \\ -\frac{44}{3651} & -\frac{5}{776} & \frac{2}{795} & -\frac{1}{13954} & \frac{1}{2912} & -\frac{1}{5044} & -\frac{1}{20022} & \frac{1}{848} & \frac{1}{7955} \end{bmatrix},$$

$$P_2 = \begin{bmatrix} \frac{1}{128} & -\frac{13}{1505} & -\frac{1}{378} & \frac{2}{3981} & \frac{1}{4781} & \frac{1}{5066} & \frac{2}{7831} & \frac{1}{14195} & \frac{1}{6642} \\ -\frac{5}{407} & -\frac{6}{965} & -\frac{1}{3434} & \frac{8}{5071} & \frac{6}{4073} & \frac{6}{5641} & \frac{2}{4175} & \frac{2}{3385} & \frac{1}{5705} \\ \frac{5}{519} & -\frac{13}{922} & \frac{4}{2587} & -\frac{5}{9374} & \frac{1}{6721} & \frac{4}{4197} & \frac{2}{2235} & \frac{1}{2877} & \frac{3}{4999} \\ -\frac{25}{832} & \frac{7}{2647} & \frac{8}{4337} & \frac{2}{901} & \frac{1}{4118} & \frac{1}{1897} & \frac{6}{7673} & \frac{4}{4383} & \frac{2}{6133} \\ -\frac{14}{1439} & -\frac{23}{1647} & \frac{5}{839} & \frac{5}{3149} & \frac{1}{1956} & \frac{1}{4436} & \frac{3}{6214} & \frac{4}{7335} & \frac{7}{8282} \\ -\frac{9}{328} & \frac{4}{1817} & -\frac{1}{13035} & \frac{12}{4127} & \frac{2}{1971} & \frac{1}{5929} & \frac{1}{5909} & \frac{2}{4415} & \frac{2}{4487} \\ -\frac{9}{689} & -\frac{7}{824} & \frac{7}{1593} & \frac{1}{3810} & \frac{3}{2497} & \frac{2}{2835} & \frac{1}{28942} & \frac{1}{10106} & \frac{3}{6800} \\ -\frac{2}{93} & \frac{1}{1050} & \frac{1}{4202} & \frac{6}{3065} & \frac{1}{24575} & \frac{1}{1493} & \frac{2}{4085} & \frac{1}{93790} & \frac{1}{7906} \\ -\frac{10}{927} & -\frac{7}{1419} & \frac{12}{4715} & \frac{1}{7027} & \frac{1}{1278} & -\frac{1}{84325} & \frac{3}{7244} & \frac{1}{2871} & \frac{1}{37282} \end{bmatrix},$$

We define

$$\begin{aligned} m_{1,1} &= \frac{106}{101} + \frac{29\,\tau}{379} + \frac{44\,\tau^2}{807} + \frac{10\,\tau^3}{369} + \frac{34\,\tau^4}{2075} + \frac{22\,\tau^5}{2993}\,, \\ m_{1,2} &= -\frac{98}{391} - \frac{36\,\tau}{139} - \frac{93\,\tau^2}{697} - \frac{39\,\tau^3}{701} - \frac{15\,\tau^4}{451} - \frac{22\,\tau^5}{1445}\,, \\ m_{2,1} &= \frac{64}{109} - \frac{25\,\tau}{154} + \frac{56\,\tau^2}{1063} + \frac{4\,\tau^3}{661} + \frac{16\,\tau^4}{4321} + \frac{5\,\tau^5}{1982}\,, \\ m_{2,2} &= -\frac{39}{47} + \frac{35\,\tau}{164} - \frac{52\,\tau^2}{343} + \frac{74\,\tau^3}{3329} - \frac{7\,\tau^4}{353} - \frac{2\,\tau^5}{2551}\,, \end{aligned} \tag{2.8.13}$$

where

$$10\rho - 6 = \tau = \frac{1}{2}\left(\eta + \frac{1}{\eta}\right), \tag{2.8.14}$$

Definition 2.8.3. We fix

$$N_1 := 30\ ,\ N_2 := 45\ , \tag{2.8.15}$$

We set

$$r_\zeta = r_\eta = \frac{13}{10}\ , \tag{2.8.16}$$

and

$$\mathcal{D}_0 := \left\{(\eta,\zeta) \in \mathbb{C}^2 : \frac{1}{r_\eta} \le |\eta| \le r_\eta \text{ and } \frac{1}{r_\zeta} \le |\zeta| \le r_\zeta\right\} \tag{2.8.17}$$

Theorem 2.8.4. Let $B = 0.3$. Let χ_0, ϵ and $\mathcal{R}$ be obtained from the quasi solution 2.8.1 and quasi inverse 2.8.2 through (2.4.61). Then, uniformly for $\rho \in [0.5, 0.7]$

$$\|\chi_0\|_1 \le 5 \times 10^{-4}\ ,\ k_1 \le \frac{2}{5}\ ,\ k_2 \le 124\ , \tag{2.8.18}$$

and Theorem 2.4.14 applies for δ in the range

$$1.07 \times 10^{-3} \le \delta \le 2.4 \times 10^{-3}, \tag{2.8.19}$$

Proof. The $\|\chi_0\|_1$ bound obtained from Corollary 2.6.28 and computation and setting $\rho = \rho_1 = 0.7$. The bound on k_1, equivalently the bound on $\|\epsilon\|_1$ is obtained by computing Lemmas 2.6.14 2.6.16 2.6.18 2.6.20 2.6.24 and Corollary 2.6.23, setting $\rho = \rho_1 = 0.7$ and applying Lemma 2.6.5. The bound on k_2 is obtained from computing Proposition 2.7.16, and the result follows from Remark 2.4.17. □

Theorem 2.8.5 (Solution to Steadily Translating Symmetric Hele-Shaw Problem)**.** Let $B = 0.3$. Then there exists a solution (Z, U) to steadily translating symmetric Hele-Shaw bubble problem(i.e. **Problem 1**) uniformly for $\rho \in [0.5, 0.7]$ in a neighborhood of (Z^0, U^0), where Z^0 predicts the bubble shape with the accuracy

$$\left|Z - Z^0\right| \le 0.195 \text{ on } |\zeta| = 1 \text{ uniformly for } \rho \in [0.5, 0.7] \tag{2.8.20}$$

Proof. The existence and uniqueness follows from Theorem 2.8.4 and the univalency prop-

erty proved in Lemma A.3. The bound is obtained in the following way: Without loss of generality we may assume $Z^0(\rho,1) = Z(\rho,1)$, as changing Z_0 only shifts the bubble and does not affect the bubble shape. Now, let $|\zeta_0| = 1$ be a fixed complex number. We may assume that $0 \le \arg(\zeta_0) \le \pi$ since the bubble is symmetric. It is observed that

$$\left|Z(\rho,\zeta_0) - Z^0(\rho,\zeta_0)\right| = \left|\int_{\zeta=1}^{\zeta_0} \left(Z_\zeta - Z^0_\zeta\right) d\zeta\right| = \left|\int_0^{\nu_0} \left(Z_\zeta - Z^0_\zeta\right) i\zeta d\nu\right| \le \int_0^{\nu_0} \left|\zeta Z_\zeta - \zeta Z^0_\zeta\right| d\nu$$

$$\le \pi \max_{|\zeta|=1} \left|\zeta Z_\zeta - \zeta Z^0_\zeta\right| \le \pi \max_{|\zeta|=1} \left|\zeta Z^0_\zeta \left(e^\Omega - 1\right)\right| \le \pi \max_{|\zeta|=1} \left|\zeta Z^0_\zeta\right| \left|e^\Omega - 1\right| \le \pi \max_{|\zeta|=1} \left|\zeta Z^0_\zeta\right| |\Omega|\, e^{|\Omega|}$$

$$\le \pi \max_{|\zeta|=1} \left|\zeta Z^0_\zeta\right| \max_{|\zeta|=1} |\Omega|\, e^{|\Omega|} \le \pi \left\|\zeta Z^0_\zeta\right\|_{\mathcal{D}_0} \max_{|\zeta|=1} |\Omega|\, e^{|\Omega|}$$

on applying the change of variables $\zeta = e^{i\nu}$, $\zeta_0 = e^{i\nu_0}$, and from inequality $\left|e^\Omega - 1\right| = \left|\sum_{j=1}^\infty \frac{\Omega^j}{j!}\right| \le \sum_{j=1}^\infty \frac{|\Omega|^j}{j!} \le |\Omega| \sum_{j=0}^\infty \frac{|\Omega|^j}{j!} = |\Omega| e^{|\Omega|}$. It is observed that

$$|\Omega| = \left((|F+w|)^2 + |E|^2\right)^2 \le \left((\|F+w\|_1)^2 + \|E\|_1^2\right)^{1/2} \le \left(\left(\frac{(1+\rho^2)}{(1-\rho^2)}\|\chi\|_1\right)^2 + \|\chi\|_1^2\right)^{1/2}$$

$$= \frac{\sqrt{2(1+\rho^4)}}{(1-\rho^2)}\|\chi\|_1 \text{ on } |\zeta| = 1,$$

These results, Theorem 2.8.4 and Lemma A.2.7 imply

$$\left|Z(\zeta_0) - Z^0(\zeta_0)\right| \le \pi \frac{4}{\pi} \frac{r_\zeta \rho_\eta}{(r_\zeta^2 - \rho_\eta^2)} \left(\sigma(1+\alpha_m^0) + t_{\mathcal{D}_0}(\alpha^0, m)\right)$$
$$\times \frac{\sqrt{2(1+\rho^4)}}{(1-\rho^2)} (1 + k_1\delta + k_2\delta^2) \|\chi_0\|_1 e^{(1+k_1\delta+k_2\delta^2)\|\chi_0\|_1}$$

for all $m \ge 0$, and the result follows from setting $m = 0$ and computing the terms. □

Chapter 3
Painlevé-1 Analysis

3.1 Introduction

Painlevé equations or transcendents, as they are sometimes referred to, arise quite often in many areas of mathematics; so much so that they have sometimes been referred to as nonlinear special functions. While some properties of solutions to Painlevé equations are known analytically, there are other properties for which only numerical evidence exists. Numerical calculations of these solutions still remain an active area of interest (see for instance [14],[15],[3]). However, as far as we are aware, numerical computations thus far have not included rigorous error analysis. Even for the P-1 equation, for the special solutions referred to as Tritronqé solutions ([4], [5]), the location of the singularity closest to the origin, while computed with apparent high precision [19], has not been justified rigorously.

The purpose of this paper is (i) to determine rigorous error bounds for the location of the first singularity for tritronqé solution to P-1, and (ii) to demonstrate more generally how a method developed earlier in the context of proof [9] of the Dubrovin Conjecture for P-1 can be extended to obtain rigorous error bounds on approximate analytical expressions for solutions in different parts of the complex plane. Note however that though the method used here is similar to [9], prior analysis did not concern accurate calculation of solutions in any part of the domain. In describing the analysis, it will also be apparent that the method generalizes to other solutions of P-1 and to other differential equations.

3.2 Problem

The tritronqueé solution[(6)] to the Painlevé-1 equation:

$$\frac{d^2y(x)}{dx^2} + 6y^2(x) - x = 0, \tag{3.2.1}$$

on $\mathbb{C}$ is the unique solution with the asymptotic behavior

$$y = \sqrt{\frac{x}{6}}\left[1 + o(x^{-5/8})\right] \text{ as } x \to +\infty. \tag{3.2.2}$$

It is well-known that any solution to P-1 is single valued and meromorphic in $\mathbb{C}$, where singularity locations (in the form of a double poles) depend on initial conditions on (y, y'). Instead of initial conditions, the solution is also completely characterized by the location of a singularity x_p and the coefficient of $\hat{a}_2$ and has the locally convergent series representation:

$$y(x) = -\frac{1}{(x-x_p)^2} + (x-x_p)^2 \sum_{j=0}^{\infty} \hat{a}_j (x-x_p)^j \tag{3.2.3}$$

where $\hat{a}_0 = -\frac{x_p}{10}$, $\hat{a}_1 = -\frac{1}{6}$, $\hat{a}_3 = 0$ and for $n \geq 4$, $\hat{a}_n$ is determined from the recurrence relation

$$\hat{a}_n = -\frac{6}{(n+5)(n-2)} \sum_{k=0}^{n-4} \hat{a}_k \hat{a}_{n-4-k} \tag{3.2.4}$$

The location of x_p closest to the origin for the tritronqueé solution is known[19] to be on the negative real axis, and numerical calculations[19] suggest its location, though this has not been confirmed rigorously. Singularties at large distance from the origin can be rigorously estimated based on adiabatic invariance of conserved quantities[25]. When x_p is not particularly large, like the first singularity of the tritronqueé solution, we are unaware of any method of rigorous analysis to confirm its location.

The main result of this paper is Theorem 3.2.3 given below. However, we first define a few quantities.

Definition 3.2.1. We define $r = \frac{7}{10}$, $x_0 = -\frac{770766}{323285} = -2.3841687675\cdots$, $L = \frac{11}{2}$, $L_0 =$

[(6)]There are actually five different Tritronquee solutions, each corresponding to different choices of anti-Stokes line where one demands $y - \sqrt{\frac{x}{6}} = o(x^{-1/8})$ however, they are all related to each other through rotation of dependent and independent variables.

$-\frac{49}{100}$, $b = \frac{4}{5}(24)^{1/4}$ and $a = \frac{5}{2}b$,

$$\tau = \tau(x) := \frac{x - (L + x_0 + r)/2}{(L - x_0 - r)/2} \tag{3.2.5}$$

and P_u and P each to be polynomials:

$$P_u(\tau) = \sum_{k=0}^{22} c_k \tau^k \ , \tag{3.2.6}$$

where $\mathbf{c} := (c_0, c_1, \cdots, c_{22})$ is given by

$$\begin{aligned}\mathbf{c} = \Big(&\frac{335867}{539062}, \frac{419712}{989125}, -\frac{352463}{3539236}, \frac{60789}{1703279}, -\frac{132842}{11825541}, \frac{43961}{54574472}, \frac{39599}{12036926}, \\ &-\frac{213665}{48625258}, \frac{61644}{14973337}, -\frac{107283}{33444500}, \frac{44761}{18892011}, -\frac{28249}{13550715}, \frac{20641}{14839893}, \frac{13459}{92774551}, \\ &-\frac{4992}{34838093}, -\frac{11771}{8149937}, \frac{24115}{27631671}, \frac{42106}{39550107}, -\frac{21163}{32637441}, -\frac{9782}{15918509}, \frac{11581}{32652169}, \\ &\frac{14692}{88640147}, -\frac{12278}{123249611}\Big)\end{aligned} \tag{3.2.7}$$

$$P(\zeta) \ = \ \sum_{k=0}^{17} a_n \zeta^n \ , \tag{3.2.8}$$

where $a_0 = -x_0/10$, $a_1 = -1/6$, $a_2 = \frac{19949}{321055}$, $a_3 = 0$ and

$$a_n = -\frac{6}{(n+5)(n-2)} \sum_{k=0}^{n-4} a_k a_{n-4-k} \ \text{ for } \ 17 \geq n \geq 4 \tag{3.2.9}$$

Further, we define

$$N_0(x) = -\frac{4412401}{98304\sqrt{6}} x^{-19/2} \left[1 - \frac{1225}{90049\sqrt{6}} x^{-5/2} + \frac{30625}{2161176} x^{-5}\right] \tag{3.2.10}$$

$$\mathcal{G}_1(x) = x^{-5/8} \exp\left[-ibx^{5/4}\right] \ , \mathcal{G}_2(x) = x^{-5/8} \exp\left[ibx^{5/4}\right] \ , \tag{3.2.11}$$

$$w_0 = \sum_{j=1}^{2} \frac{(-1)^j}{ia} \mathcal{G}_j(x) \int_{\infty}^{x} \mathcal{G}_{3-j}(y) y N_0(y) dy. = \Re\left\{\int_0^{\infty} e^{-sbx^{5/4}} \mathcal{W}_0\left(x^{5/4}, s\right) ds\right\} \ , \tag{3.2.12}$$

where

$$\mathcal{W}_0(z,s) = -\frac{4412401\sqrt{6}}{368640az^7}\left((1+is)^{-\frac{15}{2}} - \frac{1225\sqrt{6}}{540294z^2}\,(1+is)^{-\frac{19}{2}} + \frac{30625}{2161176z^4}\,(1+is)^{-\frac{23}{2}}\right) \tag{3.2.13}$$

where in (3.2.13) the principal branch is used, i. e. $(1+is)^{-\kappa} = 1$ at $s = 0$.

Remark 3.2.2. Integration by parts of the Laplace transform representation of w_0 and its derivative in (3.2.12) relates to error functions with complex arguments, which are considered known in the sense they may be calculated to any desired precision[1] For instance $w_0(L) = -1.17414\cdots\times 10^{-7}$, $w_0'(L) = 2.03367\cdots\times 10^{-7}$.

We also define domains D_j for $j = 1,\cdots,4$ with $D_1 = [L,\infty)$, $D_2 = [L_0, L)$, $D_3 = [x_0 + r, L_0)$ and $D_4 = \{x \in \mathbb{C} : |x - x_0| = r, x \neq x_0 + r\}$. We define $D = D_1 \cup D_2 \cup D_3 \cup D_4$ (See Figure 3.1).

Theorem 3.2.3. Let

$$y_0(x) = \begin{cases} \sqrt{\frac{x}{6}}\left[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w_0(x)\right] & \text{on } D_1 \\ -\frac{1}{(x-x_0)^2} + P_u(\tau(x)) & \text{on } D_2 \cup D_3 \\ -\frac{1}{(x-x_0)^2} + (x-x_0)^2 P(x-x_0) & \text{on } D_4 \end{cases} \tag{3.2.14}$$

Then the tritronqé solution y has the representation

$$y(x) = y_0(x) + E(x)\ \text{, where }\ \left|E(x)\right| \leq 2.35\times 10^{-5}\ ,\left|E'(x)\right| \leq 1.16\times 10^{-4} \tag{3.2.15}$$

Moreover, y has a unique double pole singularity at $x = x_p \in \{\zeta : \zeta \in \mathbb{C}, |\zeta - x_0| < r\}$ with $|x_p - x_0| \leq 4.1\times 10^{-6}$. This is the closest singularity of the tritronqué solution from the origin.

Note: The proof of the theorem is completed in Section 3.7 after establishing some preliminary results.

Remark 3.2.4. From evaluation of expression for y_0, y_0' at the points of discontinity, it is readily checked that $\left|y_0(L^+) - y_0(L^-)\right| \leq 5\times 10^{-14}$, $\left|y_0'(L^+) - y_0'(L^-)\right| \leq 7.5\times 10^{-14}$, $\left|y_0\left([x_0+r]^+\right) - y_0\left(x_0 + re^{i0^+}\right)\right| \leq 4\times 10^{-10}$, $\left|y_0'\left([x_0+r]^+\right) - y_0'\left(x_0 + re^{i0^+}\right)\right| \leq 7\times 10^{-8}$

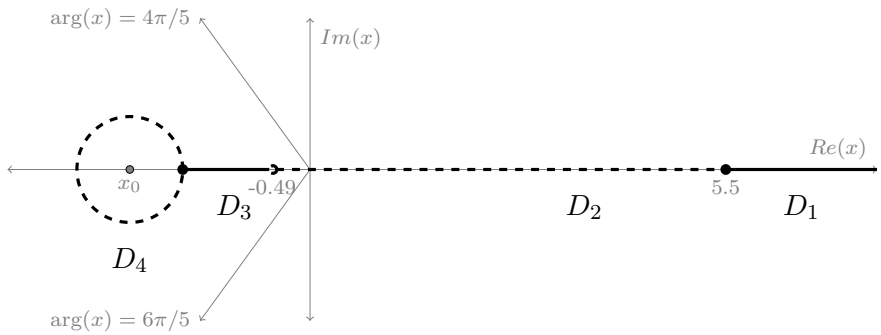


Figure 3.1: Sketch of Domain $D = D_1 \cup D_2 \cup D_3 \cup D_4$

Remark 3.2.5. The strategy we pursue is as follows: For each D_j ($j = 1, 2, 3, 4$), a choice[7] of y_0 is made to ensure that the remainder $R(x) := y_0''(x) + 6y_0^2(x) - x$ is small and the mismatches $y_0(x_e^-) - y_0(x_e^+)$, $y_0'(x_e^+) - y_0'(x_e^-)$ when boundary point x_e is approached from domains D_j and D_{j+1} is either zero or very small (see Remark 3.2.4). Then the differential equation satisfied by $E = y - y_0$, with initial condition at the end point, is transformed to a an equivalent nonlinear integral equation in the form $E = \mathcal{N}[E]$. A convenient Banach space $\mathcal{S} \subset C^0(D_j)$ is defined and a small enough ball $\mathcal{B} \subset \mathcal{S}$ centered at $E_0 = \mathcal{N}[0]$ for which $\mathcal{N}$ is contractive. The Banach contraction mapping theorem ensures existence and uniqueness of such $E \in \mathcal{B}$ satisfying the integral equation, and this result is used to estimate bounds on $|E|$ and $|E'|$. In particular, this allows for error analysis in domain D_{j+1}, once error estimates are completed in D_j since initial conditions in D_{j+1} are then determined to within small errors when continuity of $y = y_0 + E$ and its derivative is demanded at the end point x_e.

$$E(x_e^-) = E(x_e^+) + y_0(x_e^+) - y_0(x_e^-) \quad , \quad E'(x_e^-) = E'(x_e^+) + y_0'(x_e^+) - y_0'(x_e^-) \ , \qquad (3.2.16)$$

For each j, $C^2(D_j)$ regularity of E (and therefore of y) follows using the smoothness of the Kernel in the integral reformulation, and therefore E satisfies the differential equation. This implies that $y = y_0 + E$ satisfies P-1 in each domain D_j and by continuity of y and y' at common end points x_e, it is the same solution to P-1 in D. Since asymptotic condition

[7]The choice relies on asymptotic series on D_1, Chebyshev polynomial approximation on $D_2 \cup D_3$ and Taylor-like polynomial on D_4.

at ∞ has been enforced, this must be the tritronqué solution.

It is noted that although the proof of the estimates rely on a priori knowledge about local meromorphic representation, it is not essential in the error representation along any particular path in the complex-plane. For instance, if (3.2.3) were not available, then on a circle $\zeta - x_0 = re^{i\nu}$, we could have used a Chebyshev representation in $\cos(\nu)$ for $\Re(y)$ and $\Im(y)/\sin(\nu)$ to obtain a representation for solution. In the case the singularity x_0 were not a pole, but say a branch point of order 1/2, then a Chebyshev representation in $\cos(\nu/2)$ for $\Re(y)$ would be more appropriate assuming that r is small enough so that we do not run into other singularities on the Riemann surface. The Cauchy integral formula can be extended on Riemann surface to obtain singularity location (for instance see [35]). Other generalizations of the quasi-solution approach to non-integrable systems appear in [10].

Remark on Notation: The framework of the proof in each section is quite similar. Therefore, to avoid proliferation of symbols, we found it convenient to use the same notation for similar quantities in each section. Thus, for each subdomain D_j $(j = 1, 2, 3, 4)$; the approximate solution and the error terms are always denoted by y_0 and E respectively[(8)]. G_1, G_2 will always denote two independent solutions of a homogenous linear second order differential equation $\mathcal{L}E = 0$ in each domain, where $\mathcal{L}$ is typically obtained by linearizing P-1 about y_0. Sometimes it is more convenient to choose $\mathcal{L}$ as an operator close to the linearized P-1, but not quite the same. Using G_1, G_2, the differential equation for E with initial condition is transformed to an integral equation in the form $E = \mathcal{N}[E]$. We always denote $E_0 = \mathcal{N}[0]$ and $\mathcal{S}$ is the Banach space of of continuous functions in domain D_j equipped with either sup norm or weighted sup-norm, while $\mathcal{B} \subset \mathcal{S}$ is the generic notation for some small ball where $\mathcal{N}$ is contractive. Note that the actual definitions of y_0, E, G_1, G_2, $\mathcal{N}$, E_0 , $\mathcal{S}$ and $\mathcal{B}$ differ from section to section.

[(8)]In §2, it is more convenient to take $E = \sqrt{\frac{x}{6}}(w - w_0)$ and analyze w.

3.3 Analysis on $D_1 = [L, \infty)$

It is well-known (for instance see [7, §6.6a]) that if y satisfies (3.2.1) and (3.2.2), then it is uniquely expressed as[9] :

$$y(x) = \sqrt{\frac{x}{6}}\Big[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w(x)\Big] \tag{3.3.1}$$

where $w(x) = o(x^{-15/2})$. Substituting (3.3.1) into (3.2.1) and subtracting $[25w(x)]/[64x^2]$ from both sides yields:

$$\mathcal{L}w := w'' + \frac{w'}{x} + \Big[2\sqrt{6x} - \frac{25}{64x^2}\Big]w = N(w(x), x), \tag{3.3.2}$$

where

$$N(w, x) = N_0(x) - \frac{25}{64}x^{-2}\Big[1 - \frac{49}{25\sqrt{6}}x^{-5/2} + \frac{49}{12}x^{-5}\Big]w - \sqrt{6x}w^2, \tag{3.3.3}$$

where $N_0(x)$ is defined in (3.2.10). Note that $N(0, x) = N_0(x)$. It is observed that $\mathcal{G}_1$, $\mathcal{G}_2$ defined in (3.2.11) are independent solutions to the homogenous equation $\mathcal{L}w = 0$, defined in (3.3.2). From the method of variation of parameters, (3.3.2) is equivalent to

$$w(x) = \frac{1}{ia}\sum_{j=1}^{2}(-1)^j\mathcal{G}_j(x)\int_{\infty}^{x} t\mathcal{G}_{3-j}(t)N(w(t), t)dt + \alpha_0\mathcal{G}_1(x) + \beta_0\mathcal{G}_2(x), \tag{3.3.4}$$

$$=: \mathcal{N}[w](x) + \alpha_0\mathcal{G}_1(x) + \beta_0\mathcal{G}_2(x)$$

up to a convenient choice of α_0 and β_0. The condition $w(x) = o(x^{-15/2})$ is observed to imply $\mathcal{N}[w] = O(x^{-10})$ (see (3.3.9) and (3.3.10) in the ensuing), implying $\alpha_0 = 0 = \beta_0$ in (3.3.4) .

Remark 3.3.1. We look for solution $w \in C(D_1)$ satisfying

$$w(x) = \mathcal{N}[w](x), \tag{3.3.5}$$

with $w(x) = o(x^{-15/2})$ as $x \to \infty$. From regularity of kernel and a standard bootstrap argument $w \in C^2(D_1)$. Then it follows that y in (3.3.1) will satisfy (3.2.1) and (3.2.2).

[9]First four terms of an asymptotic series of y is used. It is observed that including more or less number of terms result in larger error bounds.

Definition 3.3.2. Define the Banach space

$$\mathcal{S} := \{f \in C(D_1) : \|f\|_{10} < \infty\} \text{ where } \|f\|_{10} := \sup_{x \in D_1} |f(x)|x^{10}, \tag{3.3.6}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\|_{10} \leq \epsilon\}. \tag{3.3.7}$$

We note that with w_0 defined in (3.2.12), $\mathcal{N}[0] = w_0$.

Lemma 3.3.3. $\|w_0\|_{10} \leq \frac{15}{2}$ and $\left|w_0'(x)\right| \leq (8.85)x^{-39/4}$ on D_1.

Proof. Using $x^{5/8}\mathcal{G}_j(x) = \exp\left[(-1)^j ibx^{5/4}\right]$, integration by by parts in (3.2.12) yields:

$$w_0(x) = \frac{8}{5ba}x^{-1/2}N_0(x) + \frac{4}{5ba}\sum_{j=1}^{2}\mathcal{G}_j(x)\int_x^\infty \left[t^{1/8}N_0(t)\right]' t^{5/8}\mathcal{G}_{3-j}(t)dt \tag{3.3.8}$$

From explicit computation, it may be checked that for $x \geq L$, $\left|[x^{1/8}N_0]'\right| = [x^{1/8}N_0]'$, $|\mathcal{G}_j(x)| \leq x^{-5/8}$ for $j = 1, 2$ and $x \geq L$ and therefore it may be checked that $x^{19/2}|N_0(x)|$ is monotonically increasing for $x \geq L$ and using limiting value as $\lim_{x\to\infty}$, we obtain

$$|w_0(x)| \leq \frac{16}{5ab}x^{-5/8}x^{1/8}|N_0(x)| \leq \frac{15}{2}x^{-10}, \tag{3.3.9}$$

Further, from (3.3.8), it follows from using $\mathcal{G}_j(x)\mathcal{G}_{3-j}(x) = x^{-5/4}$ that

$$w_0' = -\frac{1}{ba}x^{-3/2}N_0(x) + \frac{4}{5ba}\sum_{j=1}^{2}\mathcal{G}_j'(x)\int_x^\infty \left[t^{1/8}N_0(t)\right]' t^{5/8}\mathcal{G}_{3-j}(t)dt$$

Note from (3.2.11), $|\mathcal{G}_j'(x)| \leq \frac{5}{4}bx^{-3/8} + \frac{5}{8}x^{-13/8}$ for $j = 1, 2$. Therefore, again using property $\left|\left[t^{1/8}N_0(t)\right]'\right| = \left[t^{1/8}N_0(t)\right]'$, and $-x^{1/8}N_0(x) = \left|x^{1/8}N_0(x)\right|$ for $t, x \geq L$, it follows from (3.3) and the fact that $x^{19/2}|N_0(x)| = -x^{19/2}N_0(x)$ is increasing on D_1 and $\lim_{x\to\infty} -x^{19/2}N_0(x) = 18.324\cdots$, we obtain that

$$\begin{aligned}|w_0'(x)| &\leq \frac{1}{ab}x^{-3/2}|N_0(x)| + \frac{8}{5ab}[\frac{5}{4}bx^{-3/8} + \frac{5}{8}x^{-13/8}]\left|x^{1/8}N_0(x)\right| \\ &\leq \frac{2}{ax^{39/4}}|x^{19/2}N_0(x)|\left(1 + \frac{1}{bx^{5/4}}\right) \leq \frac{(2)(18.33)}{ax^{39/4}}\left(1 + \frac{1}{bL^{5/4}}\right) \leq 8.85x^{-39/4},\end{aligned}$$

□

Lemma 3.3.4. Let $\delta = 2 \times 10^{-3}$. Then there exists a unique fixed point w of $\mathcal{N}$ in $\mathcal{B}(w_0, \delta\|w_0\|_{10})$. Moreover, $|w(x) - w_0(x)| \leq (0.0167)\|w_0\|_{10}x^{-45/4}$ and $|w'(x) - w_0{}'(x)| \leq 0.29x^{-11}$ on D_1.

Proof. Let $w \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$. Since $\left[1 - \frac{49}{25\sqrt{6}}x^{-5/2} + \frac{49}{12}x^{-5}\right] \leq 1$ on D_1, $|\mathcal{G}_j(x)| \leq x^{-5/8}$ for $j = 1, 2$, Lemma 3.3.3 and $x \geq L$, (3.3.5) and (3.2.12) imply:

$$|\mathcal{N}[w](x) - w_0(x)| \leq \left[\frac{5(1+\delta)}{68ax^{5/4}} + \frac{16\sqrt{6}}{145ax^{35/4}}(1+\delta)^2\|w_0\|_{10}\right]|\|w_0\|_{10}x^{-10} \leq \delta\|w_0\|_{10}x^{-10}, \tag{3.3.10}$$

Multiplying by x^{10}, we immediately obtain $\mathcal{N}\left[\mathcal{B}\left(w_0, \delta\|w_0\|_{10}\right)\right] \subset \mathcal{B}(w_0, \delta\|w_0\|_{10})$. Similarly, let w_1 , $w_2 \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$. From the previous arguments, and $\|w_1\|_{10} + \|w_2\|_{10} \leq 2(1 + \delta)\|w_0\|_{10}$, (3.2.12) implies:

$$\begin{aligned} |\mathcal{N}[w_1](x) - \mathcal{N}[w_2](x)| &\leq \left[\frac{5}{68ax^{5/4}} + \frac{16\sqrt{6}}{145ax^{35/4}}2(1+\delta)\|w_0\|_{10}\right]x^{-10}\|w_1 - w_2\|_{10} \\ &\leq (0.002)x^{-10}\|w_1 - w_2\|_{10} \end{aligned}$$

Multiplying by x^{10}, shows that $\mathcal{N} : \mathcal{B} \to \mathcal{B}$ is contractive. Thus $w = \mathcal{N}[w]$ has a unique solution $w \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$ as a consequence of Banach fixed point theorem. Since $w = \mathcal{N}[w]$, it follows from estimates in (3.3.10) that $|w(x) - w_0(x)| \leq 0.0167\|w_0\|_{10}x^{-45/4}$. On observing $\left[1 - \frac{49}{25\sqrt{6}}t^{-5/2} + \frac{49}{12}t^{-5}\right] \leq 1$ on D_1 and using Lemma 3.3.3, (3.3.3) and (3.2.10) imply

$$|N(w,t) - N_0(t)| \leq \frac{25}{64t^{12}}\|w_0\|_{10}(1+\delta) + \frac{\sqrt{6}}{t^{39/2}}(1+\delta)^2\|w_0\|_{10}^2 \leq \frac{2.94}{t^{12}} + \frac{139}{t^{39/2}}$$

From this result and the bounds $|\mathcal{G}_j'(x)| \leq \frac{5b}{4}x^{-3/8} + \frac{5}{8}x^{-13/8}$ for $j = 1, 2$ for $x \in D_1$, (3.3.5), (3.2.12) imply:

$$|w'(x) - w_0{}'(x)| \leq \frac{1}{a}\sum_{j=1}^{2}|\mathcal{G}_j'(x)|\int_x^\infty |t\mathcal{G}_{3-j}||N(w,t) - N_0(t)|dt \leq 0.29x^{-11}.$$

□

Recall y_0 is defined as:

$$y_0(x) =: \sqrt{\frac{x}{6}}\left[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w_0(x)\right], \tag{3.3.11}$$

On using Lemma 3.3.4, (3.3.11) and (3.3.1) leads to the main result of this section:

Corollary 3.3.5. The tritronqué solution has the representation $y = y_0 + E$ where $|E(x)| \leq (0.00682)\|w_0\|_{10} x^{-43/4}$ and $|E'(x)| \leq (0.126)x^{-21/2}$ on D_1. In particular, $\left|E(L^+)\right| \leq 5.625 \times 10^{-10}$ and $|E'(L^+)| \leq (2.12) \times 10^{-9}$.

Proof. By construction, w satisfying (3.3.4) also satisfies the differential equation (3.3.2) with asymptotic condition $w(x) = o(x^{-5/8})$ as $x \to \infty$. On substituting $y = y_0 + \sqrt{\frac{x}{6}}(w - w_0)$, it follows y satisfies (3.2.1) with the asymptotic condition (3.2.2). Since such a solution is unique, $y = y_0 + \sqrt{\frac{x}{6}}(w - w_0)$ must be the tritronqué solution in D_1. The bounds on the error $E = y(x) - y_0(x) = \sqrt{\frac{x}{6}}(w - w_0)$ and its derivative follow from those satisfied by $w - w_0$ and its derivative, and the bounds on $\|w_0\|_{10}$. □

3.4 Analysis on $D_2 = [L_0, L)$

Remark 3.4.1. Since a double pole singularity of y at x_p close to x_0 is expected, it can potentially cause accuracy problems even on domain D_2. Hence, it is better to introduce a function u with less variation than y in D_2.

We define

$$E(x) = y(x) - y_0(x), \tag{3.4.1}$$

Substitution of (3.4.1) into (3.2.1) yields

$$\mathcal{L}E := E'' + 12y_0E = -6E^2 - R, \tag{3.4.2}$$

where R is the remainder $R(x) := y_0''(x) + 6y_0^2(x) - x$. Further, imposing C^1-transition of $y_0 + E$ at $x = L$ from left and right implies

$$E(L^-) = E(L^+) + y_0(L^+) - y_0(L^-) \quad , \quad E'(L^-) = E'(L^+) + y_0'(L^+) - y_0'(L^-), \tag{3.4.3}$$

Remark 3.4.2. The explicit form of y_0 enables exact evaluation of y_0 and all its derivatives, as well as R and its integral.

Definition 3.4.3. We define

$$\mathcal{R}(x) = \int_L^x [-R]dt \tag{3.4.4}$$

Lemma 3.4.4. In the domain $\mathcal{D}_2$,

$$\|\mathcal{R}\|_\infty \leq 3.75 \times 10^{-9} \tag{3.4.5}$$

$$y_0(x) > 0 \quad , \quad y_0'(x) > 0 \tag{3.4.6}$$

Proof. We use a straightfoward inequality (as stated in Lemma B.0.2 in the appendix) by partitioning the domain D_2 into n equal segments $\{[x_j, x_{j+1}]\}_{j=0}^{n-1}$. We note that both the integrals $\int_{x_{j-1}}^{x_j} [\mathcal{R}'(x)]^2 \, dx$ and point values $\mathcal{R}(x_j)$ can be determined explicitly. Thus checking bounds on $\|\mathcal{R}\|_\infty$ is easily facilitated resulting in (3.4.5) by choosing[10] $n = 20$. Using Lemma B.0.1, we have for $x \in [x_j, x_{j+1}]$,

$$y_0(x) > \frac{1}{2}\left[y_0(x_j) + y_0(x_{j+1})\right] - \frac{1}{2}\sqrt{x_j - x_{j-1}}\|y_0'\|_{L^2[x_{j-1}, x_j]}$$

and since the point values of y_0 as well as explicit L_2 integral can be explicitly determined, we can check condition $y_0(x) > 0$ in each subinterval. The same procedure was repeated for y_0' to obtain (3.4.6). □

From (3.4.2), the two independent solutions are denoted as G_1, G_2 satisfying

$$G_j'' + 12 y_0 G_j = 0, \quad \text{for} \quad j = 1, 2 \tag{3.4.7}$$

with initial conditions $G_1(L^-) = 1$, $G_1'(L^-) = 0$, $G_2(L^-) = 0$ and $G_2'(L^-) = 1$.

Lemma 3.4.5. $\|G_1'\|_\infty \leq 3.391$, $\|G_1\|_\infty \leq 3.775$, $\|G_2'\|_\infty \leq 1$ and $\|G_2\|_\infty \leq 1.114$ on D_2.

Proof. On multiplication by $2G_j'$, integration from L to x and using integration by parts, (3.4.7) gives

$$G_j'^{\,2}(x) + 12 y_0(x) G_j^2(x) + 12 \int_x^L y_0'(t) G_j^2(t) dt = \mathcal{G}_j'^{\,2}(L) + 12 y_0(L) \mathcal{G}_j^2(L) \tag{3.4.8}$$

[10] $n = 20$ gives a bound within about two percent of the graphically observed bound on $\mathcal{R}$; $n = 10$ gives only a 10 percent higher value.

Using $y_0, y_0' > 0$, and initial conditions on $\mathcal{G}_j$, (3.4.8) implies

$$G_1'(x)^2 + 12y_0(x)G_1(x)^2 \leq 12y_0(L), \tag{3.4.9}$$

$$G_2'(x)^2 + 12y_0(x)G_2(x)^2 \leq 1, \tag{3.4.10}$$

$|G_1'| \leq \sqrt{12y_0(L)} \leq 3.391$ and $|G_2'| \leq 1$ are immediate. To find bounds on G_1, G_2, it is convenient to partition D_2 into two intervals $[L_0, \gamma_0)$ and $[\gamma_0, L)$, where γ_0 will be chosen appropriately. Using bounds on $|G_1'|$ in (3.4.9), implies

$$\begin{aligned} |G_1(x)| &\leq \frac{\sqrt{y_0(L)}}{\sqrt{y_0(x)}} \text{ when } \gamma_0 \leq x \leq L, \\ |G_1(x)| &\leq \int_x^{\gamma_0} |G_1'(x)|dx + |G_1(\gamma_0)| \leq (\gamma_0 - x)\sqrt{12y_0(L)} + \frac{\sqrt{y_0(L)}}{\sqrt{y_0(\gamma_0)}} \text{ when } L_0 \leq x \leq \gamma_0 \end{aligned}$$

Since y_0 is monotonically increasing (see (3.4.6)), it follows from above that for any $x \in D_2$

$$\left|G_1(x)\right| \leq (\gamma_0 - L_0)\sqrt{12y_0(L)} + \frac{\sqrt{y_0(L)}}{\sqrt{y_0(\gamma_0)}} \tag{3.4.11}$$

Similarly, using (3.4.10), we obtain for any $x \in D_2$,

$$\left|G_2(x)\right| \leq \frac{1}{\sqrt{12y_0(\gamma_0)}} + (\gamma_0 - L_0) \tag{3.4.12}$$

From from explicit evaluation with $\gamma_0 = -\frac{16}{100}$, we get the bounds in the Lemma statement.

□

Using variation of parameters, it follows from (3.4.2) that

$$E(x) = \sum_{j=1}^{2}(-1)^{j+1}G_j(x)\int_L^x G_{3-j}(t)[-6E^2(t) - R(t)]dt + \alpha_1 G_1(x) + \beta_1 G_2(x) =: \mathcal{N}[E](x) \tag{3.4.13}$$

where $\alpha_1 = E(L^+) + y_0(L^+) - y_0(L^-)$ and $\beta_1 = E'(L^+) + y_0'(L^+) - y_0'(L^-)$. Corollary 3.3.5 and small mismatch of y_0 and y_0' (see Remark 3.2.4) on two sides of L immediately implies

Corollary 3.4.6. $|\alpha_1| \leq (5.63) \times 10^{-10}$ and $\beta_1 \leq (2.13) \times 10^{-9}$.

Definition 3.4.7. Define

$$E_0(x) = \mathcal{N}[0](x) = \alpha_1 G_1(x) + \beta G_2(x) + \sum_{j=1}^{2}(-1)^j G_j(x)\int_L^x G_{3-j}(t)R(t)dt \tag{3.4.14}$$

Lemma 3.4.8. $\|E_0\|_\infty \leq (1.745) \times 10^{-7}$ and $\|E_0'\|_\infty \leq (1.605) \times 10^{-7}$ on D_2.

Proof. Integration by parts and use of boundary conditions at $x = L$ in (3.4.14) leads to

$$\sum_{j=1}^{2}(-1)^j G_j(x) \int_L^x G_{3-j}(t)R(t)dt = \sum_{j=1}^{2}(-1)^j G_j(x) \int_L^x G'_{3-j}(t)\mathcal{R}(t)dt \tag{3.4.15}$$

It follows

$$\begin{aligned}
\left|E_0(x)\right| &\leq (\|G_1'\|_\infty\|G_2\|_\infty\| + \|G_1\|_\infty\|G_2'\|_\infty)(L - L_0)\|\mathcal{R}\|_\infty + |\alpha_1|\|G_1\|_\infty + |\beta_1|\|G_2\|_\infty \\
&\leq (1.745) \times 10^{-7}, \\
\left|E_0'(x)\right| &\leq \|\mathcal{R}\|_\infty + 2\|G_1'\|_\infty\|G_2'\|_\infty\|\mathcal{R}\|_\infty(L - L_0) + |\alpha_1|.\|G_1'\|_\infty + |\beta_1|.\|G_2'\|_\infty \\
&\leq 1.605 \times 10^{-7},
\end{aligned}$$

and taking supremum over $x \in D_2$ gives the result. □

The Banach space $\mathcal{S}$ is defined as

$$\mathcal{S} := \{f \in C(D_2) : \|f\|_\infty < \infty\}, \tag{3.4.16}$$

Let $f_0 \in \mathcal{S}$ and $r \geq 0$. The ball $\mathcal{B}(f_0, r) \subset \mathcal{S}$ centered at f_0 with radius r is defined as:

$$\mathcal{B}(f_0, r) := \{f \in \mathcal{S} : \|f - f_0\|_\infty \leq r\}. \tag{3.4.17}$$

Lemma 3.4.9. Let $\delta = (5.5) \times 10^{-5}$. Then there exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Moreover, $\|E\|_\infty \leq (1.75) \times 10^{-7}$ and $\|E'\|_\infty \leq (1.61) \times 10^{-7}$.

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Using Lemmas 3.4.5 and 3.4.8; (3.4.13) implies

$$|\mathcal{N}[\mathcal{E}](x) - E_0(x)| \leq 2\|G_1\|_\infty\|G_2\|_\infty (L - L_0)\, 6(1 + \delta)^2\|E_0\|_\infty^2 \leq \delta\|E_0\|_\infty$$

and taking supremum over $x \in D_2$ gives $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. From Lemmas 3.4.8 and 3.4.5, using $\|\mathcal{E}_1 + \mathcal{E}_2\|_\infty \leq 2(1 + \delta)\|E_0\|_\infty$, (3.4.13) implies:

$$|\mathcal{N}[\mathcal{E}_1](x) - \mathcal{N}[\mathcal{E}_2](x)| \leq 2\|G_1\|_\infty\|G_2\|_\infty (L - L_0)\, 12(1 + \delta)\|E_0\|_\infty \leq 1.1 \times 10^{-4}\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty$$

and taking supremum over $x \in D_2$ implies $\mathcal{N}$ is contractive. Banach fixed point theorem implies the existence and uniqueness of such E. $\|E\|_\infty \leq (1 + \delta)\|E_0\|_\infty \leq (1.75) \times 10^{-7}$ is

immediate. On using Lemmas 3.4.8 and 3.4.5, the derivative of (3.4.13) leads to

$$|E'(x)| \leq \|E_0'\|_\infty + (\|G_1'\|_\infty.\|G_2\|_\infty + \|G_2'\|_\infty.\|G_1\|_\infty)(L - L_0)6\|E\|_\infty^2 \leq 1.61 \times 10^{-7}.$$

and taking supremum over $x \in D_2$ implies the result on $\|E'\|_\infty$. □

Corollary 3.4.10. In D_2, the tritronqué has the representation $y = y_0 + E$ where, $\|E\|_\infty \leq (1.75) \times 10^{-7}$ and $\|E'\|_\infty \leq (1.61) \times 10^{-7}$ on D_2.

Proof. The solution E in Lemma 3.4.9 which satifies integral equation (3.4.13) also satisfies (3.4.2) because of smoothness of G_1, G_2. Therefore, it immediately follows that $y = y_0 + E$ satisfies P-1. Also the initial conditions (3.4.3) on E ensures that $y_0 + E$ and its derivative at $x = L^-$ match with the tritronqué at $x = L^+$, *i.e.* when L is approached from D_1. From uniqueness, y must be the tritronqué solution. Error bounds follow from Lemma 3.4.9. □

3.5 Analysis on $D_3 = [x_0 + r, L_0)$

With $y = y_0 + E$, where y_0 is given by (3.2.14), E satisfies

$$E'' - \frac{12}{(x - x_0)^2}E = -6E^2 - 12P_uE - R, \tag{3.5.1}$$

Imposing continuity of $y_0 + E$ and its derivative as $x = L_0$ is approached from the left and right implies

$$E(L_0^-) = E(L_0^+) \quad , \quad E'(L_0^-) = E'(L_0^+) \ , \tag{3.5.2}$$

since expression for y_0 is the same in domain D_3 and D_2. It is observed that

$$G_1(x) = (x - x_0)^4 \text{ and } G_2(x) = \frac{1}{(x - x_0)^3}, \tag{3.5.3}$$

are two independent solutions of the associated homogenous differential equation $G'' - \frac{12}{(x-x_0)^2}G = 0$. Using variation of parameter, (3.5.1) implies

$$E(x) = \sum_{j=1}^{2}(-1)^j \frac{G_j(x)}{7} \int_{L_0}^{x} G_{3-j}[-6E^2 - 12P_uE - R]dt + \alpha_2 G_1(x) + \beta_2 G_2(x) := \mathcal{N}[E](x), \tag{3.5.4}$$

where in order to satisfy (3.5.2),

$$7\alpha_2 = G_2(L_0)E'(L_0^+) - G_2'(L_0)E(L_0^+) \tag{3.5.5}$$

$$7\beta_2 = -G_1(L_0)E'(L_0^+) + G_1'(L_0)E(L_0^+) \tag{3.5.6}$$

Using bounds in Corollary 3.4.10, equation (3.5.5) and (3.5.6) immediately gives

Corollary 3.5.1. $|\alpha_2| \leq 9.22 \times 10^{-9} =: \alpha_{2,M}$ and $|\beta_2| \leq (9.76) \times 10^{-7} =: \beta_{2,M}$.

Lemma 3.5.2. In the domain $D_2 \cup D_3$, $P_u > 0$.

Proof. We note $x \in D_2 \cup D_3$ corresponds to $\tau \in [-1, 1)$. We consider an approximate expression

$$P_a(\tau) = \frac{335867}{539062} + \frac{835179584688}{1968351794375}\,\tau - \frac{7294680}{73240997}\,\tau^2 + \frac{60789}{1703279}\,\tau^3 \tag{3.5.7}$$

It can be checked that

$$P_u(\tau) - P_a(\tau) = \sum_{k=0}^{22} d_k \tau^k \ , \text{ with } \left|P_u(\tau) - P_a(\tau)\right| \leq \sum_{k=0}^{22} |d_k| \leq 0.039 \tag{3.5.8}$$

On the otherhand, it can be checked

$$P_a'(\tau) = \frac{182367}{1703279}\left[\left(\tau - \frac{40}{43}\right)^2 + \left(\frac{44}{25}\right)^2\right] > 0 \tag{3.5.9}$$

and therefore P_a is an increasing function implying

$$P_a(\tau) > P_a(-1) > 0.063 \tag{3.5.10}$$

Using (3.5.8) and (3.5.10), we obtain

$$P_u(\tau) > 0.063 - 0.039 > 0 \tag{3.5.11}$$

□

Definition 3.5.3. Let γ be a real number. We define weighted norm $\|.\|_\gamma$ on $C(D_3)$:

$$\|f\|_\gamma := \sup_{x \in D_3} \left|(x - x_0)^\gamma f(x)\right| \text{ for } f \in C(D_3), \tag{3.5.12}$$

We also define $E_{0,1}$, $E_{0,2}$ so that $E_0 = \mathcal{N}[0] = E_{0,1} + E_{0,2}$, with

$$E_{0,1}(x) = \alpha_2 G_1(x) + \beta_2 G_2(x) \tag{3.5.13}$$

$$E_{0,2}(x) = \frac{1}{7} G_1(x) \int_{L_0}^{x} G_2(t) R(t) dt - \frac{1}{7} G_2(x) \int_{L_0}^{x} G_1(t) R(t) dt \tag{3.5.14}$$

Remark 3.5.4. Since G_1, G_2 and R are explicit and involve only a finite number of terms in powers of $(x - x_0)$, $E_{0,2}$ and $E'_{0,2}$ can be computed explicitly for any $x \in \mathbb{D}_3$

Lemma 3.5.5. For $\gamma = 3.2$, $\|E_0\|_\gamma \leq 2.04 \times 10^{-6}$, $\|E'_0\|_\infty \leq 1.23 \times 10^{-5}$.

Proof. Using Corollary 3.5.1, and explicit representation of G_1, G'_1, G_2, G'_2 it is easy to prove that $(x - x_0)^\gamma (\alpha_{2,M}|G_1| + \beta_{2,M}|G_2|) = (x - x_0)^\gamma (\alpha_{2,M} G_1 + \beta_{2,M} G_2)$, attains its maximum value at $x = L_0$, while $\alpha_{2,M}|G'_1| + \beta_{2,M}|G'_2| = \alpha_{2,M} G'_1 - \beta_{2,M} G'_2$ attains its maximum at $x = x_0 + r$. Thus, it is readily checked $\|E_{0,1}\|_\gamma \leq 2.03 \times 10^{-6}$, $\|E'_{0,1}\|_\infty \leq 1.2245 \times 10^{-5}$. Applying Lemma B.0.2 to $(x - x_0)^\gamma E_{0,2}$, and $E'_{0,2}$ for subdivisions of the interval $[x_0 + r, L_0]$ with $n = 5$, we obtain $\|E_{0,2}\|_\gamma \leq 2.3 \times 10^{-9}$, $\|E'_{0,2}\|_\infty \leq 3.8 \times 10^{-8}$. Adding up, we get the bounds for $\|E_0\|_\gamma$, $\|E'_0\|_\infty$. given in the Lemma statement. □

Definition 3.5.6. Define the Banach space

$$\mathcal{S} := \{f \in C(D_3) : \|f\|_\gamma < \infty\}, \tag{3.5.15}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\|_\gamma \leq \epsilon\}. \tag{3.5.16}$$

Lemma 3.5.7. Let $\delta = 0.963$. Then for $\gamma = 3.2$, there exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. Moreover, $\|E\|_\gamma \leq 4.01 \times 10^{-6}$ and $\|E'\|_\infty \leq 3.76 \times 10^{-5}$ on D_3.

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. Using the fact that $G_1, G_2, P_u > 0$ in D_3, (3.5.4) implies

$$\begin{aligned} \left|[\mathcal{N}[\mathcal{E}] - E_0](x - x_0)^\gamma\right| &\leq \|E\|_\gamma \sum_{j=1}^{2} \frac{(x - x_0)^\gamma G_j(x)}{7} \int_x^{L_0} G_{3-j}(t - x_0)^{-\gamma} [12 P_u] dt \\ &+ \|E\|_\gamma^2 \sum_{j=1}^{2} \frac{(x - x_0)^\gamma G_j(x)}{7} \int_x^{L_0} 6 G_{3-j}(t - x_0)^{-2\gamma} dt \end{aligned}$$

Since

$$Q(x) = \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-\gamma}[12P_u]dt \ ,$$

$$T(x) = \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} 6G_{3-j}(t-x_0)^{-2\gamma}dt \ ,$$

and their derivatives can be explicitly calculated, and their upper bounds in D_2 determined by applying Lemma B.0.2 the interval $[x_0 + r, L_0]$ by partition it into five equal intervals ($n = 5$) results in the bounds $\Big|Q(x)\Big| \leq 0.49$ and $\Big|T(x)\Big| \leq 1$ Therefore, it follows that

$$\big|\big[\mathcal{N}[\mathcal{E}] - E_0\big](x-x_0)^\gamma\big| \leq \Big[(1+\delta)(0.49) + (1+\delta)^2\|E_0\|_\gamma\Big]\|E_0\|_\gamma \leq \delta\|E_0\|_\gamma \qquad (3.5.17)$$

and taking supremum over $x \in D_3$ implies $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. From similar arguments and $\|\mathcal{E}_1\|_\gamma + \|\mathcal{E}_2\|_\gamma \leq 2(1+\delta)\|E_0\|_\gamma$, (3.5.4) yields:

$$\begin{aligned} \big|\big[\mathcal{N}[\mathcal{E}_1] - \mathcal{N}[\mathcal{E}_2]\big](x-x_0)^\gamma\big| &\leq \Big[(0.49) + (12.2)(1+\delta)\|E_0\|_\gamma\Big]\|\mathcal{E}_1 - \mathcal{E}_2\|_\gamma \\ &\leq \frac{1}{2} \times \|\mathcal{E}_1 - \mathcal{E}_2\|_\gamma \end{aligned}$$

and taking supremum over $x \in D_3$ implies $\mathcal{N}$ is contractive in $\mathcal{B}$. Banach fixed point theorem implies the existence and uniqueness of solution to $E = \mathcal{N}[E]$ in ball $\mathcal{B}$. Further for such a solution,

$$\begin{aligned} \Big|E'(x)\Big| &\leq \|E_0'\|_\infty + \|E\|_\gamma \left\{ \sum_{j=1}^{2} \frac{\Big|G_j'(x)\Big|}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-\gamma}[12P_u]dt \right\} \\ &+ \|E\|_\gamma^2 \left\{ 6\sum_{j=1}^{2} \frac{\Big|G_j'(x)\Big|}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-2\gamma}dt \right\} \qquad (3.5.18) \end{aligned}$$

Using explicit computation of each of the two terms within $\{\cdot\}$ at $x = x_0 + r$ and and the bound on $\|E_0'\|_\infty$ in Lemma 3.5.5, $\|E'\|_\infty \leq \|E_0'\|_\infty + 6.3\|E\|_\gamma + 12.3\|E\|_\gamma^2 \leq 3.76 \times 10^{-5}$ □

The main result of this section is a consequence of 3.5.7:

Corollary 3.5.8. In D_3, the tritronqué solution has the representation $y = y_0 + E$, where $\|E\|_\gamma \leq 4.01 \times 10^{-6}$ and $\|E'\|_\infty \leq 3.76 \times 10^{-5}$. In particular,
$\Big|E(x_0 + r)\Big| \leq 1.26 \times 10^{-5}$ and $\Big|E'(x_0 + r)\Big| \leq 3.76 \times 10^{-5}$.

Proof. It is clear from the smoothness of G_1, G_2 in D_3 that the solution E to (3.5.4) guaranteed by Lemma 3.5.7 also satisfies (3.5.1), implying $y = y_0 + E$ satisfies P-1. The continuity conditions (3.5.2) implies that $y_0 + E$ matches the tritronqué at $x = x_0 + r$. Since the solution to the initial value problem is unique, $y = y_0 + E$ must be the tritronqé solution. The error bounds follow from Lemma 3.5.7 □

3.6 Analysis on $D_4 = \{x \in \mathbb{C} : |x - x_0| = r, x \neq x_0 + r\}$

It is useful to introduce $\zeta = x - x_0$. Then, from the expression for y_0 in Theorem 3.2.3, it follows that

$$Y_0(\zeta) := y_0(x_0 + \zeta) = -\frac{1}{\zeta^2} + \zeta^2 P(\zeta) \tag{3.6.1}$$

where $P(\zeta)$ given by (3.2.8). E is defined as:

$$E(\zeta) = y(x_0 + \zeta) - y_0(x_0 + \zeta) \tag{3.6.2}$$

Substitution of (3.6.2) and (3.6.1) into (3.2.1) yields:

$$E''(\zeta) + 12Y_0(\zeta)E(\zeta) = -6E(\zeta)^2 - R(\zeta), \tag{3.6.3}$$

where $R(\zeta)$ is the residual $R(\zeta) := Y_0''(\zeta) + 6Y_0(\zeta)^2 - (x_0 + \zeta)$. Requiring that as $\nu \to 0^+$ on $\zeta = re^{i\nu}$ continuity of $E(\zeta) + Y_0(\zeta)$ with solution found in D_3, we obtain conditions

$$E\left(re^{i0^+}\right) = E\left([x_0 + r]^+\right) + y_0\left([x_0 + r]^+\right) - Y_0\left(re^{i0^+}\right) \tag{3.6.4}$$

$$E'\left(re^{i0^+}\right) = E'\left([x_0 + r]^+\right) + y_0'\left([x_0 + r]^+\right) - Y_0'\left(re^{i0^+}\right) \quad , \tag{3.6.5}$$

Lemma 3.6.1. In D_4, $\|R\|_\infty \leq 1.311 \times 10^{-6}$.

Proof. Recognizing that $Y_0(\zeta)$ is a truncation of an exact series representation for a solution to P-1 up to a power of ζ^{19}, one obtains

$$R(\zeta) = \sum_{j=18}^{38} R_j \zeta^j \ , \tag{3.6.6}$$

implying from calculation of R_j that on $|\zeta| = r$,

$$\|R\|_\infty \leq \sum_{j=18}^{38} |R_j| r^j \leq 1.311 \times 10^{-6} \tag{3.6.7}$$

□

Remark 3.6.2. Now we seek to find bounds on two independent solutions of the homogeneous system $E'' + 12Y_0E = 0$ associated with (3.6.3). For this purpose, it is useful to note the geometric bound

$$\Big|a_j\Big| \leq \frac{1}{2^j} \tag{3.6.8}$$

for $0 \leq j \leq 17$ that can be verified through calculation, using recurrence relation (3.2.9).

Definition 3.6.3. Define

$$G_1(\zeta) = \sum_{n=0}^{\infty} A_n \zeta^{4+n} \, , G_2(\zeta) = \sum_{n=0}^{\infty} B_n \zeta^{n-3}, \tag{3.6.9}$$

where

$$A_0 = 1 \, , A_1 = A_2 = A_3 = 0 \, , A_n = -\frac{12}{n(n+7)} \sum_{k=0}^{\min\{n-4,17\}} a_k A_{n-4-k} \quad , \quad \text{for } n \geq 4 \tag{3.6.10}$$

$$\begin{aligned} &B_0 = 1 \, , B_1 = B_2 = B_3 = B_7 = 0 \, , \quad \text{and for } n \geq 4, n \neq 7, \\ &B_n = -\frac{12}{n(n-7)} \sum_{k=0}^{\min\{n-4,17\}} a_k B_{n-4-k} \end{aligned} \tag{3.6.11}$$

Remark 3.6.4.

$$G_1(\zeta) = \zeta^4 + \sum_{n=0}^{\infty} A_{n+4} \zeta^{n+8} \quad , \quad G_2(\zeta) = \zeta^{-3} + \sum_{n=0}^{\infty} B_{n+4} \zeta^{n+1} \tag{3.6.12}$$

Lemma 3.6.5. For any integer $n \geq 1$,

$$\Big|A_n\Big| \leq c_A (\frac{3}{4})^n \quad , \quad \Big|B_n\Big| \leq c_B (\frac{3}{4})^n \, , \tag{3.6.13}$$

where $c_A = 0.21$ and $c_B = 0.85$,

Proof. We checked the inequalities for the first twenty two coefficients $\{A_n, B_n\}_{n=1}^{22}$ through

explicit calculations involving (3.6.10) and (3.6.11). Assume the inequality holds for $n \leq n_0$ for some $n_0 \geq 22$. Then, using bounds on a_k, and noting that (3.6.10) and (3.6.11) no longer involves A_0, we obtain

$$\begin{aligned} \left|A_{n_0+1}\right| &\leq \frac{12}{(n_0+1)(n_0+8)} c_A \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \left(\frac{3}{4}\right)^{n_0-3-k} \\ &\leq \frac{36 c_A}{(n_0+1)(n_0+8)} \left(\frac{4}{3}\right)^4 \left(\frac{3}{4}\right)^{n_0+1} \leq c_A \left(\frac{3}{4}\right)^{n_0+1} \end{aligned} \tag{3.6.14}$$

So, the inequality holds for $n_0 + 1$. By induction it holds for all n. The same induction proof works for B_n after using $\frac{36}{(n_0+1)(n_0-6)} \left(\frac{4}{3}\right)^4 \leq 1$ for $n_0 \geq 22$. □

Lemma 3.6.6. In D_4,

$$\|G_1\|_\infty \leq r^4 + \frac{81 c_A r^8}{256 - 192r} \leq 0.249 \ \ , \ \|G_2\|_\infty \leq \frac{1}{r^3} + \frac{81 c_B r}{256 - 192r} \leq 3.32 \tag{3.6.15}$$

$$\|G_1'\|_\infty \leq 4r^3 + \frac{81 c_A r^7 (32 - 21r)}{64(4 - 3r)^2} \leq 1.48 \ \ , \ \|G_2'\|_\infty \leq \frac{3}{r^4} + \frac{81 c_B}{16(4 - 3r)^2} \leq 13.7 \tag{3.6.16}$$

Proof. We use the bounds in (3.6.13) in the expression for G_1, G_2 and their first derivatives obtained from (3.6.12) and use triangular inequality and the identities $\sum_{n=0}^{\infty} |z|^n = \frac{1}{1-|z|}$, $\sum_{n=0}^{\infty} n|z|^{n-1} = \left(\frac{1}{1-|z|}\right)^2$ for $|z| < 1$. □

Lemma 3.6.7. G_1 and G_2 are two linearly independent solutions to the homogenous differential equation $G'' + 12 Y_0 G = 0$ on D_4 with Wronskian $W := G_1 G_2' - G_2 G_1' = -7$.

Proof. That G_1 and G_2 satisfy $G'' + 12 Y_0 G = 0$ is clear from substituting each of the series in (3.6.12) for G and noting that the recurrence relation obtained is the one we have for A_n and B_n in (3.6.10) and (3.6.11), whose geometric decay rate assures convergence in D_4. Since the Wronskian W must be a constant, it is evident from the small ζ behavior that $G_1 = \zeta^4(1 + o(1))$, $G_1' = 4\zeta^3(1 + o(1))$, $G_2 = \zeta^{-3}(1 + o(1))$, $G_2' = -3\zeta^{-4}(1 + o(1))$, it follows that $W = -7 + o(1)$; since it is independent of ζ, $W = -7$, implying also that the two solutions G_1, G_2 are independent. □

Applying standard variation of parameter argument, it is clear that (3.6.3) implies

$$E(\zeta) = \sum_{j=1}^{2} (-1)^{j+1} G_j(\zeta) \oint_{x_0+r}^{\zeta} \frac{G_{3-j}}{7} [\,-6E^2 - R]\,dz + \alpha_3 G_1(\zeta) + \beta_3 G_2(\zeta) := \mathcal{N}[E](\zeta), \tag{3.6.17}$$

where the integration $\oint_{x_0}^{\zeta}$ is integral from x_0 to ζ over D_4 in the counterclockwise direction, α_3 and β_3 are chosen to satisfy (3.6.4), implying

$$\alpha_3 G_1(r) + \beta_3 G_2(r) = E\left([x_0+r]^+\right) + y_0\left([x_0+r]^+\right) - Y_0\left(e^{i0^+}\right), \tag{3.6.18}$$

$$\alpha_3 G_1'(r) + \beta_3 G_2'(r) = E'\left([x_0+r]^+\right) + y_0'\left([x_0+r]^+\right) - Y_0'\left(e^{i0^+}\right), \tag{3.6.19}$$

$$\tag{3.6.20}$$

Remark 3.6.8. It is to be noted that in the integral reformulation (3.6.17) of the differential equation (3.6.3), we are using the inital conditions at $\zeta = r$, inherited from the analysis of the tritronqué solution in D_3. *A priori*, we are not requiring that the solution E returns to itself when ν changes from 0 to 2π on the circle $\zeta = re^{i\nu}$. However, it is well-known that any P-1 solution is single valued, and therefore so must be the solution $y = y_0 + E$ to P-1. Since y_0 is manifestly single valued, so must be E.

Lemma 3.6.9. $|\alpha_3| \le 3.82 \times 10^{-5}$ and $|\beta_3| \le 3.76 \times 10^{-6}$.

Proof. We use (3.6.18) and (3.6.19) to solve for α_3 and β_3. From bounds on $E^+(x_0+r)$ and $E'^+(x_0+r)$ in Corollary 3.5.8, using small mismatch for y_0, y_0' at x_0+r, (see remark 3.2.4) and the bounds on G_1, G_1', G_2, G_2' from Lemma 3.6.7, we obtain the bounds α_3 ad β_3. □

Lemma 3.6.10. Let E_0 denote $E_0 = \mathcal{N}[0]$. Then, $\|E_0\|_\infty \le 2.34 \times 10^{-5}$ and $\|E_0'\|_\infty \le 1.15 \times 10^{-4}$ on D_4.

Proof. Using (3.6.17) we have

$$\left|E_0\right| \le \frac{4\pi r}{7}\|G_1\|_\infty\|G_2\|_\infty\|R\|_\infty + |\alpha_3|\|G_1\|_\infty + |\beta_3|\|G_2\|_\infty \le 2.34 \times 10^{-5} \tag{3.6.21}$$

$$\left|E_0'\right| \leq \frac{2\pi r}{7}\left(\|G_1\|_\infty\|G_2'\|_\infty + \|G_2\|_\infty\|G_1'\|_\infty\right)\|R\|_\infty + |\alpha_3|\|G_1'\|_\infty + |\beta_3|\|G_2'\|_\infty \leq 1.15\times 10^{-4} \tag{3.6.22}$$

□

Definition 3.6.11. Define Banach space

$$\mathcal{S} := \{f \in C(D_4) : \|f\|_\infty < \infty\}, \tag{3.6.23}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\| \leq \epsilon\}. \tag{3.6.24}$$

Lemma 3.6.12. Let $\delta = 2\times 10^{-4}$. There exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Moreover, $\|E\|_\infty \leq 2.35 \times 10^{-5}$ and $\|E'\|_\infty \leq 1.16 \times 10^{-4}$ on D_4.

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$.

$$\left|\mathcal{N}[\mathcal{E}](\zeta) - E_0(\zeta)\right| \quad \leq \quad \frac{24}{7}\pi r\|G_1\|_\infty\|G_2\|_\infty(1+\delta)^2\|E_0\|_\infty^2 \leq \delta\|E_0\|,$$

and taking supremum over $\zeta \in D_4$ implies $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. From similar arguments and $\|\mathcal{E}_1\|_\infty + \|\mathcal{E}_2\|_\infty \leq 2(1+\delta)\|E_0\|_\infty$, (3.6.17) yields:

$$\|\mathcal{N}[\mathcal{E}_1] - \mathcal{N}[\mathcal{E}_2]\|_\infty \leq \frac{48}{7}\pi r\|G_1\|_\infty\|G_2\|_\infty(1+\delta)\|E_0\|_\infty\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty \leq 3\times 10^{-4}\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty\ ,$$

and therefore, $\mathcal{N}$ is contractive. Banach fixed point theorem implies the existence and uniqueness of such $E \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. The following are immediate:

$$\begin{aligned}
\|E\|_\infty &\leq (1+\delta)\|E_0\|_\infty \leq 2.35\times 10^{-5},\\
\|E'\|_\infty &\leq \|E_0'\|_\infty + \frac{12\pi r}{7}\left(\|G_1'\|_\infty\|G_2\|_\infty + \|G_2'\|_\infty\|G_1\|_\infty\right)(1+\delta)^2\|E_0\|_\infty^2 \leq 1.16\times 10^{-4}.
\end{aligned}$$

□

Corollary 3.6.13. The tritronqué solution to P-1 has the following representation in D_4, $y = y_0 + E$ where where $\|E\|_\infty \leq 2.35 \times 10^{-5}$ and $\|E'\|_\infty \leq 1.16 \times 10^{-4}$ on D_4.

Proof. Smoothness of G_1 and G_2 on domain D_4 implies that the solution E satisfying (3.6.17) also solves (3.6.3). It is directly checked that $y = y_0 + E$ solves P-1 and, because of (3.6.4), matches the tritronqué at the boundary point in D_3. Since the solution to P-1 with given initial condition is unique, $y = y_0 + E$ must be the tritronqué; bounds follow from Lemma 3.6.12. □

3.7 The proof of the theorem 3.2.3

Proof of Theorem 3.2.3. Corollaries 3.3.5, 3.4.10, 3.5.8 and 3.6.13 immediate implies the error bounds on $E = y - y_0$ for the tritronqué. Since it is well-known that regular solutions of P-1 must be analytic, it follows that y is also analytic in D. Also, any solution to P-1 is meromorphic, with locally convergent series representation (3.2.3) near each singularity x_p. Cauchy integral formula implies that the integral $-\frac{1}{2\pi i}\oint_{|\zeta|=r}\zeta y(x_0+\zeta)d\zeta$ equals to the number of singularities of $y(x)$ in $|x - x_0| < r$. From this observation and corollary 3.6.13, we calculate

$$\left|1+\frac{1}{2\pi i}\oint_{|\zeta|=r}\zeta y(x_0+\zeta)d\zeta\right| \le \left|1+\frac{1}{2\pi i}\oint_{|\zeta|=r}\zeta y_0(x_0+\zeta)d\zeta\right|+r^2\|y-y_0\|_\infty \le 1.2\times 10^{-5} < 1, \tag{3.7.1}$$

implying that there is only one singulariy x_p of the tritronqué y in the region $|x - x_0| < r$. Once again, Cauchy integral formula and (3.2.3) imply $-\frac{1}{4\pi i}\oint_{|\zeta|=r}\zeta^2 y(x_0+\zeta)d\zeta = x_p - x_0$, and since $\zeta^2 y_0(x_0 + \zeta)$ is analytic in $|\zeta| \le r$, it follows that

$$|x_p - x_0| \le \left| -\frac{1}{4\pi i}\oint_{|\zeta|=r}\zeta^2 E(\zeta)d\zeta\right| \le \frac{r^3}{2}\|E\|_\infty \le 4.1\times 10^{-6}, \tag{3.7.2}$$

Also, it is known [19] that the pole of the tritronqué solution closest to the origin is on the negative real axis. Since our analysis shows solution cannot blow up on D, it follows that the singularity x_p, whose location is estimated in (3.7.2), is on the negative real axis. □

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Appendix A
Appendix for Hele-Shaw Analysis

A.1 Discretized Complex Approximations

Definition A.1.1. Let $\mathcal{D} \subset \mathbb{C}^2$ be a bounded simply connected region. We define

$$\bar{\mathbb{A}}(\mathcal{D}) := \left\{ \hat{f}(\eta, \zeta) : \hat{f} \text{ is analytic in } \mathcal{D} \text{ and } C^2 \text{ up to its boundary} \right\} \tag{A.1.1}$$

We define the functional $\| \|_{\mathcal{D}}$ on $\bar{\mathbb{A}}(\mathcal{D})$ as

$$\|\hat{f}\|_{\mathcal{D}} := \sup_{(\eta,\zeta)\in\bar{\mathcal{D}}} |\hat{f}(\eta, \zeta)| \tag{A.1.2}$$

Remark A.1.2. $\| \|_{\mathcal{D}}$ is a norm on $\bar{\mathbb{A}}(\mathcal{D})$. Moreover, let $\hat{f}, \hat{g} \in \bar{\mathbb{A}}(\mathcal{D})$, then

$$\left\| \hat{f}\hat{g} \right\|_{\mathcal{D}} \leq \left\| \hat{f} \right\|_{\mathcal{D}} \|\hat{g}\|_{\mathcal{D}}\,, \tag{A.1.3}$$

Definition A.1.3. Let $r_\zeta, r_\eta > 1$ be real numbers. We define

$$\mathcal{D}_0 := \left\{ (\eta, \zeta) \in \mathbb{C}^2 : \frac{1}{r_\eta} \leq |\eta| \leq r_\eta \text{ and } \frac{1}{r_\zeta} \leq |\zeta| \leq r_\zeta \right\} \tag{A.1.4}$$

Lemma A.1.4. Let $\hat{Z}(\eta, \zeta) \in \bar{\mathbb{A}}(\mathcal{D}_0)$, then,

$$\hat{Z}(\eta, \zeta) = \sum_{N=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} \hat{Z}_{N,j} \zeta^N \eta^j \text{ on } \mathcal{D}_0, \tag{A.1.5}$$

and

$$\left| \hat{Z}_{N,j} \right| \leq \frac{\|\hat{Z}\|_{\mathcal{D}_0}}{r_\zeta^{|N|} r_\eta^{|j|}} \tag{A.1.6}$$

for all $N, j \in \mathbb{Z}$.

Proof. Since $\hat{Z}$ is analytic in $\mathcal{D}_0$, Cauchy integral formula and suitably deforming the counters of integration imply

$$\begin{aligned}|Z_{N,j}| &= \left| \frac{1}{(2\pi i)^2} \oint_{|\zeta|=r_\zeta^{sgn(N)}} \oint_{|\eta|=r_\eta^{sgn(j)}} \hat{Z}(\eta,\zeta)\zeta^{-N}\eta^{-j}\frac{d\zeta}{\zeta}\frac{d\eta}{\eta} \right| \\ &\leq \frac{1}{(2\pi)^2}\int_0^{2\pi}\int_0^{2\pi} \frac{\|\hat{Z}\|_{\mathcal{D}_0}}{r_\zeta^{|N|}r_\eta^{|j|}} d\nu d\theta = \frac{\|\hat{Z}\|_{\mathcal{D}_0}}{r_\zeta^{|N|}r_\tau^{|j|}},\end{aligned}$$

and the proof is complete. □

Lemma A.1.5 (Orthogonality Relations)**.** Let M, N, K be integers where $K > 0$, then

$$\frac{1}{K}\sum_{k=1}^{K} e^{iM\tau_k}e^{-iN\tau_k} = \begin{cases} 1 & \text{if } K|M-N, \\ 0 & \text{if } K \nmid M-N, \end{cases} \tag{A.1.7}$$

where

$$\tau_k = \frac{2\pi k}{K} \text{ for } k = 1, \ldots, K \tag{A.1.8}$$

Proof. Note that

$$\frac{1}{K}\sum_{k=1}^{K} e^{iM\nu_k}e^{-iN\nu_k} = \frac{1}{K}\sum_{k=1}^{K} e^{i(M-N)\nu_k} = \frac{1}{K}\sum_{k=1}^{K} \zeta_K^{(M-N)k}$$

where $\zeta_K = e^{\frac{2\pi i}{K}}$. Now, if $K|M-N$, then $\zeta_K^{(M-N)k} = 1$ for all k, and if $K \nmid M-N$ then $\sum_{k=1}^{K}\zeta_K^{(M-N)k} = \zeta_K^{(M-N)}\frac{(\zeta_K^{K(M-N)}-1)}{\zeta_K-1} = 0$ since $\zeta_K^K = 1$. □

Lemma A.1.6. Let $\hat{Z}(\eta,\zeta) \in \bar{\mathbb{A}}(\mathcal{D}_0)$ and let K, L be positive integers. Let

$$\hat{Z}^d_{N,j}(K,L) := \frac{1}{KL}\sum_{k=1}^{K}\sum_{l=1}^{L} \hat{Z}(e^{i\theta_l}, e^{i\nu_k})e^{-iN\nu_k}e^{-ij\theta_l} \tag{A.1.9}$$

for each $N, j \in \mathbb{Z}$ where

$$\nu_k = \frac{2\pi k}{K} \text{ and } \theta_l = \frac{2\pi l}{L}, \tag{A.1.10}$$

then,

$$\hat{Z}^d_{N,j}(K,L) = \sum_{n=-\infty}^{\infty}\sum_{J=-\infty}^{\infty} \hat{Z}_{N+nK,j+JL} \tag{A.1.11}$$

Proof. Changing the order of summation and orthogonality relations imply

$$\hat{Z}^d_{N,j}(K,L) = \frac{1}{KL}\sum_{k=1}^{K}\sum_{l=1}^{L}\hat{Z}(e^{i\theta_l},e^{i\nu_k})e^{-iN\nu_k}e^{-ij\theta_l}$$
$$= \frac{1}{KL}\sum_{k=1}^{K}\sum_{l=1}^{L}\left(\sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty}\hat{Z}_{N,j}e^{i\tilde{N}\nu_k}e^{i\tilde{j}\theta_l}\right)e^{-iN\nu_k}e^{-ij\theta_l}$$
$$= \sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty}\hat{Z}_{N,j}\left(\frac{1}{K}\sum_{k=1}^{K}e^{i\tilde{N}\nu_k}e^{-iN\nu_k}\right)\left(\frac{1}{L}\sum_{l=1}^{L}e^{i\tilde{j}\theta_l}e^{-ij\theta_l}\right) = \sum_{n=-\infty}^{\infty}\sum_{J=-\infty}^{\infty}\hat{Z}_{N+nK,j+JL}$$

and the result is obtained. □

Definition A.1.7. Let $\hat{Z}(\eta,\zeta)\in\bar{\mathbb{A}}(\mathcal{D}_0)$ and let D_η, D_ζ, K, L be positive integers. We define

$$\hat{Z}^d(\eta,\zeta;D_\eta,D_\zeta,K,L) := \sum_{N=-D_\zeta}^{D_\zeta}\sum_{j=-D_\eta}^{D_\eta}\hat{Z}^d_{N,j}(2|N|+K,2|j|+L)\zeta^N\eta^j \tag{A.1.12}$$

and in the case when $D_\zeta = D_\eta = K = L = D$, we use the following shorter notation

$$\hat{Z}^d := \hat{Z}^d(\eta,\zeta;D,D,D,D) = \sum_{N=-D}^{D}\sum_{j=-D}^{D}\hat{Z}^d_{N,j}(2|N|+D,2|j|+D)\zeta^N\eta^j, \tag{A.1.13}$$

and we define the degree of $\hat{Z}$ as:

$$\deg(\hat{Z}^d) := D \tag{A.1.14}$$

Lemma A.1.8. Let $\hat{Z}(\eta,\zeta)\in\bar{\mathbb{A}}(\mathcal{D}_0)$, then

$$\left|\hat{Z}^d_{N,j}(2|N|+K,2|j|+L) - \hat{Z}_{N,j}\right| \le \frac{\|\hat{Z}\|_{\mathcal{D}_1}}{r_\zeta^{|N|}r_\eta^{|j|}}\frac{c_1}{r_\zeta^K r_\eta^L} \tag{A.1.15}$$

for all integers N, j, K, L with $K, L > 0$, where

$$c_1(K,L) = \frac{4}{\left(1-r_\zeta^{-K}\right)\left(1-r_\eta^{-L}\right)}, \tag{A.1.16}$$

Proof. Lemma A.1.4 and Lemma A.1.6 imply

$$\left|\hat{Z}^d_{N,j}(2|N|+K,2|j|+L)-\hat{Z}_{N,j}\right| = \left|\sum_{n,J\in\mathbb{Z},\,(n,J)\neq(0,0)} \hat{Z}_{N+n(2|N|+K),j+J(2|j|+L)}\right|$$
$$\leq \|\hat{Z}\|_{\mathcal{D}_1}\sum_{n=1}^{\infty}\sum_{J=1}^{\infty}\left(\frac{1}{r_\zeta^{|N|+n(2|N|+K)}}+\frac{1}{r_\zeta^{-|N|+n(2|N|+K)}}\right)\left(\frac{1}{r_\eta^{|j|+J(2|j|+L)}}+\frac{1}{r_\eta^{-|j|+J(2|j|+L)}}\right) \tag{A.1.17}$$

and the result is obtained from $\frac{\left(1+r_\zeta^{-2|N|}\right)\left(1+r_\eta^{-2|j|}\right)}{\left(1-r_\zeta^{-2|N|-K}\right)\left(1-r_\eta^{-2|j|-L}\right)} \leq \frac{4}{\left(1-r_\zeta^{-K}\right)\left(1-r_\eta^{-L}\right)}$ from summing the geometric series. □

Lemma A.1.9. Let $\hat{Z}(\eta,\zeta)\in\bar{\mathbb{A}}(\mathcal{D}_0)$ and let

$$\hat{Z}(\eta,\zeta)=\sum_{N=-\infty}^{\infty}\sum_{j=0}^{\infty}\hat{Z}_{N,j}\zeta^N\frac{1}{2}\left(\eta^j+\eta^{-j}\right) \tag{A.1.18}$$

where $\hat{Z}_{N,j}\in\mathbb{R}$ for all $N,j\in\mathbb{Z}$. Let

$$\hat{f}(\eta,\zeta)=\frac{1}{2}\hat{Z}(\eta,\zeta)+\frac{1}{2}\hat{Z}(\eta,\frac{1}{\zeta})\ ,\ \hat{g}(\eta,\zeta)=\frac{1}{2i}\hat{Z}(\eta,\zeta)-\frac{1}{2i}\hat{Z}(\eta,\frac{1}{\zeta}), \tag{A.1.19}$$

Then $\hat{f},\hat{g}\in\bar{\mathbb{A}}(\mathcal{D}_0)$. Let

$$f(\rho,\nu):=\hat{f}(e^{i\theta},e^{i\nu})\ ,\ g(\rho,\nu):=\hat{g}(e^{i\theta},e^{i\nu}) \tag{A.1.20}$$

where $\rho=\frac{(e^{i\theta}+e^{-i\theta}+12)}{20}=\frac{\cos(\theta)+6}{10}$, $\theta\in[\pi,2\pi]$ and $\nu\in[0,2\pi]$. Then $f(\rho,\nu)\in\mathcal{C}$ and $g(\rho,\nu)\in\mathcal{S}$. Let D,K be positive integers, and let

$$f^d(\rho,\nu;D,K):=\hat{f}^d(e^{i\theta},e^{i\nu};D,K)\ ,\ g(\rho,\nu;D,K):=\hat{g}^d(e^{i\theta},e^{i\nu},D,K) \tag{A.1.21}$$

then $f^d(\rho,\nu;D,K)\in\mathcal{C}$, $g^d(\rho,\nu;D,K)\in\mathcal{S}$ and

$$\begin{aligned}\left|F_N(f-f^d)\right| &\leq \frac{\|\hat{f}\|_{\mathcal{D}_1}r_\eta}{r_\zeta^N(r_\eta-1)}\left(1_{N\leq D}c_2(K,L,D)+1_{N>D}\right),\\ \left|F_N(g-g^d)\right| &\leq \frac{\|\hat{g}\|_{\mathcal{D}_1}r_\eta}{r_\zeta^N(r_\eta-1)}\left(1_{N\leq D}c_2(K,L,D)+1_{N>D}\right),\end{aligned} \tag{A.1.22}$$

for all $N\geq 0$ where

$$c_2(K,L,D)=\frac{c_1(K,L)}{r_\zeta^K r_\eta^L}+\frac{1}{r_\eta^{D+1}} \tag{A.1.23}$$

Proof. Since $(\eta,\zeta)\to(\eta,\frac{1}{\zeta})$ is an automorphism of $\mathcal{D}_1$, $\hat{Z}(\eta,\frac{1}{\zeta})$ is analytic in $\mathcal{D}_1$ and C^2 up to its boundary, so are $\hat{f}(\eta,\zeta)$ and $\hat{g}(\eta,\zeta)$. Evidently

$$\hat{f}(\eta,\zeta)=\sum_{N=0}^{\infty}\sum_{j=0}^{\infty}\hat{f}_{N,j}\frac{1}{2}(\zeta^N+\zeta^{-N})\frac{1}{2}\left(\eta^j+\eta^{-j}\right)$$

where $\hat{f}_{N,j}=Z_{N,j}+Z_{-N,j}\in\mathbb{R}$ for all $N,j\geq 0$, and consequently

$$f(\rho,\nu)=\sum_{N=0}^{\infty}\sum_{j=0}^{\infty}\hat{f}_{N,j}\cos(N\nu)\cos(\theta)$$

and similarly one can obtain

$$f^d(\rho,\nu;D,K)=\sum_{N=0}^{D}\sum_{j=0}^{D}\hat{f}^d_{N,j}(2N+K,2j+K)\cos(N\nu)\cos(\theta)$$

where $\hat{f}^d_{N,j}=\hat{Z}^d_{N,j}+\hat{Z}^d_{-N,j}$ for all $N,j\geq 0$. Hence, when $N\leq D$ one has

$$\begin{aligned}\left|F_N(f-f^d)\right|&=\left|\sum_{j=0}^{D}\left(\hat{f}_{N,j}-\hat{f}^d_{N,j}(2N+K,2j+K)\right)+\sum_{j=D+1}^{\infty}\hat{f}_{N,j}\right|\\&\leq\sum_{j=0}^{D}\left|\hat{f}_{N,j}-\hat{f}^d_{N,j}(2N+K,2j+K)\right|+\sum_{j=D+1}^{\infty}\left|\hat{f}_{N,j}\right|\leq\sum_{j=0}^{D}\frac{\|\hat{f}\|_{\mathcal{D}_1}}{r_\zeta^{|N|}r_\eta^{|j|}}\frac{c_1}{r_\zeta^K r_\eta^L}\\&+\sum_{j=D+1}^{\infty}\frac{\|\hat{f}\|_{\mathcal{D}_1}}{r_\zeta^{|N|}r_\eta^{|j|}}\leq\frac{\|\hat{f}\|_{\mathcal{D}_1}r_\eta}{r_\zeta^N(r_\eta-1)}\left(\frac{c_1}{r_\zeta^K r_\eta^L}+\frac{1}{r_\eta^{D+1}}\right)\end{aligned}$$

and when $N>D$, one has

$$\left|F_N(f-f^d)\right|=\left|\sum_{j=0}^{\infty}\hat{f}_{N,j}\right|\leq\sum_{j=D+1}^{\infty}\left|\hat{f}_{N,j}\right|\leq\sum_{j=0}^{\infty}\frac{\|\hat{f}\|_{\mathcal{D}_1}}{r_\zeta^{|N|}r_\eta^{|j|}}\leq\frac{\|\hat{f}\|_{\mathcal{D}_1}r_\eta}{r_\zeta^N(r_\eta-1)}$$

and the result is obtained for $\left|F_N(f-f^d)\right|$ case. The $\left|F_N(g-g^d)\right|$ case is similar. □

Corollary A.1.10. Let $\hat{f}(\eta,\zeta)\in\bar{\mathbb{A}}(\mathcal{D}_0)$ and if $\hat{f}^d$ is a discretized approximation of $\hat{f}$ of degree $\deg(f^d)$, then

$$\left|F_N(f-f^d)\right|\leq\frac{\|\hat{f}\|_{\mathcal{D}_0}}{r_\zeta^N}\left(\frac{1_{N\leq\deg(f^d)}\alpha_1}{r_\eta^{\deg(f)}}+1_{N>\deg(f^d)}(r_\zeta-1)\alpha_2\right),\tag{A.1.24}$$

for all $N\geq 0$, where

$$\alpha_1=\frac{1}{(r_\eta-1)}\left(\frac{r_\eta c_1(\deg(f^d),\deg(f^d))}{r_\zeta^{\deg(f^d)}}+1\right)\ ,\ \alpha_2=\frac{r_\eta}{(r_\eta-1)(r_\zeta-1)}\ ,\tag{A.1.25}$$

A.2 Analyticity and $\| \|_{\mathcal{D}_0}$ Bounds

Definition A.2.1. We define functionals σ_1, σ_2 on $\bar{\mathbb{A}}(\mathcal{D}_0)$ as

$$\sigma_1(f) = \max_{c_1,c_2\in\{-1,1\}} \sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} |f_{N,j}|\, r_\zeta^{c_1 N} r_\eta^{c_2 j} \text{ for } f = \sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} f_{N,j}\zeta^N\eta^j \in \bar{\mathbb{A}}(\mathcal{D}_0)\ , \tag{A.2.1}$$

$$\sigma_2(f) = 2\,|f_{0,0}| - \sigma_1(f) \text{ for } f = \sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} f_{N,j}\zeta^N\eta^j \in \bar{\mathbb{A}}(\mathcal{D}_0)\ , \tag{A.2.2}$$

Lemma A.2.2. Let $f(\eta,\zeta) \in \bar{\mathbb{A}}(\mathcal{D}_0)$, then

$$\sigma_2(f) \le \min_{(\eta,\zeta)\in\mathcal{D}_0} |f(\eta,\zeta)| \le \|f\|_{\mathcal{D}_0} \le \sigma_1(f), \tag{A.2.3}$$

Proof. Maximum modulus principle implies that the maximum of f is attained at the boundary on $\mathcal{D}_0$, i. e. when $|\eta| = r_\eta^{c_1}$ and $|\zeta| = r_\zeta^{c_2}$ for some $c_1, c_2 \in \{-1,1\}$, and since

$$|f(\eta,\zeta)| \le \sum_{N=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} |f_{N,j}||\zeta|^N|\eta|^j \le \sigma_1(f) \tag{A.2.4}$$

hold for all (η,ζ) on the boundary of $\mathcal{D}_0$, $\|f\|_{\mathcal{D}_0} \le \sigma_1(f)$ follows. Also, it is observed that

$$\sigma_2(f) = |f_{0,0}| - \sigma_1(f - f_{0,0}) \le |f_{0,0}| - |f(\eta,\zeta) - f_{0,0}| \le |f(\eta,\zeta)| \quad \text{for all } (\eta,\zeta)\in\mathcal{D}_0, \tag{A.2.5}$$

and the result follows. □

Definition A.2.3. We set

$$\alpha^0(\eta,\zeta) := \frac{\pi}{4\rho}(\zeta^2-\rho^2)Z_\zeta^0 - 1 = \frac{(\zeta^2-\rho^2)}{\rho}\left(Z_\zeta^0(\eta,\zeta) - \frac{\rho^2}{\zeta^2}Z_\zeta^0(\eta,\frac{\rho^2}{\zeta})\right) \tag{A.2.6}$$

Remark A.2.4. It is calculated that

$$1 + \alpha^0 = 1 + \sum_{n=0}^{13} \tilde{p}_n \left(\zeta^n + \frac{\rho^{2n}}{\zeta^n}\right) + \sum_{j=1}^{4} \frac{N_j}{D_j^2} \tag{A.2.7}$$

where

$$\tilde{p}_n = \begin{cases} (n-1)\mathtt{p}_{n-1} - \rho^2(n+1)\mathtt{p}_{n+1} & 0 \le n \le 13, \\ 0 & n < 0 \text{ or } n > 13 \end{cases} \tag{A.2.8}$$

and

$$N_j = n_j(d_j^2 - \rho^2)\left(\zeta - \frac{\rho^2}{\zeta}\right)^2 \ , \ D_j = (d_j - \zeta)\left(d_j - \frac{\rho^2}{\zeta}\right) \tag{A.2.9}$$

and if the choice of r_η and r_ζ ensures that $\sigma_2(D_j) > 0$ for $j = 1, \ldots, 4$ then $1 + \alpha^0 \in \bar{\mathbb{A}}(\mathcal{D}_0)$.

Lemma A.2.5. Let m be a positive integer and let

$$\begin{aligned}\alpha_m^0(\eta, \zeta) := \sum_{n=0}^{13} \tilde{p}_n \left(\zeta^n + \frac{\rho^{2n}}{\zeta^n}\right) + \sum_{j=1}^{4} n_j \Big[2 - \frac{2d_j}{d_{j,0}} \sum_{n=0}^{m} \left(\tilde{d}_j(\zeta)^n + \tilde{d}_j(\frac{\rho^2}{\zeta})^n\right) \\ + \frac{(d_j^2 - \rho^2)}{d_{j,0}^2} \sum_{n=1}^{m} n \left(\tilde{d}_j(\zeta)^{n-1} + \tilde{d}_j(\frac{\rho^2}{\zeta})^{n-1}\right)\Big]\end{aligned} \tag{A.2.10}$$

where

$$\tilde{d}_j(\zeta) = \frac{\zeta}{d_{j,0}} - \frac{(d_j - d_{j,0})}{d_{j,0}} = \frac{\zeta}{d_{j,0}} - \sum_{k=1}^{5} \frac{d_{j,k}}{d_{j,0}} \left(\eta^k + \frac{1}{\eta^k}\right) \text{ for } j = 1, \ldots, 4, \tag{A.2.11}$$

and let

$$X_j = \sigma_1\left(\tilde{d}_j(\zeta)\right) \ , \ Y_j = \sigma_1\left(\tilde{d}_j(\frac{\rho^2}{\zeta})\right) \quad \text{for } j = 1, \ldots, 4. \tag{A.2.12}$$

If $X_j, Y_j < 1$ for all $j = 1, \ldots, 4$, then

$$\|\alpha^0 - \alpha_m^0\|_{\mathcal{D}_0} \le t_{\mathcal{D}_0}(\alpha^0, m) \ , \tag{A.2.13}$$

where

$$\begin{aligned}t_{\mathcal{D}_0}(\alpha^0, m) = \sum_{j=1}^{4} \Big[\sigma_1\left(\frac{2n_j d_j}{d_{j,0}}\right)\left(\frac{X_j^{m+1}}{1 - X_j} + \frac{Y_j^{m+1}}{1 - Y_j}\right) \\ + \sigma_1\left(\frac{n_j(d_j^2 - \rho^2)}{d_{j,0}^2}\right)\left(\frac{(m+1)X_j^m}{1 - X_j} - \frac{X_j^{m+1}}{(1 - X_j)^2} + \frac{(m+1)Y_j^m}{1 - Y_j} - \frac{Y_j^{m+1}}{(1 - Y_j)^2}\right)\Big]\end{aligned} \tag{A.2.14}$$

Proof. It is observed from $X_j < 1$ that

$$\begin{aligned}\frac{(\zeta^2 - \rho^2)}{(d_j - \zeta)^2} &= 1 - \frac{2d_j}{d_j - \zeta} + \frac{d_j^2 - \rho^2}{(d_j - \zeta)^2} = 1 - \frac{2d_j}{d_{j,0}} \frac{1}{(1 - \tilde{d}_j(\zeta))} + \frac{(d_j^2 - \rho^2)}{d_{j,0}^2} \frac{1}{(1 - \tilde{d}_j(\zeta))^2} \\ &= 1 - \frac{2d_j}{d_{j,0}} \sum_{n=0}^{\infty} \tilde{d}_j(\zeta)^n + \frac{(d_j^2 - \rho^2)}{d_{j,0}^2} \sum_{n=1}^{\infty} n \tilde{d}_j(\zeta)^{n-1},\end{aligned}$$

and similarly, $\zeta \to \frac{\rho^2}{\zeta}$ and $Y_j < 1$ imply

$$\frac{(\frac{\rho^4}{\zeta^2} - \rho^2)}{(d_j - \frac{\rho^2}{\zeta})^2} = 1 - \frac{2d_j}{d_{j,0}} \sum_{n=0}^{\infty} \tilde{d}_j (\frac{\rho^2}{\zeta})^n + \frac{(d_j^2 - \rho^2)}{d_{j,0}^2} \sum_{n=1}^{\infty} n \tilde{d}_j (\frac{\rho^2}{\zeta})^{n-1},$$

and hence,

$$\begin{aligned} \alpha^0 = \sum_{n=0}^{13} \tilde{p}_n \left(\zeta^n + \frac{\rho^{2n}}{\zeta^n} \right) + \sum_{j=1}^{4} n_j \Big[2 - \frac{2d_j}{d_{j,0}} \sum_{n=0}^{\infty} \left(\tilde{d}_j(\zeta)^n + \tau_j (\frac{\rho^2}{\zeta})^n \right) \\ + \frac{(d_j^2 - \rho^2)}{d_{j,0}^2} \sum_{n=1}^{\infty} n \left(\tilde{d}_j(\zeta)^{n-1} + \tilde{d}_j (\frac{\rho^2}{\zeta})^{n-1} \right) \Big] \end{aligned} \tag{A.2.15}$$

and thus

$$\begin{aligned} &\left\| \alpha^0 - \alpha_m^0 \right\|_{\mathcal{D}_1} = \Big\| \sum_{j=1}^{4} -\frac{2n_j d_j}{d_{j,0}} \sum_{n=m+1}^{\infty} \left(\tilde{d}_j(\zeta)^n + \tilde{d}_j (\frac{\rho^2}{\zeta})^n \right) \\ &+ \frac{n_j (d_j^2 - \rho^2)}{d_{j,0}^2} \sum_{n=m+1}^{\infty} n \left(\tilde{d}_j(\zeta)^{n-1} + \tilde{d}_j (\frac{\rho^2}{\zeta})^{n-1} \right) \Big\|_{\mathcal{D}_1} \\ &\leq \sum_{j=1}^{4} \left[\sigma_1 \left(\frac{2n_j d_j}{d_{j,0}} \right) \sum_{n=m+1}^{\infty} \left(X_j^n + Y_j^n \right) + \sigma_1 \left(\frac{n_j (d_j^2 - \rho^2)}{d_{j,0}^2} \right) \sum_{n=M+1}^{\infty} n \left(X_j^{n-1} + Y_j^{n-1} \right) \right] \end{aligned}$$

and the result is obtained from a straightforward calculation of the geometric sums. □

Lemma A.2.6. Let r_η and r_ζ be such that $\sigma_2(D_j) > 0$ for $j = 1, \ldots, 4$ in $\mathcal{D}_0$. Let m be a positive integer such that $\sigma_2(1 + \alpha^0) - t_{\mathcal{D}}(\alpha^0, m) > 0$. Then $\zeta \omega_\zeta^0(\eta, \zeta) \in \hat{\mathbb{A}}(\mathcal{D}_1)$ and

$$\left\| \zeta \omega_\zeta^0 \right\|_{\mathcal{D}_0} \leq \lambda_0(m) \ , \tag{A.2.16}$$

where

$$\lambda_0(m) = \frac{\left[\sigma_1 \left(\sum_{n=0}^{13} \tilde{p}_n \left(\zeta^n + \frac{\rho^{2n}}{\zeta^n} \right) \right) + \sum_{j=1}^{4} \frac{1}{\sigma_2(D_j)^3} \sigma_1 \left(2N_j D_j \left(\zeta + \frac{\rho^2}{\zeta} \right) - d_j \left(\zeta - \frac{\rho^2}{\zeta} \right)^2 \right) \right]}{\sigma_2(1 + \alpha^0) - t_{\mathcal{D}_0}(\alpha^0, m)}, \tag{A.2.17}$$

Proof. It is calculated that

$$1 + \alpha = 1 + \sum_{n=0}^{13} \tilde{p}_n \left(\zeta^n + \frac{\rho^{2n}}{\zeta^n} \right) + \sum_{j=1}^{4} \frac{n_j \left(\zeta - \frac{\rho^2}{\zeta} \right)^2 (d_j^2 - \rho^2)}{(d_j - \zeta)^2 (d_j - \frac{\rho^2}{\zeta})^2} \tag{A.2.18}$$

It is straightforward to observe

$$\zeta\alpha_\zeta = \sum_{n=1}^{13} n\tilde{p}_n \left(\zeta^n - \frac{\rho^{2n}}{\zeta^n}\right) + \sum_{j=1}^{4} 2N_j \left(\frac{\left(\zeta + \frac{\rho^2}{\zeta}\right)}{D_j^2} - \frac{d_j\left(\zeta - \frac{\rho^2}{\zeta}\right)^2}{D_j^3}\right) \tag{A.2.19}$$

where

$$N_j = n_j(d_j^2 - \rho^2)\left(\zeta - \frac{\rho^2}{\zeta}\right)^2 \ , \ D_j = (d_j - \zeta)\left(d_j - \frac{\rho^2}{\zeta}\right) \tag{A.2.20}$$

and the result follows from Lemma A.2.2. □

Lemma A.2.7. Let r_η and r_ζ be such that $\sigma_2(D_j) > 0$ for $j = 1, \ldots, 4$ and $r_\zeta > \rho_\eta$ where

$$\rho_\eta := \frac{r_\eta + r_\eta^{-1} + 12}{20}, \tag{A.2.21}$$

then $\zeta Z_\zeta^0(\eta, \zeta) \in \hat{\mathbb{A}}(\mathcal{D}_0)$. Let m be a positive integer, then

$$\left\|\zeta Z_\zeta^0\right\|_{\mathcal{D}_0} \leq \lambda_1(m) \ , \tag{A.2.22}$$

where

$$\lambda_1(m) := \frac{4}{\pi}\frac{r_\zeta \rho_\eta}{(r_\zeta^2 - \rho_\eta^2)}\left(\sigma(1 + \alpha_m^0) + t_{\mathcal{D}_0}(\alpha^0, m)\right) \ , \tag{A.2.23}$$

Proof. $|\rho| \leq \rho_\eta$ and $|\zeta| \leq r_\zeta$ on $(\eta, \zeta) \in \mathcal{D}_1$ imply

$$\left|\zeta Z_\zeta^0(\eta, \zeta)\right| = \left|\frac{4}{\pi}\frac{\zeta\rho}{(\zeta^2 - \rho^2)}(1 + \alpha(\eta, \zeta))\right| \leq \frac{4}{\pi}\frac{|\zeta||\rho|}{(|\zeta|^2 - |\rho|^2)}\|1 + \alpha^0\|_{\mathcal{D}_0} \leq \lambda_1(m), \tag{A.2.24}$$

and the result follows. □

Lemma A.2.8. Let

$$\hat{W}(\eta, \zeta) = \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{\rho^{2n+1}}{(1 - \rho^{4n+2})}(\zeta^{2n+1} - \zeta^{-(2n+1)}), \tag{A.2.25}$$

and let r_ζ, r_η are chosen such that $r_\zeta > \rho_\eta$, then $\hat{W}(\eta, \zeta) \in \hat{\mathbb{A}}(\mathcal{D}_0)$ and

$$\left\|\hat{W}\right\|_{\mathcal{D}_1} \leq \lambda_2 \tag{A.2.26}$$

where

$$\lambda_2 := \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{\rho_\eta^{2n+1}}{1 - \rho_\eta^{4n+2}}\left(r_\zeta^{2n+1} + r_\zeta^{-(2n+1)}\right) \ , \tag{A.2.27}$$

and

Proof. and similarly, $|\rho| \leq \rho_\eta$ and $|\zeta|^{2n+1} + |\zeta|^{-(2n+1)} \leq r_\zeta^{2n+1} + r_\zeta^{-(2n+1)}$ on $(\eta, \zeta) \in \mathcal{D}_1$ imply

$$\left|\hat{W}(\eta,\zeta)\right| \leq \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{|\rho|^{2n+1}}{(1-|\rho|^{4n+2})} (|\zeta|^{2n+1} + |\zeta|^{-(2n+1)}) \leq \lambda_2 \tag{A.2.28}$$

and the proof is complete. □

Lemma A.2.9. Let r_η and r_ζ be such that $\sigma_2(D_j) > 0$ for $j = 1, \ldots, 4$ and $r_\zeta > \rho_\eta$. Let m be a positive integer and let

$$\hat{a}(\eta,\zeta) = \frac{\hat{\mathtt{a}}_1(\eta,\zeta)}{2i} - \frac{\hat{\mathtt{a}}_1(\eta,\frac{1}{\zeta})}{2i} \ , \ \hat{b}(\eta,\zeta) = \frac{\hat{\mathtt{a}}_1(\eta,\zeta)}{2} + \frac{\hat{\mathtt{a}}_1(\eta,\frac{1}{\zeta})}{2} \ , \ \hat{c}(\eta,\zeta) = \frac{\hat{\mathtt{a}}_2(\eta,\zeta)}{2i} - \frac{\hat{\mathtt{a}}_2(\eta,\frac{1}{\zeta})}{2i} \tag{A.2.29}$$

where

$$\begin{aligned} \hat{\mathtt{a}}_1(\eta,\zeta) &= -\frac{1}{B} \left(\zeta \hat{Z}_\zeta^0(\eta,\zeta)\right)^{1/2} \left(\frac{1}{\zeta} \hat{Z}_\zeta^0(\eta,\frac{1}{\zeta})\right)^{1/2} \zeta \hat{Z}_\zeta^0(\eta,\zeta), \\ \hat{\mathtt{a}}_2(\eta,\zeta) &= -\frac{1}{B} \left(\zeta \hat{Z}_\zeta^0(\eta,\zeta)\right)^{1/2} \left(\frac{1}{\zeta} \hat{Z}_\zeta^0(\eta,\frac{1}{\zeta})\right)^{1/2} \hat{W}(\eta,\zeta) \end{aligned} \tag{A.2.30}$$

Then, $\hat{a}, \hat{b}, \hat{c} \in \bar{\mathbb{A}}(\mathcal{D}_0)$, and

$$\|\hat{a}\|_{\mathcal{D}_0} \leq \frac{\lambda_1(m)^2}{B} \ , \ \left\|\hat{b}\right\|_{\mathcal{D}_0} \leq \frac{\lambda_1(m)^2}{B} \ , \ \|\hat{c}\|_{\mathcal{D}_0} \leq \frac{\lambda_1(m)\lambda_2}{B} \tag{A.2.31}$$

Moreover,

$$a = \hat{a}(e^{i\theta}, e^{i\nu}) \ , \ b = \hat{b}(e^{i\theta}, e^{i\nu}) \ , \ c = \hat{c}(e^{i\theta}, e^{i\nu}) \text{ on } (\theta,\nu) \in [\pi, 2\pi] \times [0, 2\pi], \tag{A.2.32}$$

Proof. This follows immediately from (2.4.46) and from previous Lemmas. □

Lemma A.2.10. Let r_η and r_ζ be such that $\sigma_2(D_j) > 0$ for $j = 1, \ldots, 4$ and $r_\zeta > \rho_\eta$. Let m be a positive integer, and let

$$\hat{A}_1 = \hat{a} + \hat{c} \ , \ \hat{A}_2 = 2U^0 \hat{a} + (U^0 - 1)\hat{c} \ , \ \hat{A}_3 = U^0 \hat{b} \ , \tag{A.2.33}$$

then $\hat{A}_j(\eta,\zeta) \in \bar{\mathbb{A}}(\mathcal{D}_0)$ for $j = 1,2,3$ and

$$\begin{aligned}
\left\|\hat{A}_1\right\|_{\mathcal{D}_0} &\leq \frac{\lambda_1(m)(\lambda_1(m)+\lambda_2)}{B}\ , \\
\left\|\hat{A}_2\right\|_{\mathcal{D}_0} &\leq \frac{\lambda_1(m)(2\sigma_1(U^0)\lambda_1(m)+\sigma_1(U^0-1)\lambda_2)}{B}\ , \\
\left\|\hat{A}_3\right\|_{\mathcal{D}_0} &\leq \frac{\sigma_1(U^0)\lambda_1(m)\lambda_2}{B}\ ,
\end{aligned} \tag{A.2.34}$$

Moreover,

$$A_j = \hat{A}_j(e^{i\theta}, e^{i\nu}) \text{ on } (\theta,\nu) \in [\pi, 2\pi] \times [0, 2\pi], \tag{A.2.35}$$

for $j = 1,2,3$.

Proof. This follows immediately from (2.4.45) and from the previous Lemma. □

A.3 Univalency

Lemma A.3.1. Let $\eta \in \mathbb{C}$ be fixed where $|\eta| = 1$. Let $\zeta_1, \zeta_2 \in \{|\zeta| = 1\} \cap \{\Im(\zeta) \geq 0\}$. Then

$$Z(\eta,\zeta_1) = Z(\eta,\zeta_2) \Rightarrow \zeta_1 = \zeta_2, \tag{A.3.1}$$

Proof. Let's assume that there exist $\zeta_1, \zeta_2 \in \{|\zeta| = 1\}$ where $Z(\zeta,\zeta_1) = Z(\zeta,\zeta_2)$ and $\zeta_1 \neq \zeta_2$. Denote $\zeta_j = e^{i\nu_j}$ for $j = 1,2$ where $\nu_j \in [0,\pi]$ and without losing generality assume that $\nu_1 < \nu_2$. We prove that this is impossible separately for the cases $\nu_2 - \nu_1 > \frac{2\pi}{3}$ and $\nu_2 - \nu_1 \leq \frac{2\pi}{3}$. In the case when $\nu_2 - \nu_1 > \frac{2\pi}{3}$, one has

$$\begin{aligned}
&\Re\left(Z(\eta,e^{i\nu_2}) - Z(\eta,e^{i\nu_1})\right) = \Re\left([Z-Z^0](\eta,e^{i\nu_2}) - [Z-Z^0](\eta,e^{i\nu_1})\right) \\
&+ \Re\left(Z^0(\eta,e^{i\nu_2}) - \frac{2}{\pi}\log\left(\frac{e^{i\nu_2}-\rho}{e^{i\nu_2}+\rho}\right)\right) - \Re\left(Z^0(\eta,e^{i\nu_1}) - \frac{2}{\pi}\log\left(\frac{e^{i\nu_1}-\rho}{e^{i\nu_1}+\rho}\right)\right) \\
&+ \frac{2}{\pi}\Re\left(\log\left(\frac{e^{i\nu_2}-\rho}{e^{i\nu_2}+\rho}\right) - \log\left(\frac{e^{i\nu_1}-\rho}{e^{i\nu_1}+\rho}\right)\right)
\end{aligned} \tag{A.3.2}$$

It is observed that

$$\begin{aligned}
&\left|\Re\left([Z-Z^0](\eta,e^{i\nu_2})-[Z-Z^0](\eta,e^{i\nu_1})\right)\right| \leq \left|[Z-Z^0](\eta,e^{i\nu_2})-[Z-Z^0](\eta,e^{i\nu_1})\right| \\
&= \left|\int_{\nu_1}^{\nu_2}\frac{d}{d\nu}[Z-Z^0](\eta,e^{i\nu})d\nu\right| = \left|\int_{\nu_1}^{\nu_2} i\zeta(Z_\zeta - Z^0_\zeta)d\nu\right| \leq (\nu_2-\nu_1)|i||\zeta(Z_\zeta - Z^0_\zeta)| \\
&\leq \pi|\zeta Z^0_\zeta(e^\Omega-1)| \leq \pi\frac{4}{\pi}\left|\frac{\rho\zeta}{\zeta^2-\rho^2}+\sum_{n=1}^{\infty} n\left(P_n+\sum_{j=1}^{4}\frac{n_j}{d_j^{n+1}}\right)\left(\zeta^n-\frac{\rho^{2n}}{\zeta^n}\right)\right||\Omega|e^{|\Omega|} \\
&\leq 4\left(\frac{\rho}{1-\rho^2}+\sum_{n=1}^{\infty} n\left(\sigma_1(P_n)+\sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right)\right)|\Omega|e^{|\Omega|}
\end{aligned} \tag{A.3.3}$$

where $\zeta = e^{i\nu}$. Similarly, it is observed that

$$\begin{aligned}
&\left|\Re\left(Z^0(\eta,\zeta)-\frac{2}{\pi}\log\left(\frac{\zeta-\rho}{\zeta+\rho}\right)\right)\right| \leq \left|Z^0(\eta,\zeta)-\frac{2}{\pi}\log\left(\frac{\zeta-\rho}{\zeta+\rho}\right)\right| \\
&= \frac{4}{\pi}\left|\sum_{n=1}^{\infty}\left(P_n+\sum_{j=1}^{4}\frac{n_j}{d_j^{n+1}}\right)\left(\zeta^n+\frac{\rho^{2n}}{\zeta^n}\right)\right| \leq \frac{4}{\pi}\sum_{n=1}^{\infty}\left(\sigma_1(P_n)+\sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right)
\end{aligned} \tag{A.3.4}$$

and evidently,

$$\begin{aligned}
\Re\left(\log\left(\frac{e^{i\nu_2}-\rho}{e^{i\nu_2}+\rho}\right)-\log\left(\frac{e^{i\nu_1}-\rho}{e^{i\nu_1}+\rho}\right)\right) &= \log\left(\frac{1+\rho^2-2\rho\cos(\nu_2)}{1+\rho^2+2\rho\cos(\nu_2)}\right) \\
&\quad -\log\left(\frac{1+\rho^2-2\rho\cos(\nu_1)}{1+\rho^2+2\rho\cos(\nu_1)}\right)
\end{aligned}$$

and $\nu_2-\nu_1 > \frac{2\pi}{3}$ implies $\nu_1 \in [0,\frac{\pi}{3}]$ and $\nu_2 \in [\pi-\frac{\pi}{3},\pi]$ and thus $\log\left(\frac{1+\rho^2-2\rho\cos(\nu_2)}{1+\rho^2+2\rho\cos(\nu_2)}\right) \geq \log\left(\frac{1+\rho^2+\rho}{1+\rho^2-\rho}\right)$ and $\log\left(\frac{1+\rho^2-2\rho\cos(\nu_1)}{1+\rho^2+2\rho\cos(\nu_1)}\right) \leq \log\left(\frac{1+\rho^2-\rho}{1+\rho^2+\rho}\right)$ and hence,

$$\begin{aligned}
&\Re\left(Z(\eta,e^{i\nu_2})-Z(\eta,e^{i\nu_1})\right) \geq \frac{4}{\pi}\ln\left(\frac{1+\rho^2+\rho}{1+\rho^2-\rho}\right)-\left|[Z-Z^0](\eta,e^{i\nu_2})-[Z-Z^0](\eta,e^{i\nu_1})\right| \\
&-2\left|Z^0(\eta,\zeta)-\frac{2}{\pi}\log\left(\frac{\zeta-\rho}{\zeta+\rho}\right)\right| \geq \frac{4}{\pi}\ln\left(\frac{1+\rho^2+\rho}{1+\rho^2-\rho}\right) \\
&-\frac{4}{\pi}\sum_{n=1}^{\infty}\left(\sigma_1(P_n)+\sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right) \\
&-4\left(\frac{\rho}{1-\rho^2}+\sum_{n=1}^{\infty} n\left(\sigma_1(P_n)+\sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right)\right)|\Omega|e^{|\Omega|} > 0.1 \text{ when } \rho\in[0.5,0.7]
\end{aligned}$$

which contradicts with the assumption that $Z(\eta,\zeta_1) = Z(\eta,\zeta_2)$. Now, we assume $\nu_2 -$

$\nu_1 \leq \frac{2\pi}{3}$. Let $\alpha, \beta \in \mathbb{R}$, then $Z(\eta, \zeta_1) = Z(\eta, \zeta_2)$ implies $\alpha \Re Z(\eta, e^{i\nu_2}) + \beta \Im Z(\eta, e^{i\nu_2}) = \alpha \Re Z(\eta, e^{i\nu_1}) + \beta \Im Z(\eta, e^{i\nu_1})$ and mean value theorem implies that there exists a $\nu_{(\alpha,\beta)} \in (\nu_1, \nu_2)$ where

$$\frac{d}{d\nu}\left(\alpha \Re Z(\eta, e^{i\nu}) + \beta \Im Z(\eta, e^{i\nu})\right)_{\nu=\nu_{(\alpha,\beta)}} = 0, \tag{A.3.5}$$

this is on one hand, on the other hand one has

$$\begin{aligned}
&\frac{d}{d\nu}\left(\alpha \Re Z(\eta, e^{i\nu}) + \beta \Im Z(\eta, e^{i\nu})\right) = \alpha \Re\left(i\zeta Z_\zeta(\eta, \zeta)\right) + \beta \Im\left(i\zeta Z_\zeta(\eta, \zeta)\right) \\
&= \frac{4}{\pi}\left[\alpha \Re\left(\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right) + \beta \Im\left(\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right)\right] + \alpha \Re\left(i\zeta(Z_\zeta - Z_\zeta^0)\right) + \beta \Im\left(i\zeta(Z_\zeta - Z_\zeta^0)\right) \\
&+ \alpha \Re\left(i\zeta Z_\zeta^0 - \frac{4}{\pi}\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right) + \beta \Im\left(i\zeta Z_\zeta^0 - \frac{4}{\pi}\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right)
\end{aligned} \tag{A.3.6}$$

it is observed that

$$\alpha \Re\left(\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right) + \beta \Im\left(\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right) = \frac{\alpha\rho(1+\rho^2)\sin(\nu) + \beta\rho(1-\rho^2)\cos(\nu))}{1 + \rho^4 - 2\rho^2\cos(2\nu)} \tag{A.3.7}$$

and that

$$\begin{aligned}
&\left|\alpha \Re\left(i\zeta(Z_\zeta - Z_\zeta^0)\right) + \beta \Im\left(i\zeta(Z_\zeta - Z_\zeta^0)\right)\right| \leq (|\alpha| + |\beta|)\left|i\zeta(Z_\zeta - Z_\zeta^0)\right| \\
&\leq (|\alpha| + |\beta|)|i\zeta||(Z_\zeta - Z_\zeta^0)| \\
&\leq (|\alpha| + |\beta|)\frac{4}{\pi}\left(\frac{\rho}{1-\rho^2} + \sum_{n=1}^{\infty} n\left(\sigma_1(P_n) + \sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1 + \rho^{2n}\right)\right)|\Omega|e^{|\Omega|}
\end{aligned} \tag{A.3.8}$$

by an argument similar to (A.3.3). It is also observed that

$$\begin{aligned}
&\left|\alpha \Re\left(i\zeta Z_\zeta^0 - \frac{4}{\pi}\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right) + \beta \Im\left(i\zeta Z_\zeta^0 - \frac{4}{\pi}\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right)\right| \leq (|\alpha| + |\beta|)\left|i\zeta Z_\zeta^0 - \frac{4}{\pi}\frac{i\zeta\rho}{\zeta^2 - \rho^2}\right| \\
&\leq (|\alpha| + |\beta|)\frac{4}{\pi}\left|\sum_{n=1}^{\infty} n\left(P_n + \sum_{j=1}^{4}\frac{n_j}{d_j^{n+1}}\right)\left(\zeta^n - \frac{\rho^{2n}}{\zeta^n}\right)\right| \\
&\leq (|\alpha| + |\beta|)\frac{4}{\pi}\sum_{n=1}^{\infty} n\left(\sigma_1(P_n) + \sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1 + \rho^{2n}\right),
\end{aligned} \tag{A.3.9}$$

Let $\nu^* = \frac{\nu_2 - \nu_1}{2}$. On choosing $\alpha = (1-\rho^2)\sin(\nu^*)$ and $\beta = (1+\rho^2)\cos(\nu^*)$ one obtains

$$\frac{\alpha\rho(1+\rho^2)\sin(\nu) + \beta\rho(1-\rho^2)\cos(\nu))}{1 + \rho^4 - 2\rho^2\cos(2\nu)} = \frac{\rho(1-\rho^4)\cos(\nu - \nu^*)}{1 + \rho^4 - 2\rho^2\cos(2\nu)} \tag{A.3.10}$$

$\nu_2 - \nu_1 \leq \frac{2\pi}{3}$ implies $|\nu_{(\alpha,\beta)} - \nu^*| \leq \frac{\pi}{3}$ and hence $\cos|\nu_{(\alpha,\beta)} - \nu^*| \geq \frac{1}{2}$ and hence

$$\frac{\alpha\rho(1+\rho^2)\sin(\nu) + \beta\rho(1-\rho^2)\cos(\nu))}{1+\rho^4-2\rho^2\cos(2\nu)}\bigg|_{\nu=\nu_{(\alpha,\beta)}} = \frac{\rho(1-\rho^4)\cos|\nu_{(\alpha,\beta)} - \nu^*|}{1+\rho^4-2\rho^2\cos(2\nu)} \geq \frac{\rho(1-\rho^4)}{2(1+\rho^2)^2} \tag{A.3.11}$$

on combining results, one obtains

$$\begin{aligned}
&\frac{d}{d\nu}\left(\alpha\Re Z(\eta, e^{i\nu}) + \beta\Im Z(\eta, e^{i\nu})\right)_{\nu=\nu_{(\alpha,\beta)}} \geq \frac{2\rho(1-\rho^4)}{\pi(1+\rho^2)^2} \\
&- \frac{8}{\pi}\sum_{n=1}^{\infty} n\left(\sigma_1(P_n) + \sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right) \\
&- \frac{8}{\pi}\left(\frac{\rho}{1-\rho^2} + \sum_{n=1}^{\infty} n\left(\sigma_1(P_n) + \sum_{j=1}^{4}\frac{\sigma_1(n_j)}{\sigma_2(d_j)^{n+1}}\right)\left(1+\rho^{2n}\right)\right)|\Omega|e^{|\Omega|} > 0.1,
\end{aligned} \tag{A.3.12}$$

when $\rho \in [0.5, 0.7]$ which contradicts with (A.3.5), and this completes the proof. □

Appendix B
Appendix for P1 Analysis

Lemma B.0.1. Let $y \in C^1[a,b]$. then

$$\|y - \frac{1}{2}\left(y(a) + y(b)\right)\|_\infty \leq \frac{1}{2}\sqrt{b-a}\|y'\|_{L_2(a,b)} \tag{B.0.1}$$

$$\|y\|_\infty \leq \frac{1}{2}\Big|y(a) + y(b)\Big| + \frac{1}{2}\sqrt{b-a}\|y'\|_{L_2(a,b)} \tag{B.0.2}$$

Proof. Note that $y(x) = y(a) + \int_a^x y'(t)dt$ and $y(x) = y(b) - \int_x^b y'(t)dt$. Therefore,

$$y(x) = \frac{1}{2}\left(y(a) + y(b)\right) + \frac{1}{2}\int_a^x y'(t)dt - \frac{1}{2}\int_x^b y'(t)dt$$

Therefore, it follows that

$$\Big|y(x) - \frac{1}{2}\left(y(a) + y(b)\right)\Big| \leq \frac{1}{2}\int_a^b |y'(t)|$$

The bounds (B.0.1) immmediately follow from using Cauchy-Schwartz inequality. The second part (B.0.2) follows from (B.0.1) simply from triangular inequality. □

Lemma B.0.2. Let $y \in C^1[a,b]$, then it follows that

$$\begin{aligned}\|y\|_\infty \leq \frac{1}{2}\max_{0\leq k\leq n-1}\Bigg\{&\frac{1}{2}\Big|y\left(a + \frac{k}{n}(b-a)\right) + y\left(1 + \frac{(k+1)}{n}(b-a)\right)\Big| \\ &+ \sqrt{\frac{b-a}{n}}\left\{\int_{a+\frac{k}{n}(b-a)}^{a+\frac{(k+1)}{n}(b-a)} {y'}^2(t)dt\right\}^{1/2}\Bigg\}\end{aligned} \tag{B.0.3}$$

Proof. This follows from using previous Lemma on each interval $\left(a + \frac{k(b-a)}{n}, a + \frac{(k+1)(b-a)}{n}\right)$ and then taking the maximum over the n intervals. □